Intent-Aware Logic-Constrained Neuro-Symbolic Decision Support System with Confidence Calibration and Reliance Support for SME Multi-Channel Marketing

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ACADEMIC YEAR
2025/26

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A project report submitted to the
Faculty of Computing and Information Technology
in partial fulfilment of the requirement for the
Bachelor of Software Engineering (Honours)

Department of Software Engineering and Technology
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Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.



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Abstract

Small and medium-sized enterprises often face difficulties in making effective multi-channel marketing decisions because campaign data, business objectives and operational constraints are not considered together. Existing decision-support approaches may provide predictions without sufficient explanation, intent awareness or logical safeguards, making recommendations difficult to interpret and trust.

This project aims to develop an intent-aware, logic-constrained neuro-symbolic decision support system that assists SMEs in selecting appropriate marketing actions while communicating the reasoning and confidence behind each recommendation.

The proposed methodology involves collecting and preprocessing campaign data, developing predictive models, incorporating manager-defined Survival, Retention and Growth intents, applying weighted soft-rule reasoning, and combining predictive and symbolic outputs through a structured decision framework. The system will recommend one of four actions: Increase, Maintain, Reduce or Investigate. It will also provide channel evidence, confidence information and Driver-Implication-Action-Caveat explanations to support managerial review.

The system will be implemented as a human-in-the-loop web application consisting of data management, predictive analytics, neuro-symbolic reasoning, trust and decision, and presentation layers. Its performance will be assessed through benchmark evaluation, comparison with baseline models, ablation testing, logical-conflict analysis, calibration assessment, explanation evaluation and a formative human study.

The expected contribution is a reproducible and auditable decision-support approach that combines machine-learning predictions with explicit business intent, logical constraints and interpretable explanations. The project is expected to improve the transparency, consistency and practical usefulness of marketing recommendations for SMEs, while establishing a structured foundation for future evaluation of user trust, actionability and appropriate reliance.

Acknowledgement

I would like to express my sincere gratitude to my project supervisor, Assistant Professor Dr. Low Choon Keat, for his guidance, advice, constructive feedback and support throughout this Final Year Project. His insights helped shape the direction and completion of the work.

I also thank the lecturers and faculty members of the Faculty of Computing and Information Technology, Tunku Abdul Rahman University of Management and Technology, for the academic foundation, resources and learning environment that supported this project.

My sincere thanks also go to my moderators, Dr Chin Wan Yoke and Ms Joan Hau, for their guidance, constructive feedback and support throughout the project.

I am grateful to the individuals who provided informal academic and technical feedback during prototype development. The formal participant evaluation is described separately in the methodology and ethics sections.

I am especially thankful to my family for their patience, encouragement and continued support throughout my academic journey.

I appreciate the support and contributions of everyone who assisted me throughout this project.

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Chapter 1

Introduction

1 Introduction

This chapter presents the gap between marketing evidence and action in SME multi-channel marketing and explains the decision-support approach used to address it. The problem statement, the research questions, the study objectives, scope, relevance, and the role of AI in the proposed system are also included in this chapter.

1.1 Background of the Study

SMEs are more and more relying on several digital marketing channels, such as social media, email marketing, search advertising, influencer campaigns, and online shopping. These channels include campaign-level data like impressions, clicks, leads, conversions, acquisition cost, revenue, ROI, and engagement rate. Marketing analytics assists organisations to explain this information for targeting, evaluation and budget allocation (Wedel & Kannan, 2016). However, the availability of campaign data doesn't necessarily translate into an appropriate managerial action.

Predictive analytics can offer a ballpark estimate of what's likely to happen in a campaign based on past campaign and channel data, but it's not enough on its own to guide the next step. Even if a campaign has a positive expected return, it may not be a good target for further investment due to the high cost of acquisition, shortage of cash, or short-term financial constraints. However, a campaign that is doing moderate work could still be worth continuing if it is protecting customer retention or an important market segment.

The marketing-evidence-to-action gap is defined in this thesis as the gap between the analytical evidence and an operational decision. This means that a system can measure campaign performance and predict what's likely to happen but can't convert that evidence into a bounded, defensible managerial action that is aligned with the organisation's current priorities, operational constraints and risk tolerance. This gap needs to be filled with more than predictive accuracy. It demands that the decision-making process incorporate learned evidence and explicit business reasoning so that the recommendation can be reviewed and utilised by a human decision-maker.

This is particularly a problem for small and medium-sized businesses. Typically these organisations have smaller marketing teams, less specialist data analytics capability than their larger counterparts and tighter budgets. Therefore, managers will need to consider various factors of a campaign simultaneously, such as cash flow, customer retention, growth potential and operational feasibility. Previous studies also indicate that SME performance is influenced

by strategy and data-analytics capability and that the impact of data-analytics capability varies according to the context of the firm (Wu et al., 2024; Zong et al., 2025). Hence, this project aims to solve a real and urgent problem of SME decision support.

The study uses two different evidence sources. The public campaign dataset provides campaign-level records for model development and training and the project-defined business-intent conditions and independently reviewed reference actions to evaluate against relevant campaign cases are provided in a separate audited decision benchmark.

The proposed system is meant to be used for decision-making and not for causal claims. CampaignNet estimates return on investment and provides predictive evidence for local sensitivity, and four bounded actions are generated by ActionNet under the selected business intent. A separate channel-signal component summarises historical performance. These evidence sources are then used in conjunction with explicit business safeguards, a declared business intent, and a defined risk tolerance.

The action vocabulary is restricted to Increase, Maintain, Reduce and Investigate. Additional investment or expansion is reflected by Increase when the supporting evidence is strong. Continuation of the current campaign strategy is indicated by Maintain. Reduce is a more conservative answer, when lower activity is warranted by performance, cost or strategic factors. Investigate flags when evidence is incomplete, conflicting, or operationally uncertain, requiring further review. This limited set of actions provides consistency for evaluation and prevents the creation of open-ended recommendations that are not comparable across cases.

The project has three business-intent scenarios: Survival, Retention, and Growth. Survival emphasizes cash preservation, cost control and risk reduction. Retention focuses on customer relationships, engagement quality and repeat value. Growth allows for more acquisition effort when expansion is warranted based on returns and operating conditions. These scenarios are used to assess the coherence of recommendations in the context of a stated strategic priority under controlled conditions.

A fixed set of 14 safeguards are used to model business constraints. Every safeguard is assessed as a binary condition and whether it has been violated. If a rule is violated which is relevant to that action, the score of that action is affected by the weight and priority of the rule. A smaller independent boost to the Investigate option can also be added when a non-Investigate action is violated. At the decision level, these rules are treated as soft as the actions are not excluded

from consideration but rather the scores are adjusted. So, logic-constrained is the effect of weighted rules and traceable violation scoring.

Data-driven evidence is combined with explicit symbolic reasoning to support this process. The architecture is loosely coupled neuro-symbolic, as CampaignNet and ActionNet are trained independently of the manually programmed rule catalogue. The learned components produce evidence, which is used by the symbolic layer in the action scoring process through the use of inspectable safeguards. The validation or test data are not employed to learn the rules or their weights.

The use of AI also raises concerns related to transparency, uncertainty and the reliance on humans. An explanation may help to make a recommendation understandable, but it does not guarantee its correctness. Likewise, when a model probability is high but not calibrated, too much trust may be given to the output by the user. The factors that contributed to a recommendation can be reviewed using Explainable AI (Ribeiro et al., 2016); however, it is important to ensure that the uncertainty is communicated correctly and that decisions are not made entirely by the AI, but also by the human users (Schemmer et al., 2023; Vasconcelos et al., 2023).

Validation-only temperature scaling is used to model the confidence for the Balanced profile and the observed 19-case validation distribution. The temperature-scaled output is profile- and validation-distribution-specific and is called calibrated model confidence. In addition to this qualified confidence, supporting evidence, applicable rule traces, ranked alternatives, and a structured Driver-Implication-Action-Caveat explanation are provided via the recommendation interface. The intent selected is stored as contextual metadata.

The resulting prototype is a tool that provides advisory decision support. Each recommendation is accepted, modified, deferred or rejected at the discretion of the manager. The study can explore how evidence of predictive and campaign level can be translated into bounded, context aware and auditable marketing recommendations through this human-in-the-loop boundary while maintaining managerial authority.

1.2 Problem Statement

Small and medium businesses marketing managers have to consider a number of indicators in a marketing campaign simultaneously. Analytical evidence is not sufficient to define what is most appropriate action. This study attempts to solve five related problems: How to translate

evidence into a bounded and actionable recommendation, how to clearly express strategic intent, how to show operational safeguards, how to communicate model confidence without claiming trust, and how to keep a usable record of the decision process for audit and review.

P1. Evidence-to-action gap. Predictive outputs and campaign-level indicators can provide a manager with information about the performance of the campaign, but they do not alone provide a manager with information on whether to invest more, maintain, reduce, or investigate the campaign.

P2. Low levels of strategic awareness. What is recommended for a growth-oriented SME can be inappropriate for a business that is focused on survival. Many systems are geared to general goals without taking into account whether the firm is in Survival, Retention or Growth mode. Consequently, their suggestions can be strategically off-target.

P3. Low level of integration of business constraints. Factors like budget, acceptable risk, margin protection, channel saturation, campaign-policies and operational capacity influence marketing decisions. Data-only models can generate seemingly appealing recommendations that are impractical. A more helpful decision support system, then, is one that is based on both analytical evidence and good business sense.

P4. Confidence and appropriate reliance. An explanation can make a recommendation understandable, but not necessarily correct. Model confidence should thus be calibrated independently, and human dependence should be based on the quality of the recommendation, not just confidence, or on satisfaction with the output.

P5. Lack of auditability and accountability of management. SME managers must be able to defend marketing decisions, particularly when these decisions entail a financial commitment or business risk. Without the ability to capture evidence used, constraints imposed, confidence values, and explanation provided, it will be difficult to audit the decision later if a decision-support system does not capture these.

The main issue tackled by this research is the design and evaluation of a prototype of SME marketing decision support. The prototype needs to translate evidence at the channel and campaign levels into recommendations that reflect business intent, are subject to operational safeguards, communicate calibrated model confidence, offer structured explanations, and leave room for human discretion when it comes to the final decision.

1.3 Research Gap and Rationale

Predictive marketing analytics, campaign-level evidence, constraint-aware reasoning, confidence calibration, and human-in-the-loop review are included by the literature reviewed in this study. These areas are often treated as distinct and little evidence exists on how they can be integrated into one process for translating campaign signals into a few auditable managerial actions under a clearly formulated strategic condition that is oriented towards the SME. This results in an obvious gap in the research-to-action continuum and the role of each element.

The study evaluates a four-action-decision-support method which integrates two learned evidence branches, training-derived evidence at the campaign level, predefined weighted safeguards, declared Survival, Retention or Growth context and validation-only confidence calibration and structured explanation. Baselines and component-removal conditions are used to evaluate recommendation agreement, scenario fit, logical consistency, calibrated confidence and human-review evidence. The claims that result are constrained to the public campaign source, audited benchmark, controlled intent scenarios and approved formative participant setting.

1.4 Research Questions

The project is guided by four research questions that are directly related to the objectives and evaluation plan:

RQ1. Is there more alignment with audited reference actions, actionability and rule consistency when both predictive and campaign-level channel evidence are combined with binary safeguard evaluation and weighted soft influence on candidate-action scores, versus defined partial baselines?

RQ2. What is the impact of calibrated model confidence and structured Driver–Implication–Action–Caveat explanations on confidence reliability, explanation usefulness and appropriate reliance during human review?

RQ3. Are there project-defined Survival, Retention and Growth conditions that enhance fit and usefulness of scenarios as judged by the project?

RQ4. Is the deployed prototype able to give traceable recommendations in reproducible configurations with an acceptable local response latency?

The four questions focus on quality of recommendation, confidence and reliance, intent conditioning, and operational traceability. The evidence for this is presented in chapter 5.

1.5 Research Aim and Objectives

This project involves designing, implementing, and testing an intent-aware decision-support system that transforms predictive and campaign-level channel evidence into bounded recommendations. The system is weighted, validation-distribution-specific in its calibration, and explains itself in a structured way for human review in SME-oriented multi-channel marketing scenarios.

This is operationalised through four research objectives:

RO1. To develop and freeze an audited decision benchmark based on public campaign records, project defined intent scenarios and independently reviewed reference actions.

RO2. Implementing a predictive and channel-evidence backbone, a four-action candidate space, predefined weighted safeguards and explicit intent conditioning.

RO3. To adopt validation only model confidence calibration and Driver–Implication–Action–Caveat explanations for human review.

RO4. To assess reference-action agreement, actionability, scenario fit, rule consistency, calibration, reliance, auditability and latency with respect to defined baselines and ablations.

1.6 Scope of the Study

The study develops a locally executed web prototype for SME campaign-level decision support. Its action vocabulary is limited to Increase, Maintain, Reduce, and Investigate. The public campaign dataset supports prototype modelling, but it does not provide population-level evidence about SMEs, causal channel effects, or live business impact.

The study includes leakage-controlled preprocessing, CampaignNet ROI evidence, ActionNet four-class evidence, training-derived channel indicators, binary weighted rule violations, explicit intent and risk adjustments, temperature-based confidence calibration, structured explanations, audit logging, baselines, ablations, and human-review support. The rules and their weights are predefined and frozen rather than learned from test data. Campaign cases may include one, two, or three channels. In this thesis, “multi-channel” refers to the broader

decision-support scope across several marketing channels, not to every case containing multiple channels. The project does not make customer-journey, channel-synergy, incremental-effect, or causal-attribution claims.

The project excludes autonomous campaign execution, direct integration with live enterprise systems, real-time streaming analytics, automatic legal or regulatory decision-making, unrestricted generative marketing content, and autonomous learning of business rules. Symbolic rules remain predefined and under developer control. These boundaries keep the project feasible and align the reported claims with the implemented and evaluated system.

Target users are SME owners, marketing managers and campaign executives who require assistance in interpreting campaign-level evidence across recorded channel combinations. The system does not replace these users. It presents a recommended action, supporting evidence and a confidence value for human review, while ultimate decision authority remains with the user.

1.7 Significance and AI Contribution

The project tackles an issue many small and medium-sized businesses face: turning multi-channel marketing data into meaningful and actionable steps for managers. Even then, predictive outputs are not necessarily easy to interpret by a manager, who has to decide whether to increase, maintain, reduce, or investigate a campaign, while taking into account budgets, business priorities, operational constraints and uncertainty..

1.7.1 Practical Significance

Target group: short / quick read suggestions to be given to the managers of SMEs and their campaign reviewers. The system doesn't automatically do any marketing action, but assists the manager in his decision-making process; the advice can be taken, modified, deferred or rejected.

The prototype is in charge of the execution under human control to allow the managers to observe what the prototype suggests, why and when it might be appropriate to conduct further research. It allows for more uniform, transparent and traceable decision making processes and to maintain the responsibility of the management.

The usefulness, clarity, contextuality and response time of the recommendations created will be the value of the project.

1.7.2 Technical and Research Significance

The technical significance of the project lies in combining data-driven marketing analysis with explicit business reasoning. Instead of treating predictive outputs or campaign-level channel evidence as final decisions, the proposed system uses these evidence sources to generate constraint-aware and intent-aware recommendations.

The research contribution is a bounded marketing-evidence-to-action architecture that makes the interaction between two learned evidence branches, training-derived campaign-level channel evidence, predefined weighted safeguards, explicit intent, calibrated model confidence, and structured explanation traceable and testable.

The project also evaluates the AI approach using predefined business intents and recommendation actions, enabling a controlled and measurable assessment of its contribution.

1.7.3 AI Contribution

Intent aware, Constraint aware Marketing Action Recommendation offered by AI system. Is able to provide one suggestion from the set of actions (Increase, Maintain, Reduce and Investigate) based on the data provided in a campaign and learned evidence and explicit symbolic reasoning.

The characteristics of the campaign, and the campaign performance data (marketing channel, impressions, clicks, leads, conversions, revenue, acquisition cost, ROI, engagement score and customer segment/campaign date) are put into the AI pipeline. It can also utilise indicators that are derived such as click-through rate, conversion rate, cost per acquisition, revenue per conversion, ROI band and engagement band. Moreover, it is possible to give it a declared business-intent (Survival, Retention or Growth) and business constraints can be predefined at the scoring of the recommendations.

The learned component has two models that are learned separately: CampaignNet, which has 11 campaign features as input and ROI as output; and ActionNet, which has 13 campaign features and one-hot intent as input, and 4 actions as output. The symbolic part is a catalogue of 14 activated safeguards which is separately encoded. They are not end-to-end learned rules, but rather the integration is neuro-symbolic as learned evidence and explicit rules both impact ranking.

The safeguards are implemented as soft constraints: When the safeguard is violated, the action score is decreased, but the action is still included in the set of actions to be considered. Rules are enabled/disabled and when enabled, do not have "truth values" that are scored.

Its intent-aware decision module is one of these features which distinguishes it from the conventional prediction/recommendation systems. Survival means more of an emphasis on cost control, margin protection, cash preservation and risk reduction. Retention puts a greater emphasis on the customer value, quality of engagement and repeat-purchase metrics. It's more about the power of the acquisition, momentum and performance of scalable channels. Thus, the recommendation will be dependent on evidence of the campaign and an organisation's strategic priority.

Technical assistance element "Confidence calibration". Scaling is done on the audited validation score vectors, and does not change the order of the vectors but only the "sharpening" of the probabilities. It considers the model's confidence, human confidence and their respective trustworthiness are separated.

The final explanation is created in a structured format which includes 4 components:

1. **Driver** finds the best evidence or the “rule” that was “activated” that supports the recommendation.

2. **Implication** makes connections to the impact on the business.

3. **Action** suggests management response.

4. **Caveat** expresses uncertainty, risk, assumptions or limitations.

Other components of the testable recommendation pipeline are: Predictive evidence, Supervised action evidence, Channel evidence (campaign level), Weighted safeguards, Declared intent, Calibrated confidence, Structured explanation are integrated by AI contribution. Local gradient-times-input values are NOT causal attribution values, but predictive sensitivity values.

1.7.4 Proposed Approach Overview

The proposed system considers the campaign indicators, channels, selected intent, risk tolerance and applicable business constraints. Providing campaignNet, ActionNet and evidence-based training and then using this to determine the reviewable action ranking in the rule, intent and risk stages.

Results of the system are:

1. a plan for marketing;
2. supporting decision evidence;
3. a confidence indication;
4. and a structured explanation.

It will be up to the SME manager to decide. The datasets preparation, designing the models, formulating the rules, model calibration and comparison between baseline and evaluation process is described in Chapter 3. Chapter 4 discusses the system architecture design, along with the implementation design.

1.8 Chapter Summary and Evaluation

Chapter 1 outlines the project rationale, problem statement, research questions, aims and objectives, project boundaries, and importance of AI in the project. SME marketing decision-making needs more than predictive accuracy or channel evidence. Suggestions must also come in a strategic context, have proper business logic, convey reliability, and provide necessary information to managers.

To satisfy these expectations, the proposed solution includes locally tested prototype of human-in-the-loop type. The claims are limited to audited benchmark, while planned human evaluation as autonomous implementation, causal attribution, generalisation across populations of SMEs, and implementation in production are not in the research scope.

Chapter 2 gives an organised thematic review of marketing analytics, attribution, and their application in direct digital marketing for SMEs, such as neuro-symbolic and constraint-aware analytics, explainability, confidence calibration, trust, and human reliance. It differentiates between concepts that can be used as a background for this project and those that require assessment.

Chapter 2

Literature Review

2 Literature Review

2.1 Introduction

This section encompasses the concept of a decision-support system that has an intent-aware approach, with a weighted-safeguard. This system is designed for SMEs and supports multiple channels for marketing with the ability to calibrate confidence and provide support for reliance. It delineates the differences between direct marketing and SME evidence from cross-domain architectural and human-AI evidence, then traces these design implications to Chapter 3.

The literature under review provides evidence of how prediction, attribution, symbolic reasoning, explanation, and confidence calibration all serve different purposes. It highlights that prediction and attribution can only provide an estimate of a performance or contribution but are not necessarily capable of suggesting any actions. The limits are able to be exposed by the symbolic methods, but it does not rely on the acquired knowledge. The situation can be evaluated by explanations but it does not prove that the claimed statements are correct, while confidence calibration provides reliability. The chapter emphasises that the process of merging all these aspects together while keeping the boundaries of evidence clear remains to be a major challenge if one wants to ensure the effectiveness of the system.

2.2 Literature Search and Selection Approach

The systematic thematic approach was employed in this literature review instead of systematic review of the PRISMA type. The main studies included in this review are marketing analytics and attribution, SME digital-marketing capability, prescriptive decision support, neuro-symbolic and constraint-aware methods, calibration, explanation, trust and reliance. The preferred references are articles from peer-reviewed journals, primary empirical studies, and authoritative method papers, but conference sources were maintained if they contain important methods or concerned the investigation of reliance.

In addition, backward and forward reference checking and recent journal publications from 2024 to 2026 were used. These studies were included in the review if they further the understanding of the method proposed, while it is stressed that the marketing and SME studies serve as the context, while only cross-domain AI studies inform about architecture and interactions of human and AI.

2.3 Theoretical or Conceptual Background

2.3.1 Marketing Analytics, Attribution and Prescriptive Decision Support

Marketing analytics uses various types of analyses such as descriptive, diagnostic and prescriptive analyses to come up with a set of evidence to back the decisions made regarding a campaign. The data-rich environment in marketing has resulted in customer information and signals related to campaigns being distributed across various advertising platforms, social media, customer relationship management systems, transactions and interactions taking place online; thus, utilising such signals would improve one's planning but would also add complexity to the models and governance (Wedel & Kannan, 2016). While articles that analyse the use of AI in marketing suggest that there are tremendous possibilities in terms of targeting, segmentation, customisation and optimisation, it is important to note that the success of the use of AI depends on the quality of the data and managers' ability to interpret it (Haleem et al., 2022).

The issue of how much credit should be attributed to particular channels or touchpoints that contributed to a result is one of the components of attribution modelling. Berman (2018) examines attribution modelling in a context that goes beyond the last-touch technique, Li and Kannan (2014) study conversion paths in a multichannel scenario, and Anderl et al. (2016) use graphs representing the trip structure. Through their work, the authors demonstrate how single-touch techniques miss connections that are revealed by the attribution process. Additionally, they draw attention to the fact that attribution is not the same as giving a suggestion because the predicted contribution is still predicated on modelling assumptions without accounting for a small entrepreneur's current budget, risk tolerance, or strategic objectives.

Even if the produced prototype is unable to restore customer journeys, determine causal incremental impacts, and distribute credit with reference to touchpoints, the attribution literature emphasises the importance of evidence from channels. Only at the campaign level have the campaigns been available to the general public. Consequently, the thesis outlines a more comprehensive issue of marketing-related considerations, where decision-making is based on the local prediction sensitivity technique, campaign data from campaigns, and ROI prediction.

From forecasting to making suggestions is the progression of prescriptive analytics, however, the success thereof tends to depend on how goals and limitations are presented. Lepenioti et al. (2020) assert that prescriptive analytics moves beyond forecast, but still suffers from problems regarding the incorporation of models and organisation's knowledge as well as optimising methods. The application of predictive and prescriptive analytics results in better performance in terms of product development and innovation in terms of manufacturing SMEs (Bhavikatta,

2026). Nevertheless, it is worth noting that the context of this research is limited and provides a narrower outlook on both sustainability and manufacturing.

Consequently, the available literature offers a stronger rationale for the use of AI technology in the advisory role rather than performing autonomously. The use of marketing analytics provides the necessary information regarding various alternatives outlined at all the stages above, with the manager being responsible for making his final yes/ no decision about the recommendations that have been presented as a result of the information generated. The reason behind this is that the marketing costs and opportunity costs could not be assessed based on the historical data alone since the current intent of the company cannot be inferred from this data.

On the one hand, Wu et al. (2024) conduct an empirical study of the digital-marketing strategy and performance of 148 SMEs, while Zong et al. (2025) study the big-data AI analytics in connection with the international marketing strategy of SMEs in the cultural and creative sector. Both of these research pieces show that analytics and marketing capabilities are important in the context of SMEs, but none of them provide support for the four-action recommendation, three-intent scenarios, and 14-rule recommendation catalogue proposed in this study.

The outcomes clearly distinguish between the creation of evidence and the process of making decisions. Both ROI and four-class action evidence are provided by the two branches, which are loosely connected. The ranking of the choices is influenced by the distinct rule catalogue that includes specific intent and risk offsets. Without considering the implementation as a new attribution algorithm, the arrangement permits traceability.

2.3.2 SME Strategic Intent and Human-in-the-Loop Decision Support

SMEs often have tight budgets, lack of in-house expertise in analytics and are more susceptible to failing campaigns spending. These conditions do not suggest that any one decision rule will be relevant for all SMEs; but rather that a reviewable recommendation with assumptions that are transparent will be more important. Even if the forecast is technically correct, it may be unworkable if the forecasted actions are incompatible with the cash preservation, customer-retention and the manager's risk assessment.

When adding strategic intent, everything else in the campaign evidence document becomes contextualised in a new light. If engagement is high, this could be good evidence to support spend during a growth phase, or the same evidence could be used to support the decision not to increase spend when factors like short-term liquidity are in question. Retention, in turn, could be more important to repeat purchases and customer value metrics. Hence, the thesis considers

intent as an explicit decision input instead of a model converged towards a learned objective that is assumed to be the current organisation's strategy.

The project aim for controlled implementation and evaluation is: to survival, to retention and to growth. The core principle of survival is the preservation of cash, margin and reduced risks, while the core principle of retention is repeat purchase and value of the relationship, and the core principle of growth is the allocation of more acquisition effort where there is a greater potential return. These are just operationalisations for the projects and not an affirmation that there are only three universal states of SME strategies.

These decisions can be interpreted in the context of decision-support theory, which considers them semi-structured decisions, where data, models and human judgement are all utilised (Power, 2002). Value can also be generated through complementary human and machine capabilities as opposed to total automation, in the case of hybrid-intelligence research (Dellermann et al., 2019). The term "co-rational arrangements" has been recently used to denote that decisions are negotiated between humans and AI systems, where reasoning, authority and accountability are negotiated (Hao, 2026). The review done by Hao is cross domain, it is not a validation of SME marketing directly but its governance boundary is adopted in this system, and the system provides advice, records evidence and retains management authority.

The intent is a combination of an input and evaluation condition. The intent-aware module will tune candidate priorities and scenario fit analysis will ensure that the action chosen corresponds to the scenario that is stated. It is still required to have a human review as intent categories can't be used to represent all the contextual knowledge an SME manager can have. This helps the system not to build the strategic alignment as a default value of prediction.

2.3.3 Neuro-Symbolic AI and Constraint-Aware Recommendation

Neuro-symbolic AI is an integration of data driven learning and explicit symbolic knowledge/reasoning. There are a large number of different types of systems that can be classified as “Journal surveys,” where logic and learning are less tightly coupled, such as in symbolic constraints applied to neural outputs, to systems that are more tightly coupled (Garcez & Lamb, 2023; Bhuyan et al., 2024; Marra et al., 2024). The transferable principle is that learned evidence and explicit knowledge can be applied to different aspects of a decision problem; the surveys do not suggest that any specific integration will be effective without its being evaluated specifically to the task in question and/or the SME involved.

Instead of fully-differentiable logic-learning approaches, the proposed framework is a loosely-coupled neuro-symbolic pipeline. The training of CampaignNet and ActionNet are independent of the rule catalogue. Their evidence is blended prior to 14 manually set up safeguards implement binary violations, weighted penalties and a unique Investigate boost. Logic constrained is thus to imply that logic has an impact on the ranking, but does not imply that hard constraint satisfaction is achieved.

It is important to have constraint-aware recommendation because an action may have a high predicted return, but may not be feasible to actually perform. A campaign could have a great potential for conversion but a very high acquisition cost, low margin, channel saturation or not being compatible with a survival intent. The research on constraint-based recommender systems reveals the possibilities to make requirements and preferences explicit when making recommendations (Felfernig et al., 2015). What matters here is its structure: marketing-specific constraints and weights are yet to be justified and tested.

Hard rules are very useful in cases where there are no threshold effects, but can make violation easily detectable and can deprive of a useful action at a threshold. Probabilistic soft logic is an example of the more general concept of weighted constraint influence (Bach et al., 2017) but this prototype does not use the soft logic formalism. There is a project rule in effect and either it is met or it is broken; softness is generated by a weighted violation of a project rule which affects the score but does not result in a rule being broken.

A recent public-service recommender system is based on a knowledge graph and machine-readable preconditions as well as neural processing (Konstantinidis et al., 2025). It offers a modern building block for the separation of the evidence extraction flexibility from explicit eligibility logic. The study, however, justifies the separation of responsibilities between components, but not the specific marketing rules, or the way that they are applied, in the project.

Symbolic layer's biggest advantage is the ability to audit. Every recommendation is traceable from the input features all the way to the evidence that was predicted, the scores of the candidates, the rules that were activated to determine intent, the adjustment made to the intent and the structured explanation and calibrated confidence. The results of this "traceable" process are assessed in Chapter 5 to see if there is a decrease in conflicts and an increase in the "fit" of the scenarios, as compared to the defined baselines.

2.3.4 Explainability, Confidence Calibration and Appropriate Reliance

People need to be aware of the level of trust they need to place in AI-assisted decision support; don't trust it to be as good as it can. Both overtrust and undertrust can be problematic for a manager – overtrust may result in accepting questionable advice and undertrust may lead to the failure to act upon good recommendations (Lee & See, 2004). Appropriate reliance: the user should be able to take good advice, and refuse bad advice; and remain responsible for the decision.

Explainable AI helps in the inspection of the factors, which led to the output. Ribeiro et al. (2016) proposed a way to provide local explanations for black-box predictions and Barredo Arrieta et al. (2020) summarise the key concepts, opportunities and limitations of XAI. With this thesis, you can see in the explanation if you had converted, acquired, gained revenue, engaged or had the channel behaviour. This kind of explanation explains the grounding of the output, but not reliability and/or strategic appropriateness of the output.

There is empirical evidence in HAI that backs up this significant difference. Schemmer et al. (2023) consider appropriate reliance as behavioural differences, when processing correct and incorrect AI-advice. At the same time, Vasconcelos et al. (2023) show that explanations can help to overcome overreliance, albeit under a specific experimental setting. An attitudinal trust and reliance behaviour is distinguished (see Scharowski et al., 2023) and explanation design affects these in different ways. The results showed that all of these point towards the need to consider behavioural response (and decision quality) along with subjective trust measurement.

Calibration provides a technical way of comparing the level of confidence with performance. Guo et al. (2017) show that such modern neural networks can be mis calibrated, and propose a simple post-processing method called temperature scaling. Based on this, Okamura and Yamada (2020) demonstrate that user's trust calibration can be adaptively managed in human-AI collaborative situations. With this research, it uses validation-fitted temperature scaling and ECE, which are supported by these approaches. There is, however, a basic necessity to assess the accuracy of the recommendation and appropriateness to the context, which neither of these metrics can do.

Two important qualifications have been added to this based on recent evidence. Doğru and Krämer (2025) discovered that there are different influences of expertise, trust and self-confidence on the dependence on AI advice in the game of chess. Their study helps in differentiating user confidence and trust, with their low-stakes task not necessarily being a direct indicator of user behaviour when it comes to the decision of marketing expenditure of SMEs. Hao (2026) also emphasises governance and authority sharing, and recovery in AI

decision support, but this is done using computational analysis and not in the actual implementation of AI in SME. Both sources elaborate on the purpose of the HITL concept, but do not offer direct evidence of the system being used.

The proposal is to introduce a standardised format for the explanations, with four elements: driver, implication, action and caveat. The driver finds the strongest evidence, the implication conveys it to business sense, the action offers a definite recommendation and the caveat conveys the uncertainty, missing evidence, boundary conditions, etc. This is a format that will allow for review/traceability purposes and not to ensure that managers are forced to accept the recommendation.

Explanation and calibration of confidence go hand-in-hand. The explanation provides some understanding of the system's action, and calibrated confidence provides understanding of how validation evidence qualifies the certainty of the system's action. The success or otherwise of these is an empirical issue in Chapter 5, where it examines, but don't assume, how successful they are as trust-alignment gap, behavioural overreliance, low-confidence uncritical acceptance and explanation usefulness are.

2.3.5 Literature-Derived Conceptual Framework

The framework is composed of evidence, candidate-action, reasoning, confidence-and-reliance-support and presentation layers that are derived from the literature. The trust construct is human evaluation: temperature scaling is used to “calibrate” the level of confidence in a model, and structured uncertainty/review controls are designed to facilitate the appropriate use of models.

These layers are referred to in various ways in the literature. The four areas of marketing analytics (generation of evidence), attribution (interpretation of channels), recommender systems and soft-logic techniques (explicit constraint handling), human-AI research (retained decision authority) and calibration and XAI (reliability communication and inspection). The project has included these in the framework, but it believes it's still untested when the combination will be finished with planned comparisons and ablations.

The project contributes in two fields: Architecture and Evaluation. It does not propose a new attribution algorithm or a general-purpose, neuro-symbolic formalism, but rather design and evaluate a bounded pipeline, consisting of existing predictive, reasoning, calibration and explanation approaches to the specific marketing decision problem.

2.3.6 Design Requirements Derived from the Literature

The project combines the synthesis to come up with the following 6 design requirements. Predictive and campaign level channel evidence – Evidence, NOT decisions. Secondly, the vocabulary of the output is upper bounded. Thirdly, the intent needs to be clearly stated and joint ablation is needed for the two pathways that have been implemented. Fourth, rules are binary safeguards which can be inspected and have different weighted score effects. Fifth, the calibration of confidence is not the same as an explanation and trust humans. Sixth, everything that is generated is still "human checked" and auditable.

They are requirements for CampaignNet, ActionNet, campaign level evidence based on training, four candidates, explicit intent / risk adjustments, validation-only temperature scaling and a Driver-Implication-Action-Caveat explanation. ActionNet has intent that is received directly and the decision stage puts in place intent offsets and intent specific safeguards. Chapter 3 then introduces A2: Full Method without Intent - both pathways are disabled.

The conceptual framework takes the following perspective to tackle this task: Evidence-reasoning-reliance. Models that are only data-driven are not necessarily feasible, symbolic rules cannot learn from different types of evidence in campaign, and explanations cannot be relied upon to be reliable. The proposed system is a combination of the above mechanisms, and empirically assesses their contribution.

2.4 Review of Related Work

2.4.1 Predictive, Attribution and Prescriptive Marketing Analytics

Predictive marketing analytics and attribution research are both powerful ways of making predictions and channel contribution analysis, but for different purposes. The articles by Wedel and Kannan (2016) put marketing analytics in a broader context whereas the articles by Li and Kannan (2014), Anderl et al. (2016) and Berman (2018) provide a more specific focus on multichannel attribution. These approaches to attribution are an improvement on the simple last-touch credit and can either model a path or incremental contribution, but cannot be used as a direct operational decision.

Prescriptive Analytics helps to convert the results of the estimates into recommendations for action. This change is generally achieved using organisational insights (which go beyond just the predictive model), optimisation techniques and business rules (Lepenioti et al., 2020). In Bhavikatta (2026), he presents some of the latest evidence from manufacturing small and medium enterprises (SMEs) that demonstrates the differential effects of predictive and

prescriptive analytics on organisational performance. The research confirms the value of action-orientated analytics for resource-poor organisations, but because it is in the context of sustainable product development, it is not as directly applicable to multi-channel campaign allocation situations.

The main methodological problem is therefore not about the question of the usefulness of prediction vs. attribution, but about converting their results into actions that can be taken. Marketing evidence can have an element of uncertainty, multiple channels can impact results at the same time, and high contribution channels can be expensive and/or have limited capacity. Current business goals and acceptable compromises are still not clearly defined and generic prescriptive optimisation approaches are not sufficient.

The project has a difference between evidence and recommendation framework. The evidence of a ROI in a CampaignNet, and of an action in ActionNet is both campaign-level channel evidence and is put into a bounded candidate process. Then the final ranking is decided based on the predefined safeguards, intent and risk. Attribution research becomes a conceptual basis of the approach, but the signals that is implemented is a different story.

2.4.2 Intent-Aware and Constraint-Aware Decision Support

The material in the intent-aware and constraint-aware decision support is based on a number of related, yet distinct research traditions. The traditional decision-support systems merge models and human judgement for semi-structured decision tasks (Power, 2002), and hybrid-intelligence research focuses on complementary humans and machine capabilities (Dellermann et al., 2019). Hao's (2026) co-rationality review builds upon this thought by bringing into focus authority allocation, authority repair and authority governance. All the sources are useful in maintaining the managerial control but do not offer the scoring algorithm specific to marketing.

The constraint-based recommender system explicitly states the requirements and preferences which provides clear reasons for inclusion and exclusion of certain alternatives (Felfernig et al., 2015). But there are issues when business needs are incomplete, uncertain or conflicting. When each condition is considered as a strict rule, it makes the system inflexible at the critical thresholds and the system may be unable to take actions which may be valid under other strategic approaches.

The research on soft-logic concentrates on weighted influences and traceable paths of rules, but the reality of implementations makes use of binary (not graded) rules. Weights and priorities are not outcome analysed but are previously documented and exist for a reason. It is also unclear

if these pre-determined weights adequately represent the complexities of the domain and their validity needs to be empirically tested.

In the field of Neuro-symbolic research, different approaches for the combination of neural and symbolic elements are presented (Garcez & Lamb, 2023; Bhuyan et al., 2024; Marra et al., 2024). Learned processing and semantic preconditions can be used to improve public-service recommendations, as recently done by Konstantinidis et al. (2025). This is doing a great job of comparing architectures, but there are significant differences between SME campaign management and the application domain, formal knowledge requirements and decision consequences.

The literature involved in relation helps to make it possible to make modular comparisons, but does not imply any inherent superiority. The two ablations (B2: Predictive Only and B3: Predictive plus Channel Evidence) investigate partial evidence; our proposed method's ablations then turn off rules, intent pathways and calibration components. The parity controls are introduced in Chapter 3 and the results are given in chapter 5.

2.4.3 Explainability, Calibration and Human Reliance

There are other concepts related but different to explainability: calibration, trust and reliance. XAI techniques can help to demystify how AI makes decisions, and calibration can be used to determine if a system's confidence is in line with its performance. While trust is an attitude towards the system, reliance is a behaviour that is manifested by following or rejecting advice (Scharowski et al., 2023). The ideas would become more difficult to combine into one satisfaction measure, which would prevent distinguishing between the appropriate reactions of users to some fluctuation in the quality of the recommendations.

According to the study on the AI advice, explanations are not necessarily accompanied by better decision-making. Schemmer et al. (2023) provide evidence that explanations have different impacts on various aspects of appropriate reliance, and Vasconcelos et al. (2023) illustrate how explanations can help decrease overreliance in certain decision situations. While applicable to the context of the study, these results do not necessarily lend themselves to SME marketing settings, but do provide a way of assessing whether advice is accepted or not, and not assuming that more explanation will necessarily yield a better result.

Expertise, trust and self-confidence also are clarified in decision-making situations by Doğru and Krämer (2025). They find that, in the course of conducting their chess research, their chess ability and their absolute level of trust are negatively correlated, while their self-reliance and

ability are positively correlated. This implies in the design of our system that the manager's initial judgement and his/her confidence should be separated from the confidence in the recommendation. The reason why it is maintaining the cross-task boundary is because the decision making of marketing is different from other domains with different risk profiles and different set of expertise.

This means that it can't do autonomous campaign execution with our human-in-the-loop system. The calibrated confidence indications and caveats are designed to stimulate an appropriate review process, and not to ensure acceptance. Managers are given the recommendation and supporting evidence and they document their decision – they have the authority to reject, or override, the system. Chapter 2 sets out the reasons for measuring and assessing separately from the start the interface structure's ability to facilitate appropriate reliance patterns, which will be the focus of Chapter 5.

2.4.4 Privacy, Security, Fairness and Deployment Constraints

Even if actual customer data is removed from marketing data, it may still contain other information that could be considered commercially sensitive, such as information about campaigns, costs, and revenues. The decision-support boundary includes data minimisation and controlled access. These controls help minimise exposure to the data but do not render the data risk-free, especially if there are multiple operational areas that may be linked together to draw conclusions about how well a business is doing.

There is limited evidence available to inform fairness. In certain cases, historical campaigns might be overly representative of certain channels, time of day or segments, which can lead to inconsistent performance of the model across conditions. This dataset does not give a full set of protected demographic attributes so that the study can only make claims about representation and consistency of performance, and not about fairness in terms of the full spectrum of protected demographic attributes.

Production robustness in the offline evaluation cannot be shown for all the future campaign conditions. Rule thresholds may need to be adjusted and channel costs and patterns of response may change, as may business policies. Versioned rules and evidence traces are stored in the system and the system is an advisory prototype and not a stand-alone deployment.

Useful output for an SME manager is one that does not require much effort in interpretation and yet allows the SME manager to understand the foundation of the action. The analysis of the operational choices available within the bounded action vocabulary (Increase, Maintain,

Reduce, Investigate) offers a means for comparison across systems, and a review of decisions. Investigate is also a fall-back for when evidence is lacking or contradictory, and not a sign of an uncalled for, but over-assertive allocation decision.

There is a need for "Contextual Flexibility" with consistency. The use of explicit rules can make similar cases more repeatable, intent conditioning makes it possible for legitimate changes when the organisation's stated intention changes. In this design, there is not an end to judgement, but rather a way to make the factors that result in a different recommendation apparent in the form of scores and activated-rule traces.

Explanation should be in a managerial and not just model internal language. A Reduce recommendation should include a statement that indicates if the reason for the reduction is the high acquisition cost, low conversion, saturation or intent conflict. The implication is that driver is an operational concern; and the caveat is that there is uncertainty or missing evidence. Technical audit can still be carried out on the feature value that is still available in the trace.

Confidence is seen as validation-based decision support as opposed to a guarantee. High confidence can allow quicker review, but does not mean that accountability is lifted, low confidence means there is a need for checking or the Investigate fallback. This interpretation is based on the difference in the literature between confidence calibration, and appropriate human reliance.

These restrictions further make it desirable to have a consolidated but limited framework. The evidence reviewed supports the complementary functions of prediction, attribution, symbolic constraints, intent and calibration as well as explanation. Does not say that their integration will be better than all other alternatives. The comparison made may only be put forward in the controlled evaluation (Chapter 3) and presented in the report (Chapter 5).

2.5 Comparative Analysis

The proposed framework is a synthesis of the review that retains the strengths and weaknesses of the studies included in the review. Evidence of the contribution of predictive and attribution methods, respectively; evidence of motivational output of action-orientated methods, respectively; evidence of motivational output for explicit weighted constraint influence for constraint methods, respectively; evidence of motivational output for learned–symbolic division for neuro-symbolic methods, respectively; and evidence of motivational output for

calibrated confidence, explanation, and retained managerial authority for human–AI studies, respectively.

The representative studies of each type of evidence are compared in Table 2.1 with regard to the strength of the evidence, limitations of the study, and design implication. This comparison does not look at cross-domain evidence as evidence of effectiveness in SME marketing.

Table 2.1

Critical Comparison of Related Literature and Implications for the Proposed System

Approach and representative studies	Evidence type	Transferable strength	Limitation for this study	Implication for the proposed system
Predictive marketing analytics (Wedel & Kannan, 2016; Haleem et al., 2022)	Marketing reviews	Learns patterns in campaign and customer data.	Predicted outcomes do not by themselves determine a feasible managerial action.	Use predictive output as evidence for, rather than as, the decision layer.
Multi-channel attribution (Li & Kannan, 2014; Anderl et al., 2016; Berman, 2018)	Empirical marketing studies	Estimates channel or touchpoint contribution beyond simple last-touch credit.	Contribution estimates remain sensitive to modelling assumptions and do not encode SME intent or constraints.	Treat attribution findings as background; use campaign-level channel evidence as an input to bounded candidate actions.
Prescriptive analytics (Lepenioti et al., 2020; Bhavikatta, 2026)	Review and manufacturing-SME survey	Moves from forecasts toward actions and organisational decisions.	Objectives and constraints are often assumed to be formalised; manufacturing evidence is not direct marketing evidence.	Represent intent and constraints explicitly and retain the cross-domain boundary.
Constraint and soft-rule reasoning (Felfernig et al.,	Method chapter and journal	Makes requirements explicit while	Rule quality and weights depend on defensible domain	Apply auditable binary safeguards with weighted score

2015; Bach et al., 2017)	method paper	permitting graded rule satisfaction.	encoding and maintenance.	influence and record the activated-safeguard trace.
Neuro-symbolic AI (Garcez & Lamb, 2023; Bhuyan et al., 2024; Marra et al., 2024; Konstantinidis et al., 2025)	Journal reviews and public-service framework	Combines learned evidence with explicit symbolic knowledge and preconditions.	Most evidence is cross-domain and does not validate SME marketing performance directly.	Adapt the architectural principle, then evaluate it on the project's marketing task.
Explainability and reliance (Ribeiro et al., 2016; Scharowski et al., 2023; Vasconcelos et al., 2023)	Method and human-subject studies	Supports inspection of model drivers and separates perceived trust from reliance behaviour.	Persuasive explanations can still accompany incorrect advice or encourage overreliance.	Use a structured business explanation and evaluate reliance, not satisfaction alone.
Calibration and human–AI governance (Guo et al., 2017; Okamura & Yamada, 2020; Doğru & Krämer, 2025; Hao, 2026)	Calibration method, experiments and journal reviews	Communicates reliability and preserves human authority in an advisory workflow.	Findings are task-dependent; chess and broad decision-support evidence are not direct SME-marketing validation.	Pair calibrated confidence and caveats with mandatory manager review.

The comparison indicates that the research streams that are reviewed here, cover various segments of the evidence-reasoning-reliance value chain. The most extensive domain evidence stem from marketing studies, and often only reaches a predictive stage or the level of attribution. Neuro-symbolic and recommender studies provide architectural and reasoning mechanisms, primarily not in SME marketing, and human-AI studies differentiate between trust, explanation and reliance in experimental contexts.

Comparing the studies numerically is not feasible as they vary with respect to their domain, task, and outcome. The project compares components under common data and decision

conditions: predictive performance at the channel level, campaign-level performance of evidence, fit of scenarios, conflict rate of reasoning components, calibration error of confidence components, behavioural measure of usefulness of human-review components.

2.6 Research Gaps

Gap 1 a gap between marketing and action evidenced in the literature reviewed. While existing work provides predictions, evidence at channel/touchpoint level or prescriptive concepts, there is no evidence of a specific bounded transition from campaign-level evidence to four auditable actions that is provided by the reviewed work.

Gap 2 is the lack of explicit consideration of a declared strategic scenario in the decision pipeline that is reviewed. The project identified the following conditions: Survival, Retention and Growth. It does not claim that they are a universal taxonomy.

Gap 3 is the low extent of embedding of the learning evidence in the campaign and their link to inspectable operational safeguards. The proposed binary safeguard mechanism (weighted score influence) fills this gap by allowing for score influence and traceability, but not for guaranteed hard feasibility.

Gap 4 is the gap between the communication, explanation and observed reliance of models. The proposed system has a combination of validation-calibrated, structured explanation with confidence, and Chapter 5 separately considers the use of correct and incorrect advice by the participants.

Gap 5 lack of the same combination and evaluation design of the studies reviewed as it relates to SMEs. This is not to say that there isn't some similar system out there.

2.7 Chapter Summary and Evaluation

The reviewed literature supports the marketing-evidence-to-action methodology used in this project. Marketing and SME studies define the application context. Neuro-symbolic and constraint-aware studies support the learned-rule separation. Calibration and reliance studies support distinct technical and human-centred evaluation.

The evaluation design is having the same logic. Recommendation actionability measures whether analytical evidence becomes a useful action. Scenario-fit rate measures whether recommendations match survival, retention or growth priorities. Logical conflict rate measures

whether recommendations violate predefined constraints. Expected Calibration Error evaluates whether confidence scores reflect observed reliability. Trust alignment, behavioural overreliance and low-confidence uncritical acceptance examine whether users rely on recommendations appropriately. Explanation usefulness measures whether the rationale can support managerial review.

Ablation testing is necessary because the framework integrates multiple components. If removing the intent module reduces scenario fit, the result supports the claim that strategic context matters. If removing the weighted safeguard layer increases conflict rate, the result supports the claim that explicit constraint reasoning improves feasibility. If removing calibration worsens confidence reliability, the result supports the claim that confidence adjustment is necessary for appropriate reliance. These tests make the contribution of each module observable rather than assumed.

Baseline selection reflects the defined partial technical alternatives B1: Heuristic Scorecard, B2: Predictive Only, B3: Predictive plus Channel Evidence and B4: Integrated Learned Backbone. A1: Full Method without Rules, A2: Full Method without Intent and A3: Full Method without Calibration are ablations of the proposed method rather than additional baselines.

The technical distinction lies in the ordered interaction of CampaignNet, ActionNet, campaign-level channel evidence, weighted safeguards, two-path intent conditioning, risk offsets, confidence calibration and structured explanation. Human trust is not produced by this pipeline; reliance is evaluated separately.

The project is positioned as an SME-oriented decision-support prototype, not a new attribution algorithm, universal neuro-symbolic architecture or proof of business impact. Its contribution and claims are bounded to the frozen benchmark, defined comparisons and human-review study.

The synthesis supports a focused research claim: actionable and context-aware marketing recommendations require an integrated evidence-reasoning-reliance pipeline rather than isolated prediction, attribution, explanation or rule filtering. Chapter 3 operationalises this claim through the selected dataset, baselines, proposed method, controls, calibration metrics, scenario-fit analysis and human-centred evaluation.

Chapter 3

Research Methodology

3 Research Methodology

The procedure used to evaluate intent-conscious and weighted-safeguard decision support system with the support of confidence calibration and reliance is discussed in this chapter. Public campaign source, audited decision benchmark, project-defined intent scenarios and formative participant study are separated in this project. Although Chapter 3 will establish the research design and evaluation protocol, Chapter 4 will transform these concepts into a workable development blueprint and Chapter 5 will give a report of the system.

The approach to this project will be to consider SME marketing decision support as an evidence-reasoning-reliance issue. Channel evidence at prediction, and campaign level is only used to inform the decision, but not to determine a managerial action. The regulations bring out the constraint of business. The weighted influence enables the trade-off between budget, cost of acquisition, saturation of the channel and business intent. Confidence calibration is a model certainty and structured explanation is a linkage between the recommendation and evidence to be reviewed by humans.

This chapter describes the research process, matching the research question and research objective, source segregation, audited benchmark and participant evidence, preprocessing and leakage avoidance, baselines and ablations, proposed AI technique, metrics, validation, reproducibility and ethical limit.

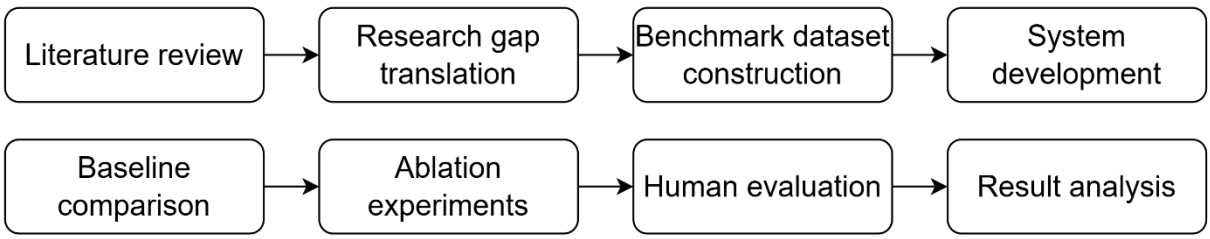


Figure 3.1 Research Methodology Framework

3.1 Research Design and Workflow

This study embraces the design science that is backed by quantitative experiments and a limited human-based assessment (Peffers et al., 2007). Evaluations of the artefact include audited reference-action agreement, consistency of rules, fit of a scenario, confidence reliability, usefulness of explanations, reliance behaviour, traceability and latency.

The research design is an experiment based, and controlled study. The dataset, the decision-case representation, business-intent taxonomy, recommendation action space, baseline configurations, the ablation variants and evaluation metrics are pre-determined prior to the final testing. Pre-correcting these aspects enhances post-hoc correction and increases reproducibility. The offline benchmark evaluation is suitable as the project does not measure the deployment of production or long-term business impact whereas human-centred evaluation is used to measure the ability of the users to comprehend and use the recommendation accordingly.

This approach is structured into modules with each module characterised by the motivation, representation of input, forward process, the role of output and evaluation. This hierarchy makes it clear what is sent to each module, how it transforms the input and what metric measures its contribution and makes the decision-support pipeline simpler to re-run and audit.

It is a loosely coupled and modular approach. CampaignNet provides evidence based on ROI, ActionNet provides evidence based on four classes conditioned on the intent chosen and training-derived evidence provides evidence based on the channel behaviour in the past. Another rule catalogue, intent offsets, risk offsets, confidence calibration and structured explanation, are post-generated evidence based, allowing component-based baselines and ablations.

The process starts with the data preparation which is controlled by leakage. CampaignNet is only fitted based on source-training records; ActionNet based on audited benchmark-training cases. The evidence branches are amalgamated prior to channel, rule, intent and risk modifications. The fitting of temperature scaling is then only applied to audited validation score vectors. The benchmark and held-out tests are not subject to fitting and design revision.

The prototype is a human-in-the-loop decision support system. It suggests, but not implements, campaigns, adjusts advertising budgets or links directly to live advertising platforms, it recommends, reports confidence and gives reasons as to why. This limit maintains accountability of the managers and ensures that spending and risk decisions are still under the human control.

Table 3.1

Research Workflow and Evidence Outputs

Design Science Research activity	Study activity	Role	Evidence output	RQ / RO link
Problem identification and motivation	Review SME marketing decision-support limitations and define the evidence-to-action problem.	Research framing	Problem statement, literature gap and initial design constraints.	RQ1-RQ4; RO1-RO4
Define objectives for a solution	Specify the audited benchmark, four actions, three intent categories, traceability and confidence/reliance requirements.	Research design	Testable objectives, acceptance gates and evaluation plan.	RQ1-RQ4; RO1-RO4
Design and development	Construct the leakage-controlled data pipeline, CampaignNet, ActionNet, fixed fusion, weighted rules, calibration, explanations and web prototype.	Artefact development	Versioned data, implemented modules, source code, editable figures and automated tests.	RQ1-RQ4; RO1-RO3
Demonstration	Execute end-to-end recommendation cases through React, FastAPI, the reasoning engine and persistence.	Integrated demonstration	Traceable recommendation packages, rule traces and recorded human-review states.	RQ4; RO2-RO3
Technical evaluation	Compare four external baselines, CampaignNet-	Experimental evaluation	Per-case predictions, seed metrics,	RQ1, RQ3, RQ4; RO4

	only, ActionNet-only, fused evidence and proposed-method ablations on frozen partitions.		calibration records, latency and reproducible run metadata.	
Human-centred evaluation	Following documented TAR UMT ethics approval or exemption, conduct the human-centred evaluation of explanation usefulness, calibrated-confidence communication and appropriate reliance.	Ethics-gated evaluation	Ethics approval or exemption reference, approved study materials, consent records and participant measures.	RQ2-RQ3; RO4
Communication	Report method, implementation, evidence, limitations and Chapter 5 findings without treating a conclusion as a research method.	Research communication	Thesis, response-to-review trace and reproducibility package.	RQ1-RQ4; RO4

3.2 Dataset or Corpus

The public campaign source has to offer adequate variation for CampaignNet and the separately audited benchmark has to offer intent and reference-action labels for ActionNet, calibration and decision evaluation. These sources have various uses and are separated using linked-identity leakage controls.

If the data set does not meet the desired configuration, the experiment is reported as a "simplified configuration", and any cross channel conclusions are considered exploratory. This way, you won't overclaim what you don't have to. The system is still considered to be a good pipeline for decision making, however, with a smaller number of channels and shorter observation times, the conclusions drawn about multi-channel interactions must be limited.

The benchmark is frozen before final evaluation, that is, the test cases, intent labels, action labels and evaluation split in the benchmark do not change after the model tuning starts. This avoids the need for repeatedly adjusting the final test cases, and enables to compare baselines, ablations and the proposed method with the same evidence.

The dataset for modelling is taken from the public Nykaa Marketing Campaign Performance Dataset from Kaggle. It comes with 55,555 campaign level records that include campaign context, channel, exposure, engagement, cost and outcome data. There is no available provenance that suggests that the records are multiple independent SMEs, and thus the source is not used as a representative sample of SMEs, but as a public prototype dataset.

The dataset is not intended to be a source for a live business deployment. The system doesn't need personally identifiable customer information, live customer tracking or social-media credentials. The fields that are only used for modeling, rule evaluation, calibration, explanation and evaluation are kept to minimise unnecessary data exposure.

Derived features are created to aid with recommendation reasoning. Some examples are click-through rate, conversion rate, cost per acquisition, revenue per conversion, ROI band, engagement band and channel saturation indicator. These features convert raw campaign records into decision situations which can be understood by both the predictive and the symbolic components. To prevent the features from being interpreted inconsistently in the test, each derived feature is defined prior to the final evaluation.

A 55,555-record public source and a 125-case audited decision benchmark are distinguished in the study. Benchmark cases are associated with source Campaign ID numbers and include project assigned intent conditions, independent reviewer labels, adjudication status and 1 audited operational reference action. This reference action is an evaluation judgement in the stated scenario and not a universal ground truth or evidence of realised business impact.

The Channel_Used field contains from 1 to 3 names of channels for each campaign. The 55,555-record source has 18,612 campaigns with one channel, 18,538 with two channels and

18,405 with three channels; the benchmark with 125 records has 45, 36 and 44 campaigns with one, two and three channels, respectively. In this way, there are over one channel mentioned in 36,943 source records and 80 benchmark cases. These counts are only for channel-combination coverage. They don't set up a touchpoint sequence or channel specific outcomes and the thesis refers to channel evidence at campaign-level instead of multi-touch or channel synergy.

Table 3.2

Data Resources and Roles

Resource	Provenance / version / access	Structure and role	Evidence boundary
Raw campaign source	Aleesha Nadeem and one collaborator; Kaggle Version 1; retrieved 15 August 2026; CC0 Public Domain. Dataset URL: https://www.kaggle.com/datasets/ishitabahamnia/nykaa-campaign-data-csv.	55,555 records by 16 observed fields. Source for CampaignNet, channel statistics and benchmark candidates.	Chronological source split; benchmark validation/test identities excluded from source training.
Processed research layer	Locally derived, versioned and hash-checked.	55,555 records by 31 fields. Lineage and reproducibility evidence; not the executable 13-feature matrix.	No reference-action or participant outcomes used for fitting.
Audited decision benchmark	Two reviewer packets, adjudication and frozen manifest.	125 cases by 26 fields; split 87/19/19. ActionNet training, validation-only temperature fitting and held-out evaluation.	Frozen before confirmatory comparison; SHA-256 3F2320FBD190B24766DB4453AB1DC7E7B212803FDFA2AC048C2437012D4D51EE.

Operational campaign copy	Deterministic startup copy from the frozen 125-case benchmark.	125 administrator-owned Campaign rows with Active status and null budget; used for workspace analysis and requests.	Does not alter or replace the versioned benchmark or its partitions.
Participant-response data	Human-participant data collection authorised under TAR UMT approval reference TARUMT/FOCS/ETHICS_APPROVAL/001, dated 21/8/2026.	Planned pseudonymous consent, assigned presentation condition, willingness to act, willingness to review, acceptance decision, actionability and explanation ratings, and decision-time fields.	Participant outcomes are not used for model fitting. Human-centred results will only be reported after collection under the approved protocol.

3.2.1 Data Collection and Sampling

A benchmark decision case is one that has been validated with a campaign record and a stated intent scenario and an independently reviewed reference action. Public source and benchmark labels as well as later participant responses are still considered evidence types. Reviewers do not have access to the system predictions, rule activations, weights or model confidence.

The Survival, Retention and Growth intent is a project-defined condition of the scenario prior to reference-action annotation, for each benchmark case. The intent and controlled action set are assigned and the definitions of the annotations are provided in the annotation packets. Reviewers independently make only the operational reference action, given the intent condition provided, but do not see the systems' outputs. Manager selected context is the same three category vocabulary after adjudication, and not considered to represent all SME strategies.

The action space is restricted to the following actions: Increase, Maintain, Reduce and Investigate. Increase means that more investment/exansion is recommended. Maintain means that the present strategy is satisfactory. Reduce means that the allocation can be reduced because performance is poor or cost is high or there is a conflict with constraints. Investigate means to examine the evidence is unclear, conflicting or lacking in certainty for confident action. With this fixed action set, the accuracy of the recommendations and scenarios fit and actionability can be measured for each configuration.

Each decision case and runtime request contains exactly one manager-declared primary intent: Growth, Retention or Survival. Mixed or weighted intent is outside the controlled study scope because it would change the ActionNet input, scenario-fit construct and annotation protocol. A request containing more than one intent is rejected for human clarification rather than being silently combined. The recommendation unit remains the operational action—Increase, Maintain, Reduce or Investigate—while campaign-level channel evidence is supporting evidence and not the primary output.

Records are included if the information to calculate the performance and cost/outcome indicators selected are available, or can be managed by a pre-defined missing-value rule. There are no duplicated records and records or cases that have not had their labels resolved after review. The sampling is done in a manner that preserves representation for channel and intent classes; and if present, the chronological order is respected for splitting. The Kaggle data is only available for access based on the rules outlined with the data source and is stored in the controlled project workspace. The benchmark is based on secondary data from the campaigns, and does not require collection of participant information or consent; separate human evaluation follows the participant information and voluntary-consent procedures.

3.2.2 Labelling and Annotation

Documented project definitions are used to prepare audited reference actions and intent labels. They are not ground truth, but rather reference actions, which are based on operationally judged and reviewed actions under a given scenario. A pilot uses and revises the rubric prior to the use of the frozen benchmark for model comparisons.

The raw-source is initially assessed according to the source-audit requirement. This pilot is done with 20-30 cases before full annotation and is used to further develop the action rubric and to eliminate ambiguous criteria.

Only the reference-action label is assigned by two reviewers independently, based on the business-intent condition, as defined by the project. Those who have only been assigned reference-action labels independently, are assessed by Cohen's kappa (Cohen, 1960) and inter-rater agreement must be at least 0.60. If there is any disagreement, it goes to a third person and if it is not resolved, it is taken out prior to the benchmark freezing.

The benchmark, partitions, action definitions, intent definitions, conflict rules and evaluation procedures are fixed prior to starting confirmatory model comparison.

The conceptually related but operationally separate annotation rubric and tested rule catalogue are used. Table 3.2A lists each of the annotation criteria, evidence for reviewers, the rule-catalogue equivalent, if one exists, if a numerical threshold exists and what control is used to prevent the leaking of labels. Without viewing model output, rule activations, numerical thresholds, weights, priorities, campaign evidence or score adjustments, reviewers use campaign evidence and generic four action definitions.

The adjudication of the reference actions is done prior to application of the rule catalogue, and the rule-only thresholds and score mechanics are then revealed in Table 3.2B.

Table 3.2A

Annotation Rubric and Rule-Catalogue Independence Matrix

Concept	Annotation rubric	Tested rule catalogue	Independence and interpretation
Action vocabulary	Reviewer selects Increase, Maintain, Reduce or Investigate using generic operational meanings.	Rules adjust scores over the same four actions but do not create reference labels.	Shared vocabulary is necessary for comparison; label creation is completed before rule evaluation.
Intent	The stated Survival, Retention or Growth scenario provides the context for the reviewed action.	Three explicit intent safeguards and fixed intent offsets affect action scores.	Intent is an acknowledged conceptual overlap; reviewers do not see safeguard activations, weights or score offsets.
Performance, cost and evidence	Reviewer considers the visible campaign indicators holistically and records a short justification.	The catalogue contains explicit ROI, conversion, CTR, engagement, cost, budget and evidence-sufficiency conditions.	The annotation packet does not expose the catalogue's numerical thresholds or weights.
Evaluation role	Adjudicated action is a reviewed operational reference judgement, not universal ground truth.	A1: Full Method without Rules, the Post-hoc Hard-Rule Filter comparator and P: Full Proposed Method test whether	Agreement is interpreted as consistency with independently reviewed judgement, not proof of business impact or causal correctness.

		the safeguard mechanism changes consistency and agreement.

The annotation standard and rule catalogue are not conceptually unrelated, but are procedurally separated. They both react in the same way to the same campaign evidence, and both contain the same four-action and three-intent vocabulary. The reviewers were given the assigned intent, the fields they could see in the campaign, generic action definitions, and the instruction to select Investigate when evidence was insufficient, ambiguous or contradictory. They were not provided with the 14 rule identifiers, numerical thresholds, weights, priorities, penalty coefficient, Investigate-boost coefficient, intent offsets, risk offsets, model output nor the rule-activation trace. This boundary is shown in Table 3.2B. The interpretation of the agreement with the audited reference action is therefore not necessarily as a confirmation of the correctness of the rule catalogue or its thresholds, but as a confirmation that it is consistent with an operational action that has been judged separately.

Table 3.2B

Annotation-Rubric and Rule-Catalogue Overlap Schedule

Overlap area	Available to annotators	Introduced only in the recommendation engine
Campaign evidence	Visible campaign row: campaign context, listed channels, impressions, clicks, leads, conversions, revenue, acquisition cost, ROI and engagement, together with the supplied intent.	The engine derives CTR, conversion rate, lead-to-conversion rate, cost per conversion, strong/weak-signal counts and rule-specific diagnostic values.
Action and uncertainty standard	Generic definitions of Increase, Maintain, Reduce and Investigate; Investigate is used when evidence is insufficient, ambiguous or contradictory.	The same action vocabulary is scored, but the engine adds weighted violations and a separate Investigate boost. No rule activation or model recommendation is shown to reviewers.
Intent	A project-defined Survival, Retention or Growth scenario is supplied before	The engine adds intent-specific safeguards—Growth/Reduce, Survival/Increase and Retention/Reduce—

	the reviewer selects the reference action.	and fixed downstream intent offsets, in addition to ActionNet's one-hot intent input.
Performance and evidence thresholds	No numerical action threshold is stated in the annotation rubric. Reviewers judge the visible evidence holistically and provide a short justification.	Rule-only thresholds are ROI < 0; conversion rate < 0.08; CTR < 0.03; engagement < 5; fewer than 500 clicks or 50 conversions; strong signals ROI > 2, conversion > 0.20, CTR > 0.10 and engagement > 20; and exceptional performance ROI > 4 with all three other strong conditions.
Cost, channel and budget thresholds	Reviewers see acquisition cost, revenue, conversions, impressions and listed channels, but receive no rule threshold, budget-feasibility formula or channel-saturation flag.	Rule-only checks compare cost per conversion with revenue per conversion; define saturation as impressions > 50,000 with CTR < 0.05; and require a supplied budget to cover 50 additional conversions at the estimated cost per conversion.
Weights and score mechanics	No weights, priorities, penalties, boosts, fusion coefficients, intent offsets, risk offsets or calibrated confidence are available during annotation.	The rule catalogue uses weights 0.70–0.95, priorities 1–3, a 0.20 penalty coefficient and a 0.10 Investigate boost, followed by separately specified intent and risk score adjustments.
Timing and blinding	Two reviewers label independently; disagreements are adjudicated; unresolved cases are excluded before the benchmark is frozen.	The rule catalogue is frozen and evaluated separately. System predictions, rule traces and test results are unavailable during reference-label creation.

3.2.3 Preprocessing

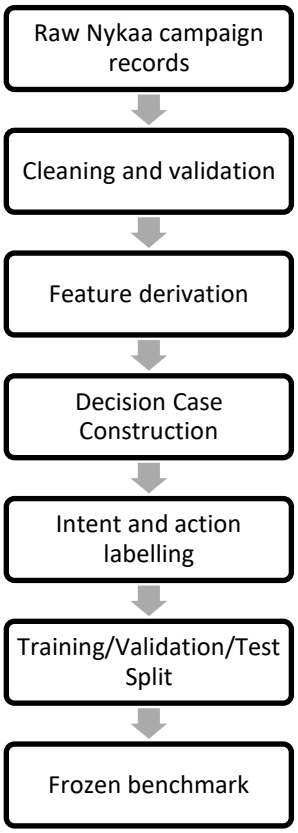


Figure 3.2 Data Preparation Workflow

The first step of the pre-processing methodology is to conduct a raw campaign record audit. Exact duplicate rows or duplicate campaign identifiers, unparseable dates or dates with inconsistencies, numbers stored as text, missing performance measures, zero denominators in derived ratios, extreme values that are not plausible and inconsistencies in the spelling of categories are examples of input problems. These are all regarded as data-quality issues and not evidence of a model. The steps from the raw-record preservation up to the frozen benchmark are summarised in Figure 3.2; the detailed partitioning protocol is given separately in Section 3.3.

The sequence keeps an unchanged raw layer, validates the schema, identifier and date fields, removes exact duplicate rows, standardises and parses date fields, sorts date fields chronologically, coerces designated measure to numeric, partitions the data into training, validation and test sets, applies transformations fit on training data, imputes missing numeric data, creates zero-safe ratios, standardises model inputs and validates the final feature structure. If a date format is mixed or not supported, then validation fails rather than the date is interpreted silently. Also, a numerical field that has no training value that can be used fails the pipeline. In

ratio features the value for the feature with a denominator of zero will be set as a non-fitted fallback to prevent the propagation of a division-by-zero into the model.

The reason for using median imputation is that it is less sensitive to highly skewed marketing measures than is mean imputation. Outlier treatment is not done via general row deletion as this may be operationally important campaigns that are unusual; rather, any model-target clipping boundary is estimated from the training partition. The predictive path contains only numbers that are the results of numerical campaigns measures, the fixed business-intent vocabulary is encoded only in the supervised action component and the channel information is kept separately as evidence signal and not expanded to a high dimensional campaign-category matrix. Validation and feature construction is done with python and the pandas and NumPy libraries, while feature standardisation is done with scikit-learn. The result is an ordered and validated modelling frame that includes the 13 modelling features that are numerical and a trace of evidence that connects the model with the source identifier from the records that were processed.

Only training records are used to fit all the transformations, whose values are learned from data. This includes numerical imputation medians, standardisation means and standard deviations, model-target clipping quantiles, target-normalisation statistics and channel-normalisation statistics. Deterministic rules, such as schema checks, date-format rules, identifier checks and zero-denominator formulas are applied in the same way to all partitions. Any statistics are not re-estimated using validation or test data. The complete splitting order, validation boundary and test-set restrictions to enforce this control are defined in section 3.3.

3.3 Data Splitting and Leakage Prevention

The decision benchmark that has been independently labelled is annotated in the manner as described in Section 3.2.2 before it is partitioned. An interrater agreement is calculated for the independently assigned reference-action labels with Cohen's kappa. Any disagreements are taken to a body for adjudication and if that cannot be settled, the case may not be included prior to the benchmark being frozen. This benchmark is kept unchanged when comparing with models, to avoid changing labels due to model errors or experimental findings.

After ordering by date and Campaign_ID, both the public campaign source and the audited benchmark have a 70/15/15 split, where 70% is the split for the top 70 campaigns, 15% is the split for the top 15 campaigns, and 15% is the split for the remaining 15 campaigns. Identities are linked before fitting: Any source row that is used in a benchmark validation or test case is

not included in the CampaignNet and channel-evidence training partition. The resulting source partitions are 38,888/8,333/8,334 and the benchmark partitions are 87/19/19.

Only source-training records can be used for source imputation statistics, CampaignNet scaling, ROI clipping and normalisation, CampaignNet parameters and channel normalisation. ActionNet and its stand-alone scaler are only suitable for the 87 audited benchmark-training cases. The single temperature parameter of locked score vectors can be used for the 19 cases audited for validation. The predefined rule catalogue, the weights, priorities, intent vectors, risk offsets and fusion constants are not optimised on validation or test data.

These two held out source and benchmark test partitions will not be used until Chapter 5. They are not adopted for preprocessing, model/threshold selection, rule revision, calibration and explanation-template selection or code-based benchmark annotation. A split manifest is used to keep track of membership in the Campaign_ID and to verify the isolation of the linked-source before confirmatory evaluation.

The chronological split results in the following audited-action support in the fixed order: Increase/Maintain/Reduce/Investigate: training 15/33/23/16, validation 0/8/6/5 and test 3/7/5/4. In all four actions, training and held-out testing is done in ActionNet, whereas validation is missing and the test supports are small. Thus, the four-by-four confusion matrix and per-class precision, recall and F1 with class support, macro averages and uncertainty intervals must be reported in Chapter 5. The calibration results are limited since the 19-case validation set is not able to evaluate the class Increase and the three training seeds cannot be used to solve the problem of label support.

3.4 Proposed Approach

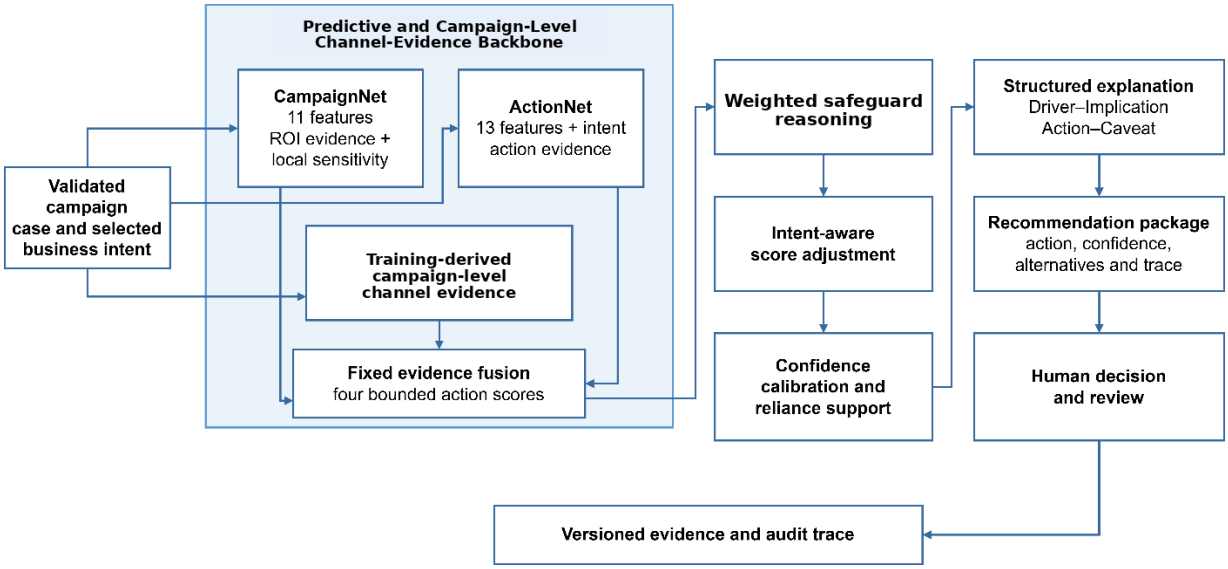


Figure 3.3 Implementation Pipeline for Decision Support System

Figure 3.3 presents the planned method workflow. Validated campaign data first enter two learned evidence branches: an 11-feature CampaignNet for ROI evidence and a 13-feature-plus-intent ActionNet for four-class action evidence. Their outputs are fused, smoothed and combined with training-derived campaign-level channel evidence. Predefined binary rule violations, intent and risk adjustments construct the final score vector; validation-only temperature scaling and a structured explanation then support human review.

Intent enters through two explicit paths: ActionNet's one-hot input and the downstream intent pathway comprising fixed intent offsets plus intent-specific safeguards. A2: Full Method without Intent must disable both paths. This control is necessary to attribute any scenario-fit difference to intent conditioning rather than to only one retained pathway.

The final recommendation package contains the selected action, calibrated confidence, activated reasoning evidence and structured explanation. This package supports auditability because a recommendation can be traced from the input case to predictive evidence, safeguard status, intent adjustment, confidence calibration and explanation. This trace also supports later testing in Chapter 5 because each stage produces measurable outputs.

Table 3.3

End-to-End Method Pipeline

Stage	Input	Processing	Output	Evaluation Link
Data preparation	Raw campaign records	Clean, validate, derive features	Frozen benchmark cases.	Dataset completeness

		and construct decision cases.		and reproducibility.
Predictive and campaign-level channel-evidence backbone	Prepared source-training records and audited benchmark-training cases	Generate CampaignNet ROI evidence, ActionNet action evidence, training-derived campaign-level channel evidence and local predictive sensitivity.	Fused learned-evidence vector and supporting evidence trace.	Reference-action agreement, baseline comparison and evidence-branch ablation.
Candidate action generation	CampaignNet, ActionNet and campaign-level channel evidence	Maintain four bounded candidates for later ranking.	Candidate action list.	Recommendation accuracy and actionability.
Weighted safeguard reasoning	Candidate actions and rule base	Score budget, cost, risk, saturation and feasibility constraints.	Constraint-aware ranking.	Logical conflict rate.
Intent-aware decision module	Action ranking and intent label	Adjust priorities for Survival, Retention or Growth.	Intent-consistent recommendation.	Scenario-fit rate.
Confidence and reliance-support layer	Final recommendation and evidence trace	Calibrate confidence and generate driver, implication, action and caveat.	Recommendation package for human review.	ECE, trust alignment and explanation usefulness.

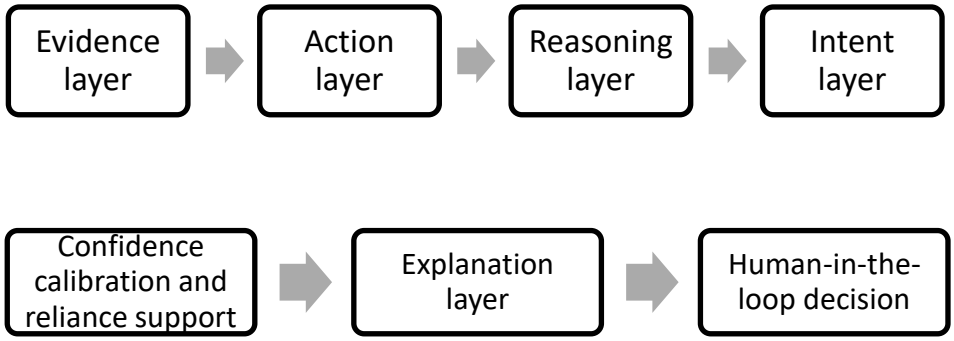


Figure 3.4 AI Component Architecture

The AI methodology is organised using a consistent module structure: motivation, module design and technical advantage. This structure makes the method easier to understand and evaluate because each component explains why it is needed, how it operates and which metric should show its contribution. Figure 3.4 summarises the main AI component architecture.

3.4.1 Predictive and Campaign-Level Channel-Evidence Backbone

The evidence backbone contains CampaignNet and ActionNet. CampaignNet maps 11 leakage-controlled campaign features to a standardised ROI estimate and an ROI-derived four-action vector. ActionNet maps 13 campaign features plus a three-value one-hot intent vector to probabilities over Increase, Maintain, Reduce and Investigate. A separate channel signal is computed only from source-training ROI and conversion evidence.

For explanation support, CampaignNet also provides normalised gradient-times-input local predictive sensitivity. These values describe which inputs influenced the local predictive logit and are neither causal effects nor customer-journey attribution. The two learned branches and the channel signal are fused through fixed implementation coefficients that are documented and stress-tested rather than described as learned optimal weights.

The separation supports a fair comparison among B1: Heuristic Scorecard, B2: Predictive Only, B3: Predictive plus Channel Evidence, B4: Integrated Learned Backbone and the subsequent reasoning ablations. Chapter 5 reports the results of these comparisons.

3.4.2 Candidate Action Generation

Candidate action generation addresses the marketing-evidence-to-action gap by retaining the same four candidates, Increase, Maintain, Reduce and Investigate, throughout scoring. Investigate is both a legitimate audited action for ambiguous cases and a runtime fallback whose

score can be boosted by violated safeguards; those two roles are reported separately in evaluation traces.

The module receives prediction scores, local predictive-sensitivity drivers and derived performance indicators. It then generates eligible actions from the four-action set: Increase, Maintain, Reduce and Investigate. Strong performance and acceptable cost may make Increase eligible. Stable moderate performance may support Maintain. Weak performance or excessive acquisition cost may support Reduce. Conflicting evidence or low confidence may support Investigate. Multiple candidates can be generated, allowing the reasoning layer to rank alternatives instead of forcing an immediate decision.

A bounded action generator provides a consistent basis for evaluation. All baselines and ablation variants use the same action labels, allowing recommendation accuracy, actionability and scenario-fit rate to be compared directly. The fixed vocabulary also strengthens auditability because each recommendation category has a clear managerial interpretation.

3.4.3 Weighted Safeguard Reasoning

Safeguard reasoning tests whether candidate actions are permissible or merely less desirable under SME business constraints. Each applicable safeguard is evaluated as satisfied or violated. Six safeguards are specified as hard constraints—evidence sufficiency, the Growth, Retention and Survival intent safeguards, budget feasibility when a budget is explicitly supplied, and materially conflicting evidence. A hard violation removes the affected action from the feasible set. The remaining eight safeguards are soft: their violations reduce an action score by the predefined weight, priority and penalty coefficient and may add the separate Investigate boost.

The catalogue contains 14 predefined safeguards with weights from 0.70 to 0.95 and priorities from one to three. Hard constraints use their binary violation to exclude an action; their stored weights and priorities remain trace metadata and are not used to rescue an infeasible action. Soft violations modify ranking without rejecting a candidate. When Increase, Maintain and Reduce are all excluded, the system returns Investigate with the decision state NO SAFE RECOMMENDATION. This is a reachable manual-review outcome and not an instruction to execute a campaign change.

Safeguard correctness is evaluated primarily by reviewed action sets rather than exact agreement with one primary label. For each audited case, the four actions are partitioned into acceptable, conditionally acceptable and unacceptable sets. An intervention is correct when the unconstrained action is unacceptable and the constrained result is acceptable or conditional. A

NO SAFE RECOMMENDATION result is correct only when Increase, Maintain and Reduce are all unacceptable and Investigate remains an acceptable manual fallback. Exact primary-label agreement is retained as a secondary diagnostic.

The annotation-rule independence boundary specified in Tables 3.2A and 3.2B applies to this catalogue. Annotators may use the underlying campaign evidence and the generic action and intent vocabulary, but the rule identifiers, numerical thresholds, weights, priorities, penalty coefficient, Investigate-boost coefficient, intent offsets, risk offsets, model outputs and activation traces are engine-only information introduced after reference-action adjudication. The tables therefore document both the conceptual overlap and the procedural controls that prevent the rule catalogue from generating or revising the audited reference labels.

Logical conflict rate, set-based intervention correctness, unresolved-case rate and selective coverage provide the main evaluation of this module. Removing safeguards is expected to increase conflicts; the full method must reduce those conflicts without presenting excluded actions as automatically executable.

3.4.4 Intent-Aware Decision Module

The intent-aware decision module adjusts action ranking according to the selected strategic context. The same campaign evidence can justify different actions depending on whether the business is trying to survive, retain customers or grow. Without intent awareness, the system may optimise a generic performance target while producing recommendations that are misaligned with the current business objective.

Intent conditioning operates through two pathways. ActionNet receives the one-hot intent during supervised evidence generation. After rule reasoning, fixed action-specific intent offsets and three intent safeguards provide a second, explicit path. A2: Full Method without Intent removes the ActionNet intent input, fixed offsets and intent safeguards together.

The main evaluation metric for this component is scenario-fit rate. Scenario-fit measures whether the selected action is compatible with the assigned strategic intent. If A2: Full Method without Intent performs worse on scenario-fit, the result supports the claim that intent conditioning improves contextual relevance.

3.4.5 Confidence Calibration and Reliance-Support Module

The Confidence Calibration and Reliance-Support Module aligns reported model confidence with audited validation correctness for the Balanced profile and the observed 19-case validation

distribution. Its calibrated status is profile- and validation-distribution-specific; it does not establish Increase-class calibration or general four-action calibration because no Increase reference case occurs in the validation partition. Temperature scaling does not calibrate a manager's psychological trust. In this thesis, reliance support is operationalised through this qualified calibrated model confidence, structured uncertainty communication and separate behavioural evaluation.

The module applies temperature scaling as a post-processing calibration method. Temperature scaling adjusts predicted probabilities using a validation set without changing the ranking order of recommendations. Expected Calibration Error is then computed by grouping predictions into confidence bins and comparing average confidence with observed correctness in each bin (Guo et al., 2017). Lower ECE indicates better alignment between confidence and actual reliability.

The output of this module is a calibrated confidence score and a confidence category. High-confidence recommendations can support faster review, medium-confidence recommendations require normal review and low-confidence recommendations should trigger caution or investigation. The module is evaluated through ECE, trust-alignment gap, behavioural overreliance rate and low-confidence uncritical-acceptance rate. Removing this component should mainly affect confidence reliability rather than the action label itself.

3.4.6 Structured Explanation Generation

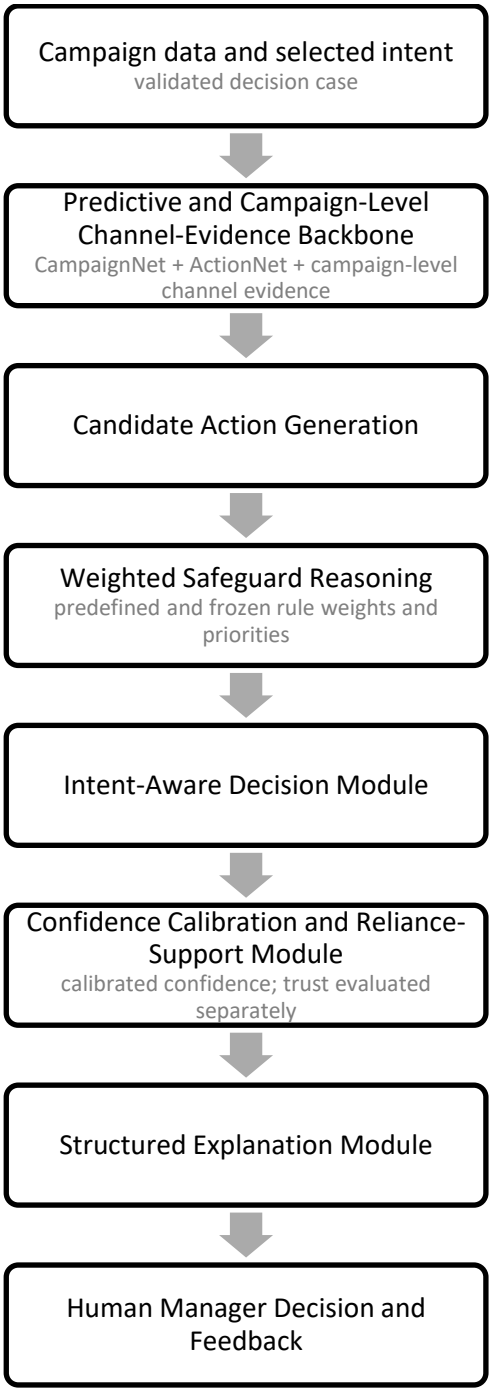


Figure 3.5 System Component Architecture

The explanation module generates a fixed explanation package consisting of Driver, Implication, Action and Caveat. The Driver states the main evidence behind the recommendation. The Implication translates the evidence into business meaning. The Action states the suggested response. The Caveat communicates uncertainty, risk or assumptions. This format is suitable for SME decision support because it avoids purely technical explanations while still preserving auditability. The selected Intent is reported separately as contextual metadata and is not treated as a fifth explanation element.

Given the final recommendation, the module extracts the strongest evidence drivers, activated rules, selected intent and calibrated confidence. It then produces a concise explanation that links data evidence to the suggested action. For example, a recommendation to reduce investment may be explained by high acquisition cost, weak conversion and a survival intent. A recommendation to investigate may be explained by inconsistent evidence or low confidence.

The explanation module is evaluated through explanation usefulness and trust alignment. The goal is not to make users accept every recommendation. The goal is to help users understand why a recommendation was produced and how much caution is required before acting. This supports appropriate reliance and human accountability.

3.5 Baseline Selection and Justification

The proposed system integrates CampaignNet, ActionNet, training-derived campaign-level channel evidence, predefined weighted safeguards, two-path intent conditioning, risk offsets, confidence calibration and structured explanations. Baselines test distinct alternative methods; ablations remove a component from the proposed method while keeping the remaining implementation fixed.

All systems receive the same prepared decision cases, use the same four-action output space of Increase, Maintain, Reduce, and Investigate, and are evaluated using identical frozen test cases. Model parameters and decision thresholds are selected using the training or validation partitions, while calibration parameters are fitted using validation data only. The rule weights and priorities are predefined implementation values, documented and frozen before confirmatory evaluation; they are not learned or optimised using the validation or test partitions. No system is permitted to access test labels or additional test-set information during development.

The comparisons are organised according to the five experiment blocks described in Section 3.7.

3.5.1 Block 1 Baselines: Decision Recommendation Performance

Block 1 compares decision-performance baselines using the same benchmark splits and four-action output schema.

The following systems are compared:

1. **B1: Heuristic Scorecard.** This baseline uses manually specified thresholds for indicators such as return on investment, conversion rate, acquisition cost and engagement. It represents a transparent but relatively inflexible form of managerial decision support. Its limitation is that manually selected thresholds may not capture learned relationships among heterogeneous campaign characteristics.
2. **B2: Predictive Only.** A nearest-class-centroid classifier over the 13 prepared features produces four-action probabilities without campaign-level channel evidence, intent, rules or calibration. It is a separately specified baseline, not CampaignNet with modules silently disabled.
3. **B3: Predictive plus Channel Evidence.** The B2 logits are adjusted by the same training-derived campaign-level channel signal used for parity, but no higher-level rule, intent or calibration component is applied.
4. **B4: Integrated Learned Backbone.** CampaignNet and ActionNet evidence are fused using the specified fixed coefficients and smoothing, while downstream campaign-level channel adjustment, safeguard reasoning, intent offsets, risk offsets and calibration are disabled as specified by the runner configuration.

These baselines test progressively richer evidence configurations. Four supplementary reference comparators are also evaluated without replacing the authoritative B1-B4 sequence: B0: Majority-Class Popularity establishes a frequency-only lower bound; the Intent-Only Deterministic Baseline tests the declared intent without learned evidence; the Weighted Deterministic Baseline combines fixed intent and campaign-signal scores without learned parameters; and the Unconstrained AI Baseline retains the learned evidence backbone while disabling safeguard reasoning and feasibility constraints. The unconstrained condition is a safety reference rather than a target for unrestricted deployment. Proposed-method ablations separately remove rules, both intent pathways or calibration so that comparator and component-removal evidence remain distinguishable.

3.5.2 Block 2 Baselines: Confidence Calibration, Explanation and Appropriate Reliance

Block 2 evaluates whether calibrated confidence and structured explanations improve appropriate reliance rather than merely making recommendations appear more persuasive.

The following variants are compared:

1. H1: Raw-Uncalibrated.

2. H2: Calibrated-Confidence Only.

3. H3: Structured-Uncalibrated.

4. H6: Complete Calibrated DIAC.

The raw explanation condition serves as a control for the possibility that participants respond positively simply because explanatory text is present. The structured explanation condition uses a fixed Driver–Implication–Action format without a Caveat, while the complete condition adds a Caveat communicating confidence-related uncertainty. Comparing these variants distinguishes the effect of calibration from the effect of explanation wording. Block 2 therefore compares H1, H2, H3 and H6 to separate the effects of confidence calibration and structured explanation.

3.5.3 Block 3 Baselines: Logic-Constrained Consistency

Block 3 evaluates whether integrated binary safeguard evaluation with weighted soft influence reduces logically inconsistent recommendations.

The following systems are compared:

1. **A1: Full Method without Rules.** The full learned evidence and intent pathways are retained, while all operational and intent-specific rule penalties and Investigate boosts are disabled.

2. **Post-hoc hard-rule filter.** The initial action is selected first, after which fixed rules reject or replace actions that violate declared constraints.

3. **P: Full Proposed Method.** The complete method applies the predefined weighted safeguards during candidate-action scoring, allowing partially competing considerations to influence the ranking before the final recommendation is selected.

The Post-hoc Hard-Rule Filter comparator tests whether the benefits of the proposed system could be obtained through simple late-stage rule filtering. The weighted safeguard method differs by applying binary safeguard violations as weighted score changes directly to action ranking instead of applying rules only after an action has already been selected.

3.5.4 Block 4 Baselines: Business-Intent Conditioning

Block 4 evaluates whether explicitly representing Survival, Retention, and Growth produces recommendations that better fit the declared business scenario.

The following policies are compared:

1. **A2: Full Method without Intent.** ActionNet receives a neutral intent representation and the downstream intent offsets plus intent-specific rules are disabled; all other eligible components remain fixed.

2. **A3: Full Method without Calibration.** Both intent pathways and the rule catalogue remain enabled, but T = 1 is used and the output is identified as uncalibrated.

3. **Full Proposed Method.** Both learned evidence branches, training-derived campaign-level channel evidence, rules, both intent pathways, risk offsets, validation-fitted temperature scaling and structured explanation are enabled.

Using the same backbone across these configurations controls for predictive performance and allows differences in scenario fit to be attributed more directly to intent conditioning.

3.5.5 Block 5 Explanation Controls

Block 5 compares the following three Table 3.5A conditions:

1. H4: Terse-Calibrated.

2. H5: Structured-Calibrated No Caveat.

3. H6: Complete Calibrated DIAC.

This supporting experiment determines whether explanation format materially changes perceived clarity, usefulness, willingness to act, and willingness to review a recommendation before acting. It is treated as a design-refinement study rather than the main scientific contribution.

Table 3.4 distinguishes partial baselines from proposed-method ablations across the five experiment blocks. Source audit, annotation and benchmark freeze form Preparatory Gate P0

rather than a comparison condition. Runs R002-R006 are executed only after that gate and the linked split-isolation checks pass.

Table 3.4

Summary of Baseline and Ablation Systems

Experiment block	Compared systems	Main component isolated
Block 1	B1: Heuristic Scorecard; B2: Predictive Only; B3: Predictive plus Channel Evidence; B4: Integrated Learned Backbone	Evidence-branch and channel-evidence contribution
Block 1 supplementary references	B0: Majority-Class Popularity; Intent-Only Deterministic Baseline; Weighted Deterministic Baseline; Unconstrained AI Baseline	Frequency-only, declared-intent-only, deterministic-score and unconstrained-safety reference points
Block 2	H1: Raw-Uncalibrated; H2: Calibrated-Confidence Only; H3: Structured-Uncalibrated; H6: Complete Calibrated DIAC	Calibration and explanation structure
Block 3	A1: Full Method without Rules; Post-hoc Hard-Rule Filter; P: Full Proposed Method	Integrated constraint reasoning
Block 4	A2: Full Method without Intent; A3: Full Method without Calibration; P: Full Proposed Method	Business-intent conditioning
Block 5	H4: Terse-Calibrated; H5: Structured-Calibrated No Caveat; H6: Complete Calibrated DIAC	Explanation-template design

3.6 Evaluation Metrics and Acceptance Criteria

The evaluation combines recommendation performance, logical consistency, calibration, human reliance, explanation and operational metrics because the artefact is a decision-support system rather than only a predictive model.

3.6.1 Recommendation-Performance Metrics

Macro F1 calculates the F1 score independently for each of the four action classes and then averages the class-level scores. It is used because class frequencies may be unequal and overall accuracy could conceal poor performance on less frequent actions.

Top-2 agreement is the proportion of decision cases for which the audited reference action appears among the two highest-ranked actions produced by the evaluated system.

Actionability score is the mean of participant ratings for usefulness, specificity, and feasibility. Each item is measured on a five-point scale, with higher values indicating that the recommendation is easier to understand and apply.

3.6.2 Scenario and Logical-Consistency Metrics

Scenario-fit rate is the proportion of recommendations judged compatible with the declared Survival, Retention, or Growth intent.

Conflict rate is the proportion of recommendations that violate at least one predefined budget, channel-cap, or intent-compatibility rule. A recommendation with several simultaneous violations counts as one conflicting case in the overall rate, while individual violation types are recorded separately.

Recommendation reversal rate is the proportion of cases for which the selected action changes after the constraint mechanism is applied. This metric shows how frequently the reasoning layer alters the initial evidence-based recommendation.

3.6.3 Calibration and Reliance Metrics

Expected Calibration Error measures the discrepancy between predicted confidence and empirical recommendation correctness across five predefined confidence bins. Lower values indicate better agreement between confidence and observed correctness.

For participant p and trial t, willingness to act W_{pt}, measured on a five-point scale, is normalised to the interval [0,1] as:$ptW_{\mathrm{pt}}[0,1]$

$$r_{\mathrm{pt}} = (W_{\mathrm{pt}} - 1) / 4$$

Recommendation correctness is defined as:

y_{pt} = 1, if the recommendation matches the audited reference action; y_{pt} = 0, otherwise.

$$y_{pt} = \begin{cases} 1, if\ the\ recommendation\ matches\ the\ audited\ reference\ action, \\ 0, otherwise \end{cases}$$

Let $\mathscr{E}$ denote the set of eligible trials and let N = $|\mathscr{E}|$. The trust-alignment gap is calculated as:

$$G = (1/N) \sum_{(p,t)} \in \mathcal{E} |r_{pt} - y_{pt}|.$$

The corresponding trust-alignment score is:

$$A = 1 - G.$$

A lower value of the trust-alignment gap indicates closer correspondence between willingness to act and recommendation correctness, while a higher alignment score indicates stronger alignment. Confidence intervals and condition comparisons use participant-level clustering so that repeated trials completed by the same participant are not treated as statistically independent participants.

Behavioural overreliance rate uses only trials containing an incorrect recommendation. Its denominator is the number of eligible incorrect-recommendation trials, and its numerator is the number of those trials for which the participant selects 'Accept without checking'. Low-confidence uncritical-acceptance rate uses only trials with displayed confidence below 0.50, regardless of recommendation correctness. Its denominator is the number of eligible low-confidence trials, and its numerator is the number of those trials for which the participant selects 'Accept without checking'. The two measures therefore use different eligibility rules and shall be reported with separate numerators and denominators.

Willingness to act records the extent to which a participant follows the recommendation. This metric is interpreted together with correctness and confidence rather than being treated as a standalone indication of system quality.

3.6.4 Explanation Metrics

Explanation usefulness measures whether the explanation helps participants understand and review the recommendation.

Perceived clarity measures whether the explanation communicates its evidence and intended action in an understandable manner.

Willingness to review before acting measures whether the explanation and confidence information encourage participants to perform an appropriate level of verification before accepting the recommendation.

3.6.5 Operational Metrics

Operational latency is measured from submission of a decision case until the complete recommendation package is returned. Both p50 and p95 latency are reported.

Decision time is the elapsed time between presentation of the recommendation and submission of the participant's evaluation or decision.

Results are reported with 95% bootstrap confidence intervals (Efron, 1979). Model training uses three predefined random seeds, with the same seed set applied to every comparable configuration. Holm correction (Holm, 1979) is applied to repeated pairwise comparisons to control the family-wise error rate. Statistical significance is interpreted together with effect size and practical importance.

Table 3.5

Metrics and Acceptance Criteria

Metric	Interpretation	Acceptance criterion
Macro F1	Balanced agreement with the four audited action classes	At least 0.60 and at least 0.05 above the heuristic baseline for confirmatory downstream claims
Top-2 agreement	Audited action appears in the two highest-ranked actions	B4: Integrated Learned Backbone should match or outperform B1-B3
Actionability score	Mean usefulness, specificity, and feasibility rating	P: Full Proposed Method should outperform the corresponding partial or ablated configuration

Scenario-fit rate	Compatibility with declared business intent	P: Full Proposed Method should outperform A2: Full Method without Intent
Conflict rate	Violation of budget, channel-cap, or intent rule	The weighted safeguard layer in P: Full Proposed Method should reduce conflicts relative to A1: Full Method without Rules and the Post-hoc Hard-Rule Filter comparator
Recommendation reversal rate	Action changes after constraints are applied	Reported together with conflict reduction and actionability
Expected Calibration Error	Difference between confidence and empirical correctness	Calibrated condition should have lower ECE than the uncalibrated condition
Trust alignment gap	Difference between reliance and actual correctness	Calibrated structured-explanation condition should reduce the gap
Behavioural overreliance rate	Proportion of eligible incorrect-recommendation trials in which the participant selects 'Accept without checking'.	The calibrated structured-explanation condition should reduce behavioural overreliance relative to the raw condition
Low-confidence uncritical-acceptance rate	Proportion of eligible trials with displayed confidence < 0.50 in which the participant selects 'Accept without checking'.	The calibrated structured-explanation condition should reduce Low-confidence uncritical-acceptance relative to the raw condition
Explanation usefulness	Helpfulness of explanation for review	Structured templates should outperform raw or terse explanations
p50 latency	Typical system response time	Reported for all principal configurations

p95 latency	High-percentile system response time	Below two seconds in Block 2; rule layer should not add more than 20% without reducing conflict
Decision time	Time required for participant judgement	Compared across intent and explanation conditions

3.6.6 Testing, Validation and Statistical Plan

Testing and validation consist of benchmark validation, model validation, logic validation, calibration validation, human evaluation, software verification, and reproducibility checks.

3.6.7 Benchmark and Annotation Validation

Benchmark validation applies the source audit, annotation pilot, independent labelling, Cohen's kappa threshold, adjudication and benchmark-freezing procedure defined in Section 3.2.2. Validation confirms that the benchmark, partitions, action and intent definitions, conflict rules and evaluation procedures are locked before confirmatory model comparison begins.

3.6.8 Model and Data Validation

Training, validation, and test partitions are separated by time or business unit where possible to minimise leakage. Preprocessing parameters are learned only from the training partition. Model hyperparameters and decision thresholds are selected using only the validation partition, and the temperature-scaling parameter is fitted using that partition. The predefined rule weights and priorities remain frozen; validation evaluates their behaviour but does not optimise them.

The test partition is used once for final confirmatory comparison. The same test cases, features, action labels, computational environment, and metric implementations are used for every comparable configuration.

3.6.9 Logic and Conflict Validation

The business-rule implementation is tested using controlled cases representing valid and invalid combinations of budget, channel capacity, evidence strength, and business intent.

For every generated recommendation, the system records whether a conflict occurred and which rule categories were violated. Multiple violations are retained in the audit log even though the case contributes only one conflict to the overall conflict-rate calculation.

Representative rule activations, predefined and frozen rule weights, firing frequencies, and action changes are manually inspected to verify that the reasoning behaviour remains consistent with the declared domain logic.

3.6.10 Calibration Validation

Temperature scaling for the formal evaluation is fitted only on the 19-case audited validation partition using the locked Balanced-profile four-action score vectors after CampaignNet and ActionNet evidence fusion and smoothing, training-derived campaign-level channel adjustment, binary weighted safeguard penalties and Investigate boosts, and downstream intent adjustment. Runtime risk offsets are excluded from temperature fitting. Because the Balanced risk vector is zero, this score state is numerically equivalent to the Balanced runtime score vector before any non-Balanced risk adjustment. The validation action support is Maintain = 8, Reduce = 6, Investigate = 5 and Increase = 0. The fitted temperature is therefore evaluated only as an aggregate calibration adjustment for the observed validation distribution. It does not establish class-conditional calibration for Increase or general calibration across all four actions. Temperature scaling changes probability sharpness but not action ranking, and test outcomes remain untouched during fitting. Conservative and Aggressive outputs are exploratory and remain labelled uncalibrated unless separately calibrated and evaluated.

The selective-decision thresholds are chosen on the validation partition only. Automatic RECOMMEND requires calibrated confidence of at least 0.35, a top-two score margin of at least 0.02, a non-Investigate action, no hard exclusion of the selected action and no unresolved feasibility condition. Lower confidence or margin produces CHECK ADVISED; Investigate, a missing budget for Increase or another unresolved feasibility condition produces HUMAN REVIEW REQUIRED; and exclusion of all three executable actions produces NO SAFE RECOMMENDATION. Chapter 5 reports coverage, accepted-case accuracy, automatic conflict rate and unresolved rate at this locked operating point.

Calibration reporting uses the following five pre-specified confidence intervals:

$$I_1 = [0.0,0.2), \quad I_2 = [0.2,0.4), \quad I_3 = [0.4,0.6), \quad I_4 = [0.6,0.8), \quad I_5 = [0.8,1.0]$$

For each bin, the corresponding validation set contains all cases whose reported confidence falls within that interval. The bin membership, eligible-case count and correct-case count are defined as follows:

$$V_b = \{i : c_i \in I_b\}, \quad n_b = |V_b|, \quad k_b = \sum_{i \in V_b} \mathbb{1}(\hat{y}_i = y_i)$$

For every non-empty bin, the mean confidence, observed accuracy and absolute calibration gap are:

$$conf_b = \left(\frac{1}{n_b}\right)\left[\sum_{i \in V_b} c_i\right], \qquad acc_b = \frac{k_b}{n_b}, \qquad g_b = |conf_b - acc_b|, \qquad n_b > 0$$

Expected Calibration Error is calculated across the five bins as:

$$ECE = \sum_{b=1, n_b>0}^{5} \left(\frac{n_b}{N}\right) |conf_b - acc_b|, \qquad N = 19$$

Empty bins are retained in the reliability table with zero eligible cases; their mean confidence, observed accuracy and calibration gap are reported as not estimable, and they contribute zero to the weighted ECE sum. Negative log-likelihood and the multiclass Brier score are calculated as:

$$NLL = -\left(\frac{1}{N}\right)\sum_{i=1}^{N} \ln\left(p_{i,y_i}\right)$$

$$BS = \left(\frac{1}{N}\right)\sum_{i=1}^{N}\left\{\sum_{a \in A}\left[p_{i,a} - \mathbb{1}(y_i = a)\right]^2\right\}$$

These measures and the reliability table are interpreted as descriptive small-sample evidence with an explicit zero-Increase limitation; no population-level or Increase-class calibration claim is made. Chapter 5 reports the fitted Balanced-profile temperature, the eligible and correct counts in all five pre-specified confidence bins, and 95% percentile bootstrap intervals from 2,000 case-level resamples of the 19 validation cases for temperature, ECE, negative log-likelihood and Brier score. As a binning-sensitivity check, ECE is also recomputed descriptively with four and ten equal-width bins. Empty bins and non-convergent bootstrap fits are counted and disclosed rather than silently removed; none of these analyses establishes four-class or Increase-class calibration.

3.6.11 Human Evaluation Protocol

The human evaluation is pre-specified as a formative within-participant study targeting 18-24 participants, including at least eight SME owners, marketing practitioners or campaign executives. The target is pragmatic rather than a confirmatory power calculation: 18-24 complete participants yield 162-216 trial records, but repeated observations do not equal 162-216 independent participants. Population-level effectiveness claims are therefore prohibited.

The target of 18–24 complete participants is a feasibility rationale rather than a confirmatory power calculation. Eighteen complete participants permit at least three participants in each of the six cyclic counterbalancing groups, while 24 permit four per group. This size is intended to test the study procedure, estimate effect directions and produce uncertainty intervals; it is not sufficient for a population-level effectiveness claim. If fewer than 18 participants complete all required procedures, condition comparisons are reported descriptively and are not presented as confirmatory inferential evidence.

Each participant reviews nine frozen benchmark cases: three Growth, three Retention and three Survival cases. Six presentation conditions are rotated across the nine trials using the identifiers and attributes in Table 3.5A: H1 Raw-Uncalibrated, H2 Calibrated-Confidence Only, H3 Structured-Uncalibrated, H4 Terse-Calibrated, H5 Structured-Calibrated No Caveat and H6 Complete Calibrated DIAC. Every participant encounters all six conditions, with three conditions occurring twice; the duplicated conditions vary by cyclic counterbalancing group so exposure remains balanced across participants.

Table 3.5A

Authoritative Human-Study Condition Matrix

Condition identifier	Calibration / confidence display	Explanation / Caveat	Exposure, research question and outcome role
H1: Raw-Uncalibrated	Uncalibrated (T = 1); raw model confidence shown.	Raw explanation text; no structured Caveat.	1 or 2 exposures, balanced by cyclic counterbalancing; RQ2; primary reference arm for trust-alignment contrast and secondary reliance outcomes.
H2: Calibrated-Confidence Only	Validation-calibrated Balanced-profile confidence shown.	No structured explanation template; no Caveat.	1 or 2 exposures; RQ2; secondary calibration/reliance comparison.
H3: Structured-Uncalibrated	Uncalibrated (T = 1); raw model confidence shown.	Driver-Implication-Action; no Caveat.	1 or 2 exposures; RQ2; secondary explanation-versus-

			calibration comparison.
H4: Terse-Calibrated	Validation-calibrated Balanced-profile confidence shown.	Terse feature list; no Caveat.	1 or 2 exposures; RQ2 / Block 5; secondary explanation-format outcome.
H5: Structured-Calibrated No Caveat	Validation-calibrated Balanced-profile confidence shown.	Driver-Implication-Action; no Caveat.	1 or 2 exposures; RQ2 / Block 5; secondary explanation-format outcome.
H6: Complete Calibrated DIAC	Validation-calibrated Balanced-profile confidence shown.	Driver-Implication-Action-Caveat; Caveat included.	1 or 2 exposures; RQ2; primary intervention arm for trust-alignment contrast and secondary explanation/reliance outcomes.

Table 3.5A is the single authoritative condition map. Block 2 compares H1, H2, H3 and H6 to separate calibration from structured-explanation effects. Block 5 compares H4, H5 and H6 to isolate explanation format while holding calibrated confidence constant. The six-condition participant protocol rotates H1-H6 across nine cases; therefore, the four Block 2 conditions and three Block 5 explanation variants are overlapping subsets of one participant-condition design rather than separate studies.

Recruitment proceeds after formal ethics authorisation through supervisor-approved contacts and relevant university, entrepreneurship, industry-outreach or alumni channels. Before the first recorded response, each participant receives the information sheet and provides informed consent. A pseudonymous participant code and a practitioner/non-practitioner group indicator are retained; names, email addresses and customer identifiers are not stored in the analysis dataset.

Condition identity remains blinded during participation. A deterministic hash of the participant code selects one of six cyclic condition offsets, while a separate deterministic shuffle varies case order. The participant is the sampling and clustering unit; one participant-case response is the trial unit. Duplicate participant-case submissions are rejected, and the same participant group must be retained when a session is resumed.

For each trial, the participant reviews the campaign context, stated intent, recommendation presentation and available confidence or explanation information for the assigned condition. The study records actionability, specificity, feasibility, explanation usefulness, willingness to act, willingness to review, user acceptance, managerial decision and decision time. The current instrument does not collect a separate unaided action; consequently, the study evaluates comparative presentation and reliance support rather than a pre-post change from unaided judgement.

For participant p and trial t, the primary trial-level outcome is the trust-alignment gap between normalised willingness to act and audited recommendation correctness:

$$G_{pt} = r_{pt} - y_{pt} \vee$$

The primary estimand is the model-based marginal mean difference in this trial-level gap between H6: Complete Calibrated DIAC and H1: Raw-Uncalibrated. It is estimated from the pre-specified mixed-effects model and averaged equally across the three supplied intent levels:

$$\Delta_G = E_m(G_{pt} \vee C_{pt} = \text{Complete}) - E_m(G_{pt} \vee C_{pt} = \text{Raw})$$

A negative value of the primary contrast favours H6 because a smaller trust-alignment gap indicates closer correspondence between willingness to act and recommendation correctness; zero indicates no estimated difference, and a positive value favours H1. The participant-case response remains the trial-level unit of analysis, while participant and benchmark case are modelled as clustering factors. The estimand is a model-based marginal contrast for the formative sample, not a participant-level mean difference or a population-level causal effect. Behavioural overreliance rate, low-confidence uncritical-acceptance rate, explanation usefulness, actionability, willingness to review and decision time remain secondary endpoints.

Trial-level analysis uses presentation condition and supplied intent as fixed effects, with random intercepts for participant and benchmark case to account for repeated observations of both people and cases. Binary uncritical-acceptance outcomes use a mixed-effects logistic model; ordinal ratings use a cumulative-link mixed model where estimable; and approximately continuous ratings or log-transformed decision time use a linear mixed model. Model convergence requires a successful optimiser status, finite coefficients and standard errors, no singular random-effects fit and no variance estimate fixed at a numerical boundary. If the cross-classified model fails any of these checks, the pre-specified fallback uses case fixed effects with participant-clustered bootstrap intervals. Effect estimates and 95% confidence intervals accompany p-values. Holm correction controls the family-wise error rate across the pre-specified secondary pairwise condition contrasts; the single primary contrast is identified and interpreted separately.

Eligible observations require recorded consent, a valid pseudonymous participant code, a frozen benchmark case, an assigned presentation condition and the outcome required for the relevant analysis. Preview or test submissions, duplicate participant-case submissions, records outside the approved procedure and records lacking the primary endpoint are excluded before analysis. Missing responses are not imputed: an observation missing a secondary item is excluded only from that item-specific analysis. The primary analysis uses all otherwise valid trials, and a complete-participant sensitivity analysis requires all nine trials. No observation is removed because the participant disagrees with the recommendation, because the recommendation is incorrect or because its response time is inconvenient. All findings remain formative unless recruitment, balance, model convergence and data-completeness conditions are satisfied.

3.6.12 Software and Reproducibility Testing

Functional tests verify campaign input, preprocessing, CampaignNet and ActionNet evidence, campaign-level channel evidence, candidate generation, exact binary rule activation, both intent pathways, calibration, explanation and recommendation logging.

Integration tests verify that the complete recommendation package transfers correctly between the backend, database, and user interface. Performance tests record p50 and p95 latency. Security and privacy tests verify access control, data minimisation, and exclusion of unnecessary personally identifiable information.

Each principal model experiment is executed using three predefined random seeds. Experiment identifiers, software versions, dataset version, partition identifiers, configuration parameters, outputs, and errors are recorded automatically.

Each run produces the following reproducibility artefacts:

a. results/<run_id>.json

b. logs/<run_id>.log

c. refine-logs/EXPERIMENT_RESULTS.md

3.7 Experimental Setup

Five experiment blocks evaluate distinct research claims using the same frozen decision-case benchmark where applicable. This structure makes the contribution of each system component observable instead of reducing the evaluation to a single overall performance value.

Preparatory Gate P0 freezes the source and benchmark versions, linked split manifest, annotation rubric, intent and action definitions, 14-rule catalogue, fixed score coefficients, calibration procedure, explanation templates and primary metrics. After P0 passes, runs R002-R006, corresponding to Blocks 1-5, are executed.

3.7.1 Block 1: Source Audit and Baseline Decision Benchmark

Block 1 establishes whether the public source and audited decision benchmark satisfy their separate gates and compares the defined partial decision baselines under shared splits, labels and metric code.

The source-audit gate requires at least 500 campaign-period records, at least three marketing channels, at least six monthly periods, and at least one downstream business outcome. If the source does not satisfy these requirements, the project is reduced to a two-channel configuration and any cross-channel synergy claim is treated as exploratory.

A frozen benchmark of between 120 and 150 decision cases is constructed from the source data. Each case is assigned one reference action from Increase, Maintain, Reduce, and Investigate. Reference actions are produced through a supervisor-reviewed annotation rubric and relevant historical campaign context.

Before full annotation, a pilot inter-rater study is conducted using between 20 and 30 cases. Two reviewers then independently label all benchmark cases. Cohen's kappa must be at least 0.60 before the benchmark can be frozen. Disagreements are submitted to a third reviewer, and cases that remain unresolved are excluded.

The authoritative progression is B1: Heuristic Scorecard, B2: Predictive Only, B3: Predictive plus Channel Evidence and B4: Integrated Learned Backbone. Supplementary reference comparators are B0: Majority-Class Popularity, Intent-Only Deterministic Baseline, Weighted Deterministic Baseline and Unconstrained AI Baseline. Proposed-method component removals are reported separately as ablations. All systems preserve the canonical action order and use the same held-out cases and metric functions.

The primary metrics are macro F1, top-2 agreement, actionability score on a held-out subset, and p50 response latency. B4: Integrated Learned Backbone must at least match or outperform B1-B3 while maintaining interactive response time. If macro F1 is below 0.60 or fails to improve upon B1: Heuristic Scorecard by at least 0.05, subsequent logic, confidence-calibration and reliance findings are presented as exploratory rather than confirmatory.

3.7.2 Block 2: Confidence Calibration and Appropriate Reliance

Block 2 tests whether calibrated confidence and structured explanations support appropriate reliance.

Temperature scaling is fitted using only the validation partition. Confidence values are grouped into the following five predefined bins:

a. [0.0, 0.2)

b. [0.2, 0.4)

c. [0.4, 0.6)

d. [0.6, 0.8)

e. [0.8, 1.0]

As defined in Table 3.5A, the compared conditions are H1: Raw-Uncalibrated, H2: Calibrated-Confidence Only, H3: Structured-Uncalibrated and H6: Complete Calibrated DIAC. Block 2 therefore separates the effects of confidence calibration and structured explanation without introducing alternative condition aliases.

The primary calibration metric is Expected Calibration Error for the Balanced profile, accompanied by the five-bin support table and the validation action-support statement. The human-centred metrics are trust-alignment gap, behavioural overreliance, low-confidence uncritical acceptance, explanation usefulness and p95 response latency. Every reliance comparison uses participant-level clustering. Results do not describe Increase as validated by calibration, because no Increase reference case occurs in the validation partition, and all calibration conclusions remain limited to the observed 19-case validation distribution.

If calibration does not reduce the trust-alignment gap or the relevant uncritical-acceptance measures, the negative result is reported. In this case, the system may retain explanation support but must not claim that calibrated confidence improves appropriate reliance.

3.7.3 Block 3: Weighted-Safeguard Consistency

Block 3 tests whether the weighted safeguard layer reduces logically inconsistent recommendations.

A recommendation is classified as conflicted when it violates at least one predefined budget, channel-cap, or intent-compatibility rule. Multiple violations within the same case count as one conflicting recommendation when calculating the conflict rate, though each violation type is recorded separately for diagnostic analysis.

The compared systems are A1: Full Method without Rules, the separately named Post-hoc Hard-Rule Filter comparator, and P: Full Proposed Method using the predefined and frozen weighted safeguard catalogue.

The primary metrics are conflict rate, scenario-fit rate, actionability score, and recommendation reversal rate after constraint application. Predefined and frozen rule weights, rule-firing rates, and representative test-set activations are also reported.

The weighted safeguard layer reduces conflict rate by a practically meaningful margin without causing a substantial reduction in actionability. If the predefined rule-weight profile produces behaviour inconsistent with the intended domain logic, the claim concerning logic-constrained reasoning is reduced accordingly. If the rule layer increases p95 latency by more than 20% without reducing conflict rate, it is simplified or removed from the final claimed contribution.

3.7.4 Block 4: Intent-Conditioned Recommendation Study

Block 4 evaluates whether business-intent conditioning improves contextual relevance.

The evaluation uses held-out benchmark cases balanced where feasible across project-defined Survival, Retention and Growth conditions. A2: Full Method without Intent disables both the ActionNet intent input and downstream intent logic; P: Full Proposed Method retains both pathways.

The compared systems are A2: Full Method without Intent, A3: Full Method without Calibration, and P: Full Proposed Method.

The primary technical metrics are per-intent recommendation accuracy, Macro F1, scenario-fit rate and logical-conflict rate. The A2-to-P comparison tests whether removing intent conditioning changes outcomes within Survival, Retention and Growth. A2 is a technical ablation rather than a participant-study presentation condition; therefore, human actionability and willingness-to-act effects are not attributed specifically to intent conditioning.

3.7.5 Block 5: Explanation Template Variant

Block 5 is a supporting experiment used to select the final dashboard explanation format.

As defined in Table 3.5A, the compared formats are H4: Terse-Calibrated, H5: Structured-Calibrated No Caveat, and H6: Complete Calibrated DIAC.

The participant-level metrics are actionability, explanation usefulness, trust alignment and willingness to review before acting. All measures are reported separately because a format may improve perceived usefulness while reducing review willingness. A template is selected for the final interface only when its benefits and reliance-related trade-offs are explicitly reported.

3.7.6 Run Order and Decision Gates

Table 3.6 maps run identifiers to preparatory gates and experiment blocks. R001 corresponds to Preparatory Gate P0 and is not an additional experiment block; R002-R006 correspond to Blocks 1-5 respectively.

Table 3.6

Run Order and Decision Gates

Run	Experiment	Decision gate
R001	Preparatory Gate P0: source audit, rubric pilot, reference-action agreement and benchmark freeze	Does the source satisfy the audit gate, and is Cohen's kappa at least 0.60?
R002	Block 1: baseline decision-performance comparison	Does the integrated backbone reach macro F1 of at least 0.60 and improve upon the heuristic baseline by at least 0.05?

R003	Block 2: confidence calibration and appropriate-reliance study	Does calibration reduce the trust-alignment gap and the two uncritical-acceptance measures while keeping p95 latency below two seconds?
R004	Block 3: logic-constrained consistency experiment	Do weighted safeguards reduce conflicts without unacceptable actionability or latency loss?
R005	Block 4: intent-conditioning experiment	Does intent conditioning improve scenario fit and judged usefulness?
R006	Block 5: explanation-template experiment	Is one explanation format sufficiently better to justify adoption in the final interface?

The full audited benchmark requires at least 120 eligible cases before the 70:15:15 split. This serves as a benchmark eligibility gate, not a requirement for the validation subset to contain 120 cases. With 125 cases, the validation subset contains 19 cases and is used exclusively to fit temperature. When fewer than eight relevant practitioners participate, human findings are reported as formative and not generalized to the wider SME population.

3.7.7 Software, Hardware and Versions

Chapter 3 records the planned technology categories and reproducibility controls. Exact as-executed package versions, hardware and database verification boundaries are reported once in Chapter 4 to prevent contradictory version claims.

Table 3.7

Development Environment

Category	Technology	Purpose
Frontend Framework	React	Interactive user interface
Build Tool	Vite	Fast frontend development and production build

Programming Language	TypeScript	Type-safe frontend development
Backend Framework	FastAPI	RESTful API and AI service integration
Programming Language	Python	AI implementation and backend logic
Database	MySQL	Persistent storage of campaigns, recommendations, and evaluations
ORM	SQLAlchemy	Database access and object-relational mapping
Deep Learning Framework	PyTorch	Neural predictive model implementation
Data Processing	pandas	Data cleaning, preprocessing, and feature engineering
Machine Learning Utilities	scikit-learn	Dataset splitting, preprocessing, baseline models, and evaluation metrics
Version Control	GitHub	Source code management
IDE	Visual Studio Code	Software development
Operating System	Windows 11	Development environment
Hardware	Intel Core i5 Processor, 24 GB RAM	Experimental platform

3.7.8 Software Versions

Dependency lockfiles, a Python requirements file, fixed seeds, and versioned data and configuration artifacts are retained. Table 4.7 specifies the selected development environment; Chapter 5 records the actual environment used for formal evaluation.

Table 3.8

Software Versions

Software	Version
Python	3.11.9
React/React DOM	19.2.7
Vite	8.1.4
TypeScript	7.0.2
FastAPI	0.139.0
SQLAlchemy	2.0.46
MySQL	8.4
PyTorch	2.7.1+cu118
pandas	2.1.4
scikit-learn	1.7.2
Git	2.51.2
Visual Studio Code	Version recorded in the project environment capture
Windows	Windows 11

The project follows standard dependency management practices to maintain consistent development environments across all setups.

Frontend dependencies are managed using pnpm, while backend Python packages are managed using pip and recorded in a requirements.txt file. This approach allows the complete software environment to be recreated on another machine with identical package versions.

Version control is maintained using GitHub, where source code revisions are tracked throughout the development lifecycle.

3.8 Ethics, Privacy and Reproducibility

3.8.1 Ethical Considerations

It's an advisory system and not an automatic marketing system. The proposed study to involve human participants was granted TAR UMT ethics approval with reference number TARUMT/FOCS/ETHICS_APPROVAL/001 signed on 11/10/2026. This report details an approved study which is a formative comparison of counterbalanced recommendation-presentation conditions for 18–24 participants including at least 8 SME owners, marketing practitioners, campaign executives or those with marketing decision experience. Current instrument does not provide an unaided action. The study documentation includes information about consent, pseudonymous participant code, presentation condition, willingness to act, willingness to review, acceptance decision, usefulness, specificity, feasibility, and clarity ratings and decision times. No observer and customer level information is collected that can be used to identify the participant or the customer.

The approved participant information sheet and informed consent procedure are followed when recruiting participants and collecting data. This is an optional participation; they can withdraw from participation as per the approved protocol. Persons are anonymised when responding and the analysis-ready data does not contain personal information.

The API has a built-in technical ethics gate that will not allow case and response rules to be activated for human-study unless: ETHICS_APPROVAL_REFERENCE is set to the official reference TARUMT/FOCS/ETHICS_APPROVAL/001 and the audited-benchmark eligibility requirement is met. This control will not be sufficient to prevent any accidental recruitment before appropriate authorisation, but will not replace institutional review, consent processes and data-governance requirements.

3.8.2 Privacy and Data Governance

The project is a data minimisation one, keeping only the attributes that are needed for the model, evaluation, explanation and calibration of the rules. No personal information of customers is included in the data set. The results of each benchmark are reported in an aggregated form in order to further protect the individual privacy. These practices are consistent with the principles of privacy provided for in the Personal Data Protection Act 2010 of Malaysia including with regard to purpose limitation and privacy protection of personal data.

3.8.3 Experimental Reliability and Reproducibility

The project ensures reliability by a number of means. It freezes all the parameters for random number generation. It documents all preprocessing steps; It has a version control system; It records all the experimental configurations. All test cases are used in the same way in all baselines and ablation variants. Further, rules, intent labels, action labels and metric formulas are defined prior to final evaluation to ensure the results can be reproduced and the results compared with others.

3.8.4 Study Limitations and Validity

This study remains at the prototype stage, which inherently constrains its evaluation scope. The dataset cannot possibly encompass every SME industry or campaign context, while our action space is restricted to just four options: Increase, Maintain, Reduce and Investigate. Similarly, our intent taxonomy covers only Survival, Retention and Growth. Such boundaries strengthen experimental control but necessarily limit generalizability. Moreover, the evaluation is offline and benchmark-based, it cannot assert long-term business impact without future live deployment.

It maintained internal validity through a frozen test set, identical comparison cases, and component-removal experiments, though it acknowledges that benchmark labeling and rule definitions may still introduce judgment bias. Construct validity received support from mapping each objective to explicit technical and human-centered metrics, while recognizing that bounded actionability and scenario-fit measures cannot capture every managerial judgment. External validity faces limitations due to the use of one secondary benchmark and three simplified SME intents. Conclusion validity benefits from confidence intervals, multiple-comparison correction, and practical-significance interpretation, yet findings remain contingent on benchmark size and participant coverage.

3.8.5 Reproducibility and Data Availability

To enhance the reproducibility of the experimental results, it has implemented several key controls throughout the planned evaluation.

Initially, the benchmark dataset will be frozen prior to any model training. Once the training, validation, and test partitions are established, they will remain fixed throughout all experiments. This ensures that every baseline model, ablation study, and the proposed method undergoes evaluation using precisely the same data partitions.

Second, it will use deterministic random seeds during model training and dataset splitting to reduce variations caused by random initialization and sampling.

Third, source code must be version-controlled. The execution gate will require a clean tagged commit, dependency lockfiles, data hashes, and configuration hashes before confirmatory runs; a working tree containing uncommitted implementation changes will not qualify as a frozen repository state. Chapter 5 will document the achieved source-freeze commit and tag, along with the exact source identity used by each executed evaluation run.

It automatically log evaluation metrics, prediction outputs, and experimental configurations after each experiment. This approach enables fair comparisons between baseline models, ablation studies, and the proposed neuro-symbolic framework.

Before beginning experimentation, it establishes fixed business intent definitions, recommendation rules, benchmark datasets, and evaluation procedures. This prevents post-hoc modifications to the evaluation process, ensuring methodological rigor.

3.8.6 Reproducibility Measures

Table 3.9

Reproducibility Measures

Aspect	Measure
Dataset	Frozen benchmark dataset
Dataset Split	Fixed training, validation, and testing partitions
Randomness	Fixed random seeds for dataset splitting and PyTorch model training
Software	Fixed package versions
Source Code	GitHub version control
Dependencies	npm and pip package management
Evaluation	Standardised metrics and benchmark procedures

Logging	Prediction results and experiment configurations recorded

3.9 Objective–Method–Evidence Mapping

To every research objective and question, the project relates it to a particular method, source of evidence and a measurable acceptance criterion. This will make the assessment not to be filled with unclear statements on the usefulness of the system or recommendation intelligence. Rather, all major assertions should be supported by tangible data, predetermined benchmarks, relevant metrics, ablation options, or human-focused evaluations.

There are four evidence routes that are identified by the mapping. Research Question 1 explores the agreement in reference and consistency of rules. Research Question 2 explores model-confidence calibration, usefulness of explanations, and dependence. Project-defined intent conditions are tested in Research Question 3. Research Question 4 assesses traceability, reproducibility, and latency. These concerns cannot be tackled by predictive accuracy alone.

Table 3.10 summarises the links between the research goals, questions, method of implementation, experimental procedures, and the evaluation measures in the course of this study.

Table 3.10

Alignment of the research objectives, questions, and evaluation methods

Research Objective	Research Question	Method / System Component	Evaluation Evidence	Acceptance Criterion
RO1. To develop and freeze an audited decision benchmark based on public campaign records, project defined intent scenarios and independently reviewed reference actions.	RQ1 and RQ3	Source audit, leakage-controlled preprocessing, intent-scenario assignment, independent action review and benchmark freeze.	Completeness, provenance, duplicate/leakage audit, split manifest, label support and reviewer evidence.	The benchmark is frozen before confirmatory evaluation and all 125 cases satisfy the required schema and action vocabulary.
RO2. Implementing a predictive and channel-evidence backbone, a four-action candidate space, predefined weighted safeguards and explicit intent conditioning.	RQ1 and RQ3	CampaignNet, ActionNet, campaign-level channel evidence, bounded actions, hard feasibility checks and weighted soft safeguards.	B0/B1-B4 comparisons, safety references, branch tests, rule and intent ablations under shared held-out cases.	The implemented components produce traceable outputs and the full method improves the relevant quality/safety trade-off against the declared partial baselines.
RO3. To adopt validation only model confidence calibration and Driver–Implication–Action–Caveat explanations for human review.	RQ2	Validation-only temperature scaling, five-bin reliability analysis, selective review states and structured DIAC explanations.	ECE, NLL, Brier score, reliability support, explanation faithfulness, human-study contrasts and inappropriate-reliance diagnostics.	The components are implemented and evaluated; any failure of calibration or reliance improvement is reported as a partial rather than successful objective outcome.

RO4. To assess reference-action agreement, actionability, scenario fit, rule consistency, calibration, reliance, auditability and latency with respect to defined baselines and ablations.	RQ1-RQ4	Frozen confirmatory runner, baseline parity assertions, ablations, selective-policy audit, human study and reproducibility capture.	Accuracy, Macro F1, top-two agreement, scenario fit, conflict rate, calibration, human-centred measures, trace checks and latency.	The planned evidence is reported with support counts, uncertainty, implementation boundaries and explicit negative or inconclusive findings.

3.10 Chapter Summary and Evaluation

Chapter 3 defined the evidence-reasoning-confidence approach, defining the scopes of public-source and audited-benchmark data, learned branches of evidence, weighted protection, dual intent pathways, risk compensation, confidence calibration, human-review procedures, baselines, ablations, and pre-test gates. Chapter 4 now shows how these methodological decisions are translated into real software architecture, database structures, algorithms and interface designs that are required to be implemented.

The framework of the methodology, implementation pipeline, data preparation workflow, component architecture, evaluation workflow, and requirement gathering process ensure the presence of a clear visual traceability between the proposed AI approach, system requirements, and future system design. This traceability allows a stable methodology-implementation and testing transition all through the development lifecycle.

This chapter describes the intended processes without providing any experimental data or ultimate performance assertions. Such information will be found in Chapter 5. Rather, Chapter 4 is devoted to the translation of the suggested approach into the tangible software architecture, database structures, algorithms, and interface designs that need to be implemented.

Chapter 4

System Development Blueprint

4 System Development Blueprint

This chapter takes the agreed to Chapter 3 methodology into the final system-development blueprint – coding. It completely specifies the architecture, data processing and evidence pipeline, method interfaces, equations and algorithms, baseline parity, selected environment, trace outputs, constraints and human-decision boundary. The coding, evidence from the repository, run time logs, tests and comparisons actually are included in Chapter 5 but are not evidence of completion of Chapter 4.

Throughout the chapter the language and terminology of the stage are utilised. Proposed is when a system or method proposed in the study is coded; specified is when a system or method specified for coding is coded sufficiently enough to be coded; developed or implemented is when the system or method developed or implemented in the study is coded; tested is when a system or method checked in the study is coded; and evaluated is when the evidence evaluated for a research question or objective is coded. Chapter 4 therefore, has no claims about implementation completion as a measure of the success of research, but only uses the language of specifications.

4.1 Overall Architecture or Framework Design

The artefact to be proposed is a local web system, that will be organised around three operational zones, and that will be used via the browser. The presentation/service zone is where the interface to the service (React/Typescript) is, as well as the FastAPI authentication code and Pydantic validation code and routing/persistence orchestration. The zone for recommendations consists of: 1 RecommendationEngine in progress (RecommendationEngine for Learned Evidence, Campaign level channel evidence, Weighted safeguard reasoning, Intent and risk adjustments, Validation fitted temperature scaling and structured explanation). The Persistence and research zone will be home to SQLAlchemy operational records, and versioned CSV and JSON research artefacts. The persistence target for the operations is MySQL 8.4, meaning that SQLite can only be used as a standalone test, using the same ORM contract. No indication is given whether the production deployment is to public, cloud or autonomous deployment.

The entire proposed local system and the extent of the communication is illustrated in figure 4.1. Content and evidence of a campaign submitted by a manager/administrator via React. Then, React accesses FastAPI via the REST/JSON interface to validate the request, load the records of the campaign and active-rules, call RecommendationEngine and persist the records of the recommendation and rule traces. The 14 default rules are also idempotently copied to the administrator Campaign workspace (Active and null budget) at start up, as are the 125

benchmark cases, which have been previously audited and frozen. Versioned benchmark is the last version to be assessed – another copy of the database can be used for campaign analysis and the requests for recommendations is called operational version. On the one hand, an update of the advertising-platform is not made automatically, on the other hand no marketing action is made.

For each recommendation request, the engine constructs the 13 runtime features, obtains CampaignNet evidence from 11 features, obtains ActionNet evidence from 13 features plus the one-hot selected intent and adds training-derived campaign-level channel evidence. The four candidates—Increase, Maintain, Reduce and Investigate—remain in canonical order. The frozen catalogue first applies hard feasibility exclusions and then applies weighted soft score adjustments, followed by intent and risk offsets. Validation-fitted temperature scaling calibrates model confidence rather than human trust. The selected intent is singular and manager supplied; mixed intent is rejected for clarification.

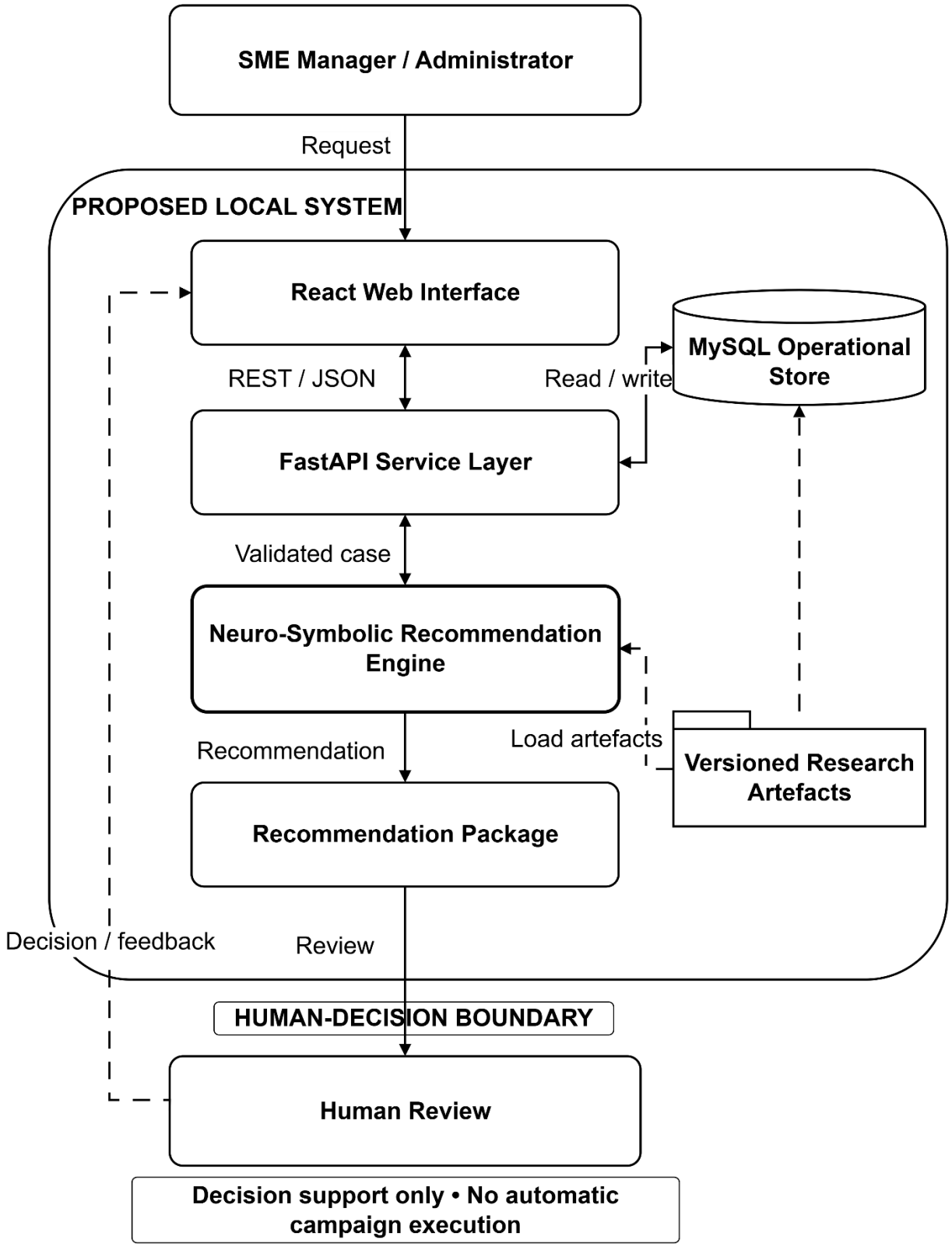


Figure 4.1 Overall Architecture and Boundaries of the Proposed System

Table 4.1

Specified System Scope and Verification Plan

Artefact	Specified design	Planned verification
Architecture and modules	Five specified layers with explicit Python, REST and database interfaces.	Repository-structure, import and interface tests shall verify the layer mapping.
Web prototype	React interface, FastAPI service, campaign workflow, recommendation review, rule administration and evaluation views.	Frontend production-build and API-integration checks shall verify the interface contract.
Decision method	Predictive evidence, local predictive sensitivity, four actions, binary safeguards with weighted soft influence, intent ranking, confidence calibration and structured explanation.	Unit, integration and locked-runner checks shall verify component behaviour and trace outputs.
Data and evidence	Raw, processed and audited benchmark layers plus application and experiment-record schemas.	Source, schema, provenance and split-isolation procedures shall verify the evidence boundaries.
Deployment	Local workstation only; MySQL 8.4 is the target operational database and SQLite is an isolated-test fallback.	Verification shall not be interpreted as public, cloud or autonomous production deployment.

As the project results in a software artefact, the project is documented in the form of needs of the implementation in a concise and testable format, but not the methodological rationale (as provided in chapter 3). Each requirement has an observable verification evidence and performance outcomes which is reserved in Chapter 5.

Table 4.2

Testable Functional and Non-Functional Requirements

ID	Requirement	Planned verification
FR1	Authenticate users, maintain signed sessions and restrict administrative rule functions by role.	Login, registration, password-change and forbidden-

		administrator API tests shall verify role and session controls.
FR2	Create, edit, delete, list and idempotently import campaign records after validating supported channels and numeric fields.	Campaign CRUD, duplicate-import and invalid-channel API tests shall verify record validation.
FR3	Generate one ranked recommendation package from Increase, Maintain, Reduce and Investigate for a manager-selected intent.	Engine and end-to-end recommendation-flow tests shall verify the four-action output contract.
FR4	Return raw and calibrated confidence, ranked alternatives, predictive evidence, activated rules and a structured explanation.	Engine-package and recommendation-detail tests shall verify confidence, alternatives, evidence, rules and explanation fields.
FR5	Persist recommendation history, managerial decision and evaluation feedback without automatically updating model or rule weights.	Database-backed recommendation, decision and feedback tests shall verify persistence without online learning.
FR6	Execute baselines, ablations and the proposed method on the locked benchmark with recorded configurations and exports.	The confirmatory runner and run-register checks shall verify configuration and export completeness.
FR7	Collect blinded, pseudonymous human-study responses and prevent duplicate participant-case submissions.	Counterbalancing, token, duplicate-submission and export tests shall verify the human-study data contract.
NFR1	Keep p95 recommendation latency below the two-second acceptance threshold defined in Chapter 3.	Per-case latency shall be captured for Chapter 5 p95 analysis.
NFR2	Reproduce a run from fixed data versions, split rule, seed, method identifier and configuration.	Run metadata, seed summaries and prediction exports shall support reproduction checks.

NFR3	Protect credentials and minimise identifiable participant or customer data.	Password-hash, signed-token, role and pseudonymisation checks shall verify data protection controls.
NFR4	Expose uncertainty and retain human approval for every business action.	Confidence caveat, Investigate fallback and accept/defer/reject checks shall verify retained human authority.
NFR5	Allow model, rule, intent, calibration, explanation and interface components to be changed independently.	Module-boundary, API-contract and rule-record checks shall verify replaceability without changing external contracts.

4.2 Data Pipeline Implementation

The preprocessing and leakage-control decisions in Sections 3.2.3 and 3.3 are translated into an exact executable pipeline centred on preprocessor.py. prepare_frame shall load, validate, de-duplicate and chronologically order batch data, while case_features shall reconstruct the same 13 runtime features from an API-validated case. Figure 4.2 specifies the separate public-source and audited-benchmark routes, their 38,888/8,333/8,334 and 87/19/19 partitions, the training-only and validation-only boundaries, and the separate startup route that copies 125 frozen cases into operational Campaign rows without changing the benchmark evidence.

The specified batch flow is deterministic. pandas.read_csv loads the file and drop_duplicates removes exact duplicate rows. The pipeline rejects campaign frames containing annotation or reference-action fields, trims date text, replaces slash separators and accepts only a consistently applied D-M-YYYY or YYYY-M-D format. Mixed or invalid formats raise an error. Records are then sorted by Date and Campaign_ID, the nine non-derived model measures are coerced to numeric values, and medians are calculated from the earliest 70% only. A field whose training portion contains no usable numeric value causes the pipeline to fail; otherwise, the frozen medians fill missing values in all three partitions. NumPy constructs click-through rate, conversion rate and lead-to-conversion rate with a zero fallback, while cost per conversion falls back to Acquisition_Cost when Conversions is zero. The returned frame stores the fit-row count and median dictionary in DataFrame.attrs and exposes the validated 13-feature model structure.

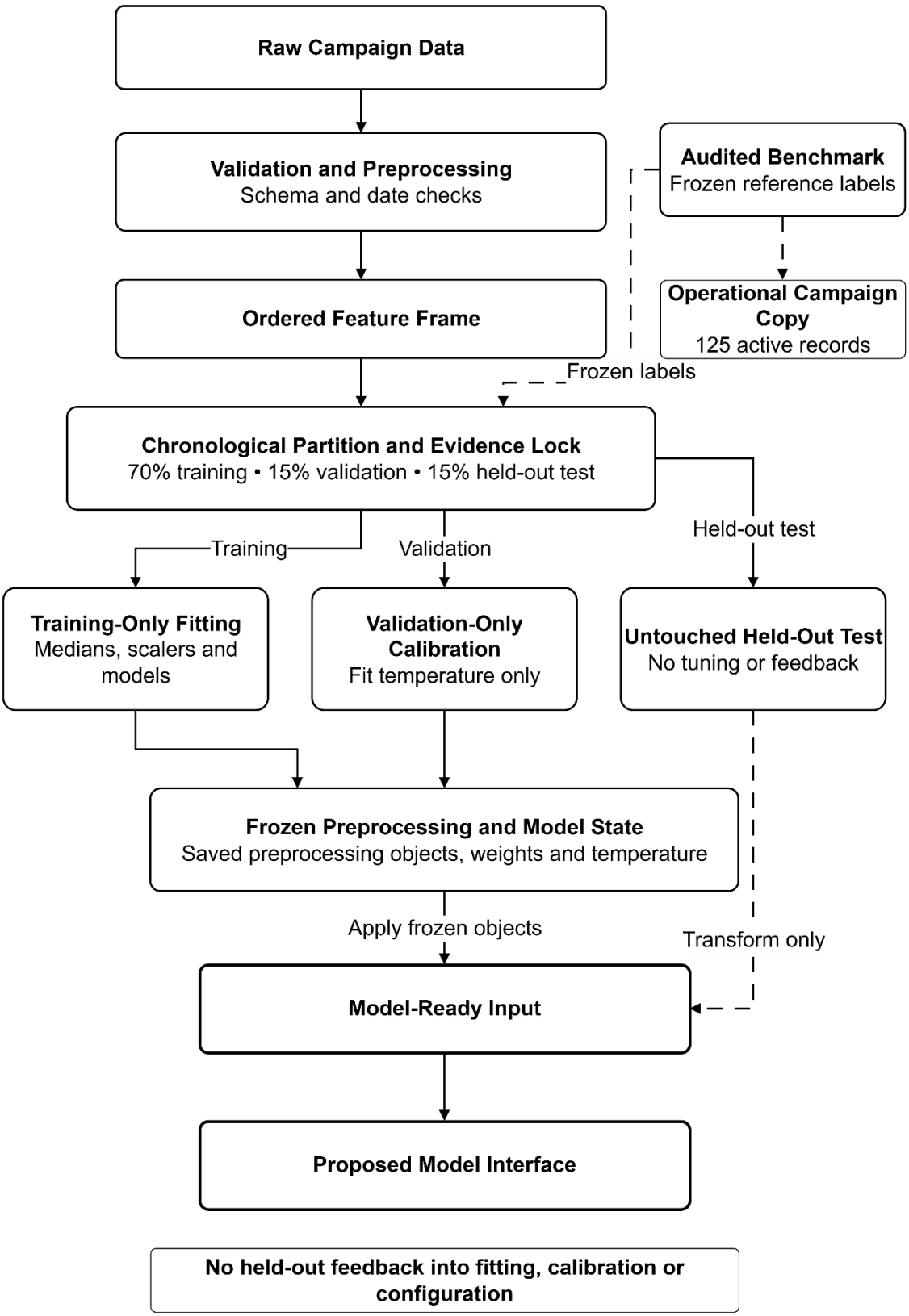


Figure 4.2 Specified Data-Processing and Evidence Pipeline

The immutable raw source, data/layers/01_raw_dataset_55555x16.csv, contains 55,555 records and 16 original fields. A separate versioned research layer, data/layers/02_processed_model_data.csv, contains 55,555 records and 31 fields: the original 16 fields plus 15 derived audit and research fields. It is retained for lineage and exploratory evidence and is not presented as the executable 13-feature model matrix. The specified pipeline shall construct CTR, conversion rate, lead-to-conversion rate and cost per conversion in memory from validated source fields through prepare_frame and case_features. This distinction prevents the persisted research layer from being confused with the runtime model interface.

The approved benchmark contains 125 records and 26 fields, and every case is linked to a public-source Campaign_ID. Before formal evaluation, the split-isolation procedure shall verify that the 87 benchmark-training identifiers link only to permitted source-training records and that none of the 19 benchmark-validation or 19 benchmark-test identifiers occurs in the 38,888-row source-training partition. The procedure shall fail the evaluation gate if prohibited overlap is detected. The executed isolation result is reported in Chapter 5.

The calibration eligibility condition $|B| \geqslant 120$ refers to the complete, fully audited benchmark B before splitting. It does not assert that 120 cases are available for calibration. Temperature T is fitted only from the 19-case audited validation subset; the 19-case benchmark test subset remains untouched until Chapter 5.

Train-inference parity is specified for feature order and branch-specific scalers. CampaignNet shall select 11 predictors; ActionNet shall transform all 13 campaign features and append a three-element intent vector. Budget handling shall remain consistent across the API, database, CSV importer, React form and engine: an omitted or blank budget remains null, is displayed as 'Not provided', and makes the budget-feasibility rule non-applicable. An explicitly entered zero remains a real zero and is evaluated by the rule, as is a positive amount. Planned API, database, importer, interface and engine tests shall verify these three cases; their executed outcomes belong in Chapter 5.

The engine shall reconstruct preprocessing and model state deterministically at start-up rather than depend on a single serialised pipeline object. Versioned source, benchmark, rule and configuration files shall provide the durable inputs, while run metadata, predictions, confidence, rule traces and persistence records shall provide the required evidence outputs. Before formal evaluation, the source configuration and research artefacts shall be locked under identifiable commits or archived snapshots. Chapter 5 shall report the actual source identifier, tag, executed run identifier and verification outcome.

The blueprint selects two bounded choices relative to the broader Chapter 3 plan. General campaign-category encoding is excluded from the numerical backbone: the three-value intent is one-hot encoded only inside the supervised action model, while channel is handled through a separate training-derived evidence lookup. General outlier removal is also excluded; only the ROI training target shall be percentile-clipped so that unusual campaign rows are not discarded. These choices reduce fitted preprocessing state and keep the transformation trace auditable, while requiring runtime fallback values and in-memory preprocessing state to be recreated consistently.

Table 4.3

Persisted Data Structures and Evidence Traces

Store or entity	Key fields / indexes	Implementation purpose
Raw and processed CSV layers	Campaign_ID; 16 source fields; four derived ratios; date order.	Immutable source lineage and repeatable feature construction.
Audited benchmark	Reviewer labels, adjudicated intent/action, notes and Annotation_Status.	Locked human reference cases and validation/test partitions.
User, Intent and Campaign	Unique indexed email; campaign PK; user and intent FKs; campaign metrics.	Authenticated ownership, selected intent and operational input.
BusinessRule	Unique rule code; optional intent FK; condition, weight, priority and active state.	Versionable symbolic knowledge used by the rule engine.
Recommendation and RecommendationRule	Method, raw/calibrated confidence, evidence, ranking, latency; triggered rule detail.	Recommendation-level explanation and audit trace.
Evaluation and HumanStudyResponse	Recommendation FK; pseudonymous participant index; unique participant-case pair.	Feedback, blinded condition response and duplicate prevention.
Experiment outputs	Run ID, seed, split, versions, predictions, confidence, conflicts and latency.	Reproduction of Chapter 5 calculations from stored evidence.

The application database and versioned research files have distinct operational and evidential roles, but shared identifiers preserve traceability between them. The database supports interactive transactions, whereas CSV and JSON artefacts preserve frozen data and run outputs. Recommendation rows store the selected method, raw confidence, calibrated confidence, ranked actions, evidence, explanation and latency; RecommendationRule rows retain the rule, satisfaction state and diagnostic detail. Human-study records use a participant code rather than a name, enforce one response per participant-case pair, and exclude free-text comments from the analysis-ready export.

Human-study presentation-condition identifiers shall be stored and displayed exactly as defined in Table 3.5A: H1: Raw-Uncalibrated; H2: Calibrated-Confidence Only; H3: Structured-Uncalibrated; H4: Terse-Calibrated; H5: Structured-Calibrated No Caveat; and H6: Complete Calibrated DIAC. Block 2 uses H1, H2, H3 and H6, while Block 5 uses H4, H5 and H6. No Chapter 4 alias or alternative condition name is permitted.

4.3 Proposed Method Implementation Design

The proposed method accepts a validated campaign, selected intent, risk tolerance and active frozen rule catalogue and produces a bounded recommendation package. CampaignNet supplies ROI-derived evidence and local predictive sensitivity; ActionNet supplies intent-conditioned four-class evidence. Fixed fusion and smoothing are followed by channel adjustments, binary weighted rule penalties, an independent Investigate boost, intent offsets and risk offsets. Temperature scaling converts the final score vector into model probabilities used for ranking, confidence communication and explanation.

4.3.1 Model Architecture and Components

Figure 4.3 presents a compact three-stage view of the proposed method. Stage I combines CampaignNet, ActionNet and training-derived campaign-level channel evidence into learned evidence. The selected intent conditions ActionNet and also follows a second path into the neuro-symbolic reasoner, while risk preference enters only the reasoning stage. Stage II integrates the learned evidence with the predefined safeguard catalogue and contextual adjustments to produce ranked action scores. Stage III calibrates model confidence, assembles the structured recommendation package and presents it to the SME manager for review. The system remains advisory and does not execute campaign or budget changes automatically. Figures 4.4, 4.5 and 4.6 provide the detailed internal views omitted here.

Figure 4.4 expands Stage I of Figure 4.3. CampaignNet receives 11 predictive features and produces ROI-derived action evidence together with local predictive-sensitivity information.

The sensitivity values describe local model influence and are not causal attributions. ActionNet receives the 13 campaign features and a three-element one-hot representation of the selected intent and produces evidence for the four bounded actions. Training-derived campaign-level channel evidence remains a separate source. The learned-evidence package records whether ActionNet evidence is audited, development-only, or unavailable, in which case the CampaignNet-only fallback is identified explicitly.

Figure 4.5 expands Stage II of Figure 4.3. CampaignNet and ActionNet evidence are combined using the specified fixed fusion coefficients, smoothed and adjusted by training-derived campaign-level channel evidence. Fourteen predefined and frozen safeguards then evaluate the candidates. Six hard safeguards exclude infeasible actions; eight soft safeguards apply weighted score adjustments. If no executable action remains, Investigate is returned with NO SAFE RECOMMENDATION. Intent and risk offsets are then applied while preserving the canonical action order: Increase, Maintain, Reduce and Investigate.

The selected intent follows two distinct paths. First, its one-hot representation conditions ActionNet. Second, intent-specific safeguards and downstream intent offsets modify the candidate-action scores. An intent-removal ablation must disable both paths. Investigate also has two traceable roles: it can be the audited reference action for a labelled case, and it can be the runtime fallback when the evidence for other actions conflicts with the active safeguards.

Figure 4.6 expands Stage III of Figure 4.3. Validation-fitted temperature scaling converts the final action-score vector into model probabilities for confidence communication and ranking. The recommendation package contains the selected action, calibrated model confidence, ranked alternatives, activated-safeguard evidence, a Driver-Implication-Action-Caveat explanation, and the selected intent as contextual metadata. Scores, evidence provenance, safeguard traces, calibration status, and managerial feedback are retained for audit. The SME manager may accept, modify, defer, or reject the recommendation; the system does not execute campaign or budget changes automatically.

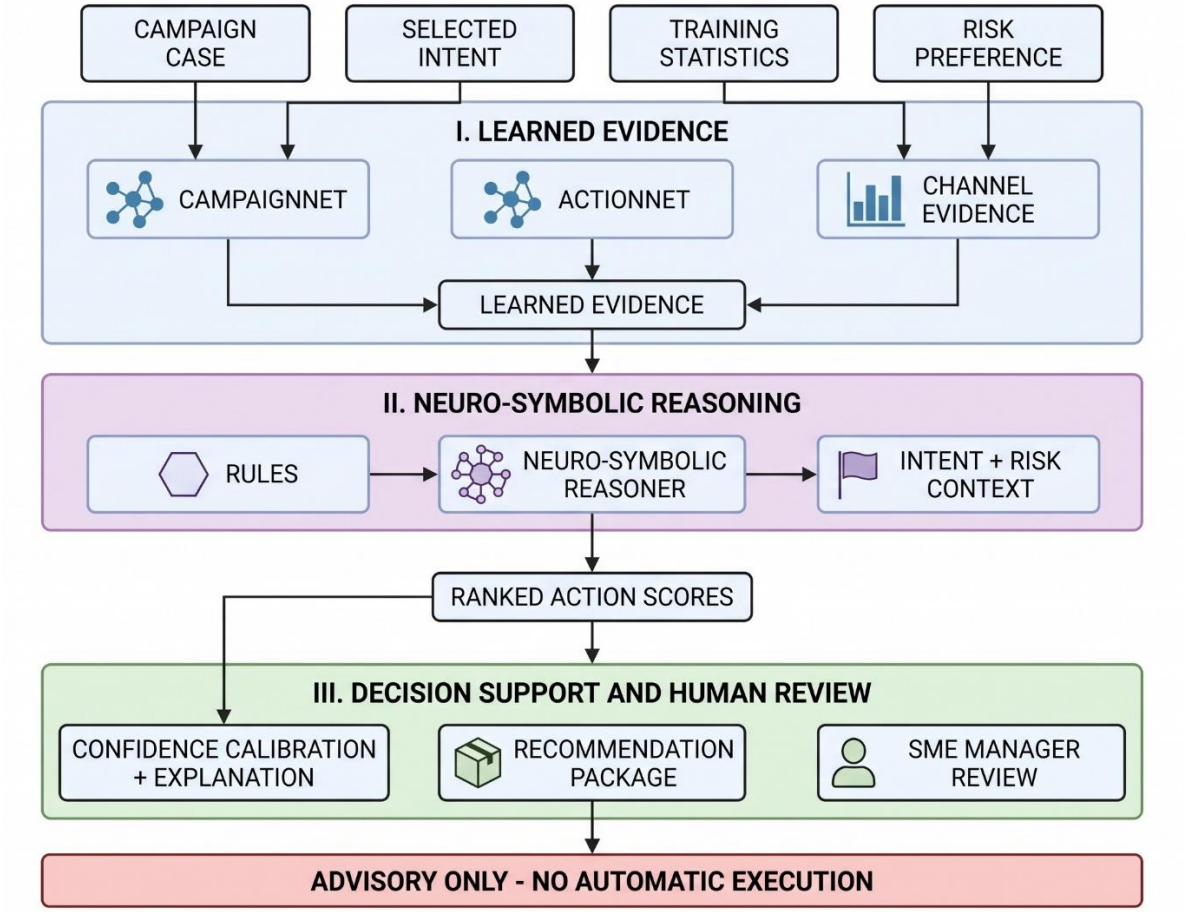


Figure 4.3 General Architecture of the Proposed Neuro-Symbolic Method

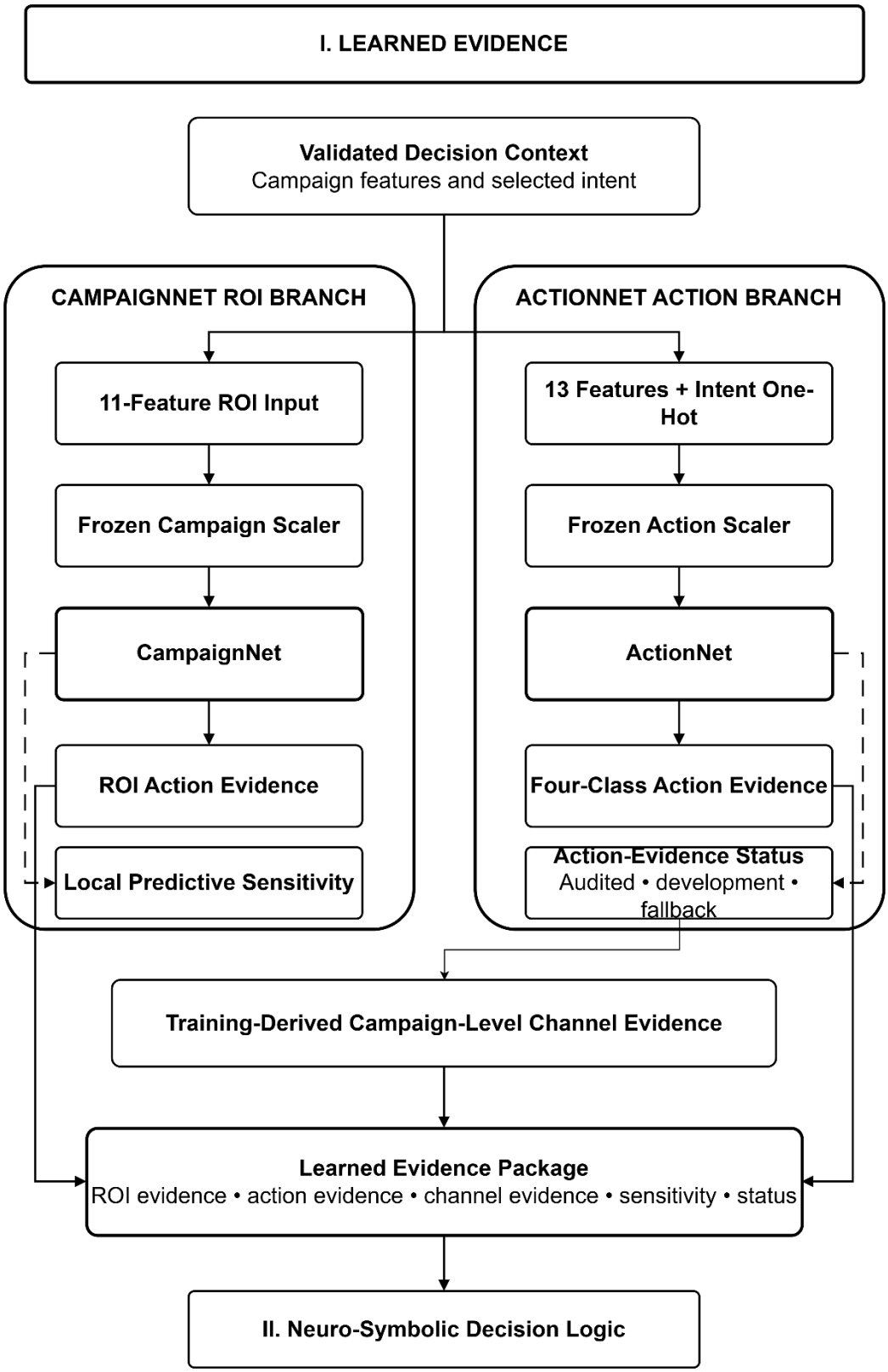


Figure 4.4 Learned Evidence Architecture

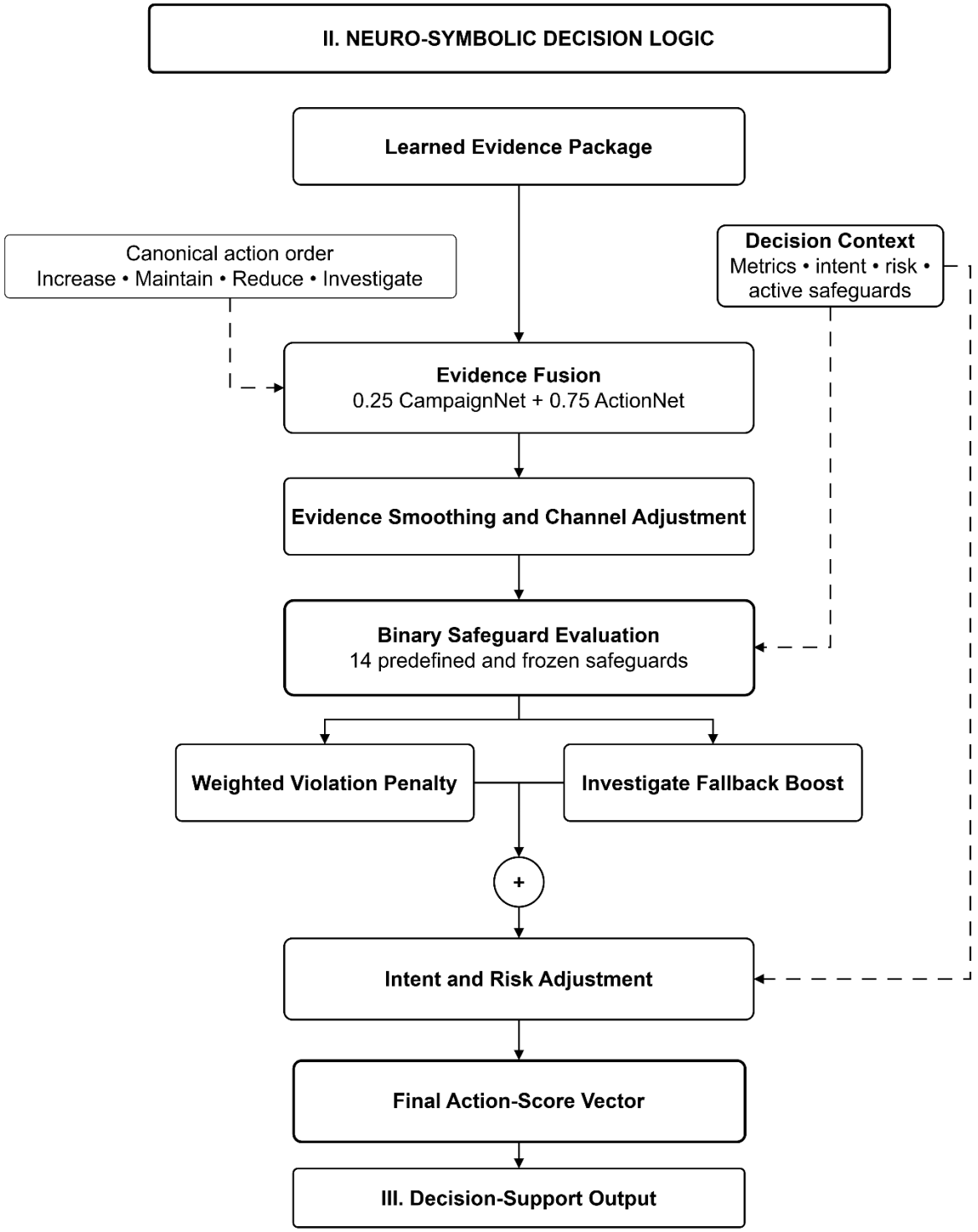


Figure 4.5 Neuro-Symbolic Decision Logic

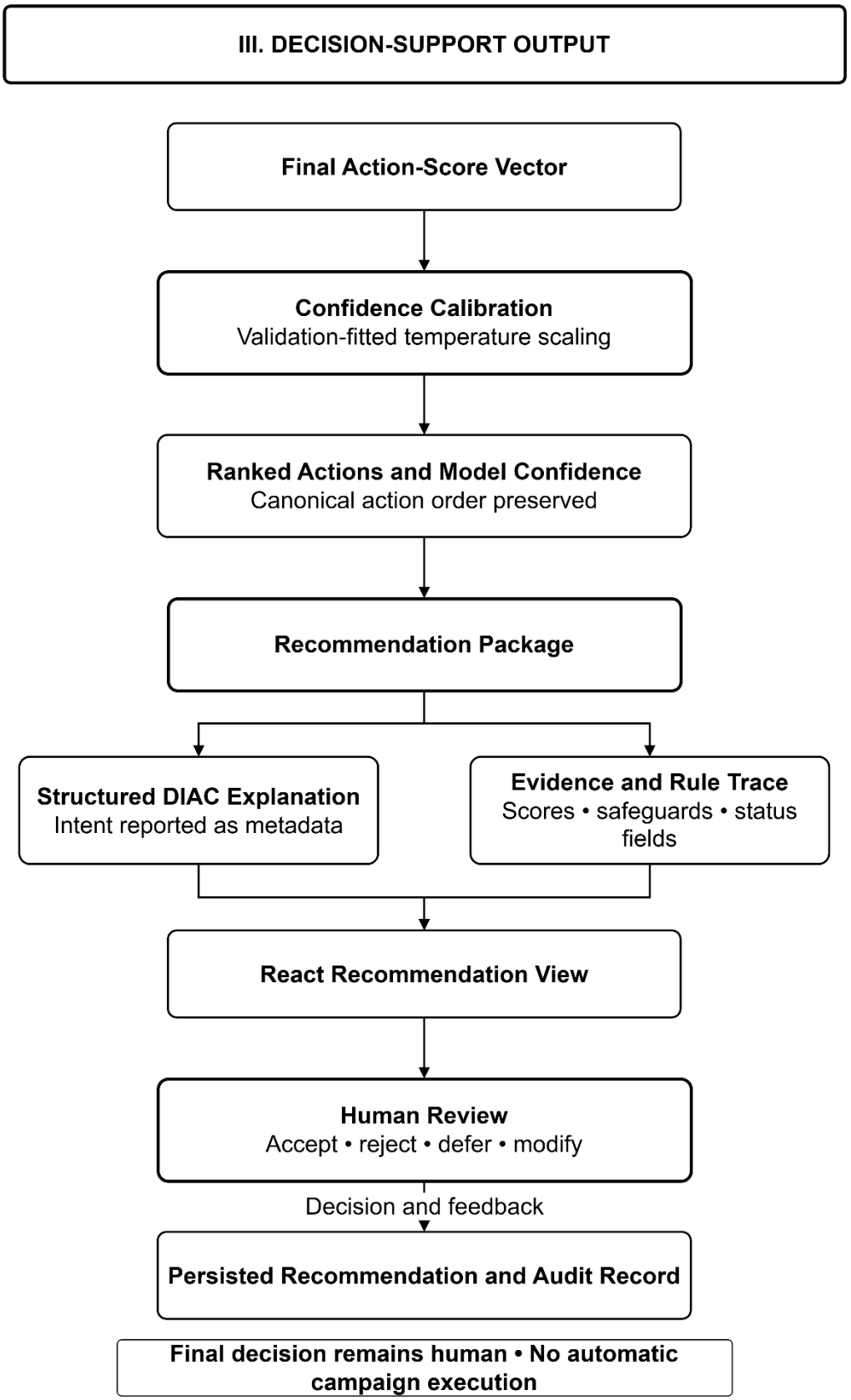


Figure 4.6 Decision-Support Output

Table 4.4

Specified Model and Software Components

Component	Input and dependency	Output / responsibility
Campaign preprocessor	CSV or API campaign fields; pandas and NumPy.	Validated 13-feature vector, ordered frame and four derived ratios.
ROI predictive model	Eleven leakage-controlled numeric predictors; StandardScaler and PyTorch.	Normalised ROI estimate and four initial action evidence values.
Action evidence model	Thirteen features plus three-value intent encoding from audited training cases.	Four-class supervised evidence for Increase, Maintain, Reduce and Investigate.
Campaign-Level Channel Evidence and Local Predictive Sensitivity	Source-training campaign records, selected channels and gradient-enabled CampaignNet input.	Normalised training-derived campaign-level channel evidence and the five largest absolute gradient-times-input local predictive-sensitivity values.
Candidate generator	Four action scores.	Bounded action dictionary and ranked softmax list.
Weighted rule reasoner	Features, optional budget, selected intent and active rules.	Score penalties, Investigate boost and per-rule satisfaction traces.
Intent ranker	Selected Survival, Retention or Growth intent.	Action-specific intent adjustments and allowed-action consistency.
Calibration and explanation	Final logits, validation temperature, local predictive sensitivity and triggered safeguards.	Calibrated confidence plus Driver-Implication-Action-Caveat explanation plus selected-Intent context.
FastAPI, React and SQLAlchemy	Validated REST payloads, signed session and ORM entities.	Human-review interface, persistent audit trail and evaluation export.

The ROI network is designed to be small, matching the CPU environment used, and is a network that has only eleven units in the input layer, 16 units in the hidden layer (ReLU) and

one unit in the regression output layer. The training target shall be clipped to its 1st and 99th percentile, standardised and fitted for 80 epochs with Adam (learning rate 0.015, weight decay 0.0001) on the public-source training frame. The action network shall be given the 13 specified features as well as a three element one hot intent vector and have a hidden layer with 24 units of ReLU activation, and output four logits. It should be fitted on the audited training partition for 220 epochs with Adam, learning rate of 0.012 and weight decay of 0.0001.

The actions that are used throughout the system and baselines are: Increase, Maintain, Reduce and Investigate. Capping the output will help free-form recommendations not circumvent the evaluation rubric. The campaign level channel evidence is based on the mean ROI and conversion data from the training set for each channel reported and normalised to [0,1]. The Local Predictive Sensitivity is the absolute product of the standardised input and gradient of the selected predictive logit, normalised and only the top 5 taken. It is an evidence for explanation, rather than a causal attribution claim.

A total of 14 default safeguards are found in the specified catalogue and these include negative ROI, low conversion, low click-through rate, low engagement, insufficient observations, strong or exceptional signals, acquisition cost, channel saturation, budget feasibility and conflicting evidence and the three intent policies. The weights and priorities will be manually defined in DEFAULT_RULES and will be locked as rule-catalogue-v1 for confirmatory evaluation. The values should not be learnt during model training and validation, and should only be investigated during validation as a result of the models' behaviour.

The chosen DEFAULT_RULES catalogue has weights ranging from 0.70 to 0.95 and priorities of 1, 2 or 3. It has a configuration identifier rule-catalogue-v1 in data/rule_catalogue_v1.json. During normal use, administrators may keep evidence of the rules but, if a change is made to the catalogue, a new configuration identifier and a new experimental run will be used instead of “changing” evidence.

Pydantic validation and SQLAlchemy persistence give clear input/output limits and typed REST endpoints isolate the web interface from the model objects. Pydantic validates the fields that are required for the campaign and supported channels. This recommendation, rule evidence and user decision is persisted by SQLAlchemy. The evidence_json also includes action_evidence_status which is audited_actionnet, development_actionnet or campaignnet_only_fallback, the fallback will be included in the explanation that is returned. The interface needs to be explicit, meaning a user needs to request a recommendation and there are accept, defer or reject controls; it doesn't actually make changes to the ads. Trust alignment

is computed in a second step based on the human behaviour, and is therefore displayed separately from the confidence.

SQLAlchemy will generate the necessary tables at start-up and on an empty database, the three intents, three demonstration campaigns and default rule catalogue will be seeded. The recommendation engine will be lazily initialised on the first call to it to avoid redoing model fitting for each endpoint call. Recommendation request will load selected campaign and active rules in the authenticated user's scope, execute the shared component engine, and commit the recommendation, rule traces, calibration status, and ActionNet evidence status as one workflow. The detail endpoint should be used to recreate the evidence package instead of silently recomputing it in case a subsequent development run alters the rules or model parameters and thus affects the evidence package created in the first run.

The schema of the application's transactions is shown in Figure 4.7. User, Campaign and Intent contain information about the decision-maker and the context of the decision, Recommendation contains the information about the action selected, the confidence and the explanation of why the selected action was chosen, and RecommendationRule stores information about the rule evidence used in the decision making process, Evaluation stores information about manager feedback. The human-study response structure is not included in the operational core diagram, but is rather evaluation-support data.

The relationships also contain an evidence trail from a user owned campaign to the intent selected, the recommendation that was generated, the rules that were activated and the evaluation that occurred. The use of a separate associative entity, Storing RecommendationRule, enables the multiple RuleOutcome records to be stored for a single Recommendation record without having to duplicate the Recommendation record. This structure facilitates historical reconstruction – a displayed recommendation derives from its campaign context, intent, confidence and explanation and rule evidence without the need to rerun the model.

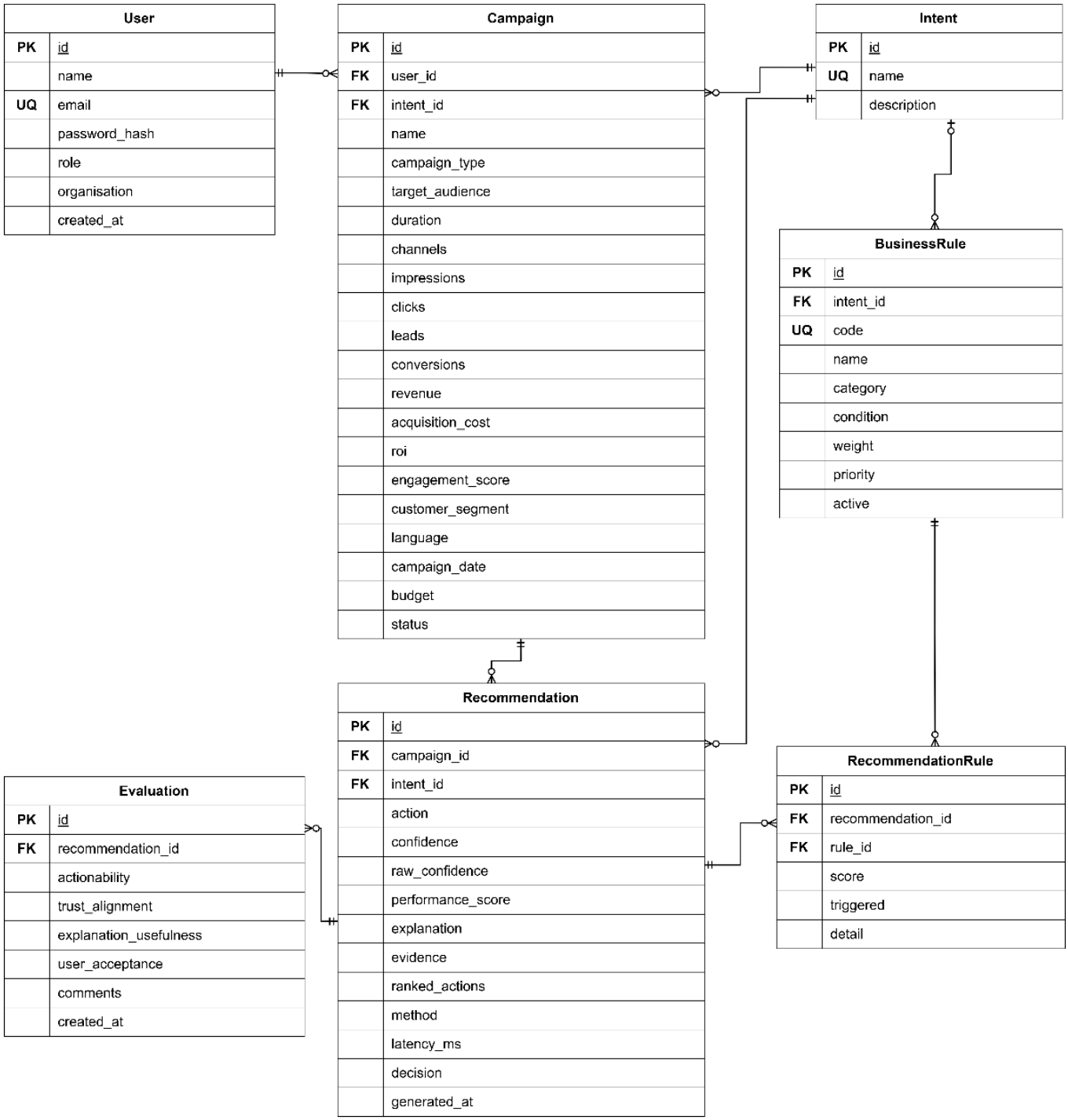


Figure 4.7 Application Entity Relationship Diagram

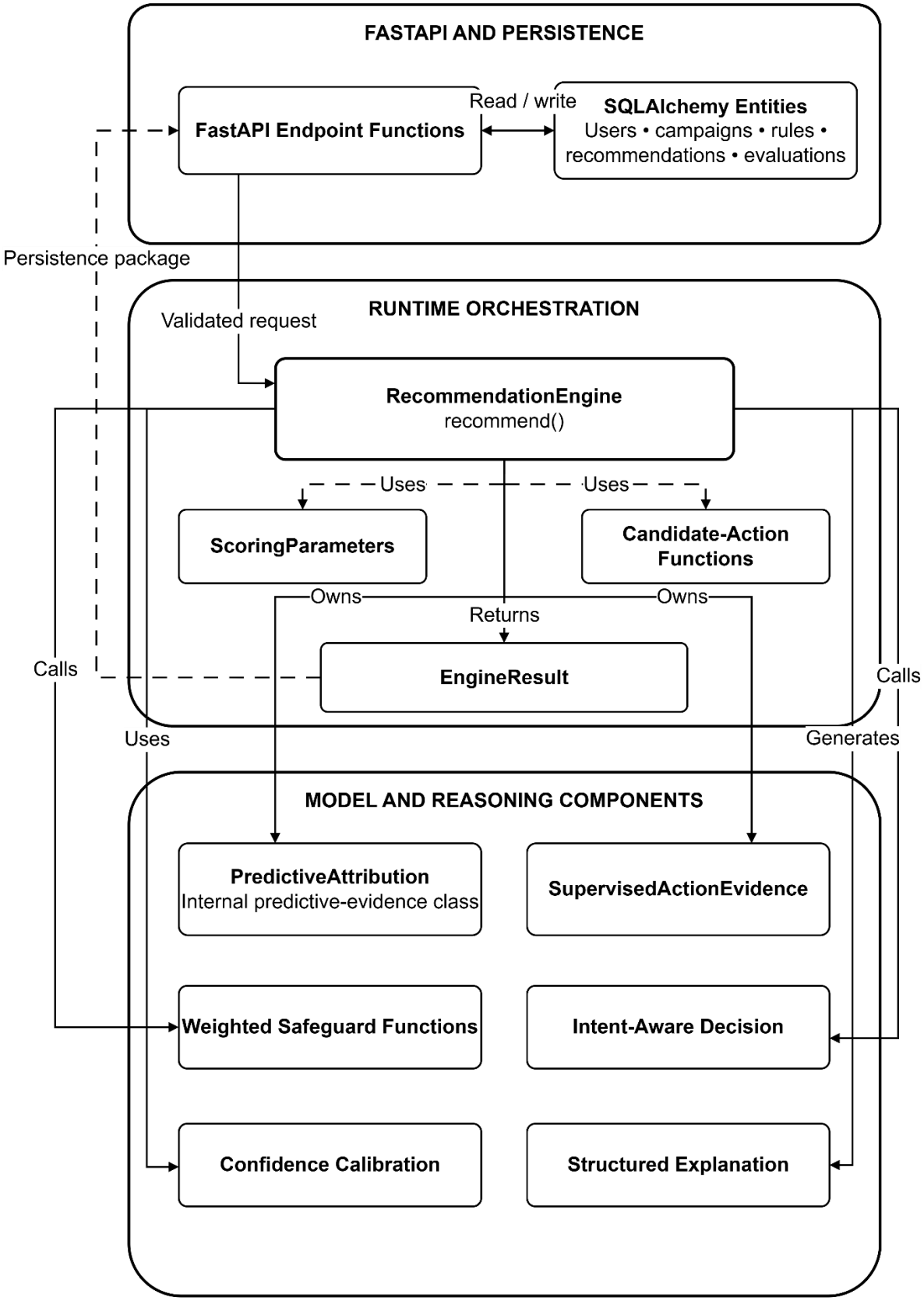


Figure 4.8 Model and Service Class Diagram

Figure 4.8 is constructed of responsibilities of classes and interfaces that are specified. The FastAPI endpoint functions validate the requests, access SQLAlchemy entities and call RecommendationEngine. RecommendationEngine owns the ROI predictive-evidence (the legacy class identifier is PredictiveAttribution), the optional SupervisedActionEvidence object, ScoringParameters, the temperature and active rule data and calls the following functions: candidate-action, weighted-safeguard, intent, calibration, explanation and returns an EngineResult. The internal ID does not refer to customer-journey or causal attribution. Then, the record of EngineResult is mapped to Recommendation and RecommendationRule records. The diagram thus indicates responsibilities and interfaces that are defined.

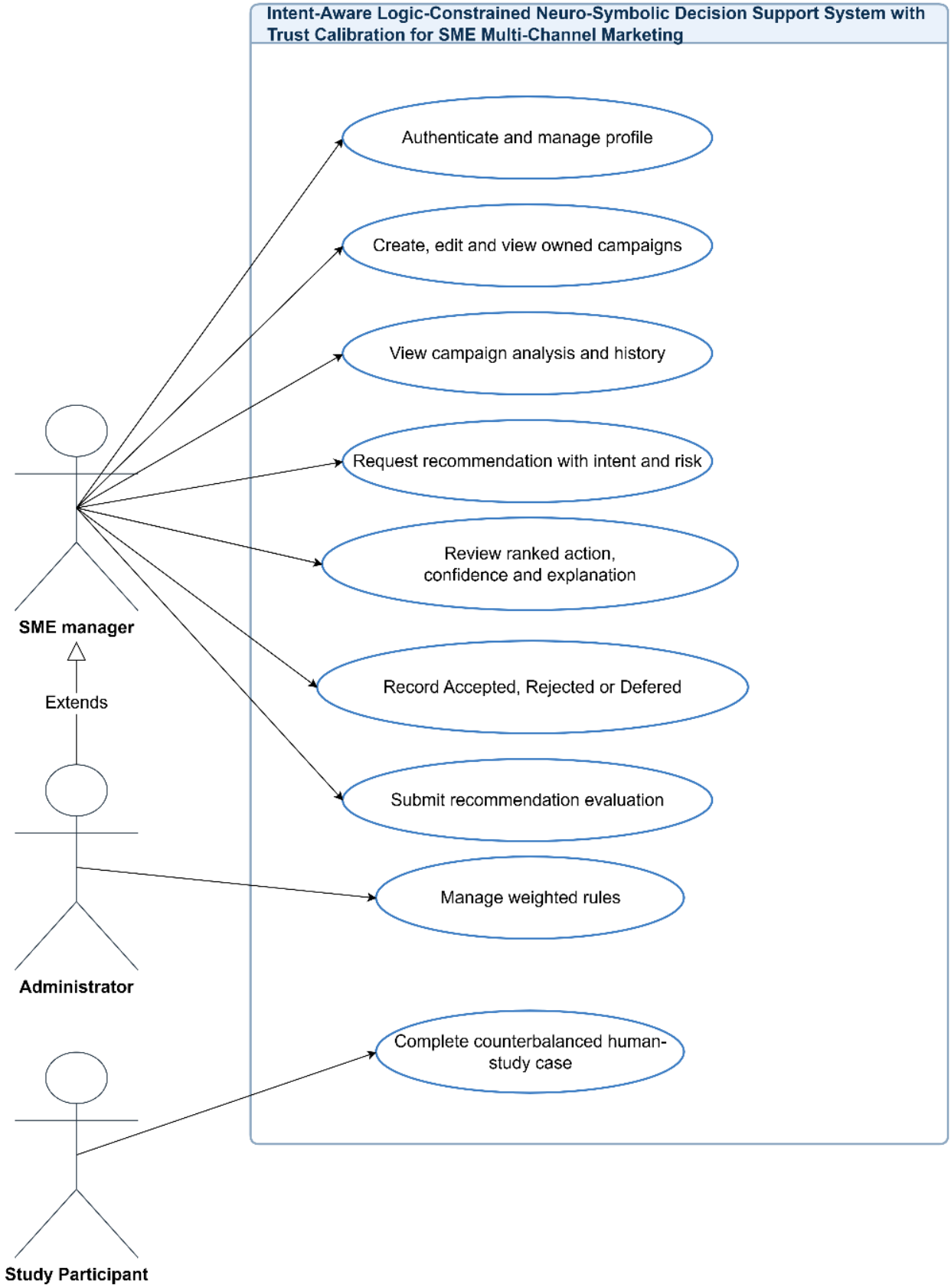


Figure 4.9 Use Cases

The use cases that are specified are still support functions. A manager keeps track of the context of a campaign, asks for and reads a recommendation, writes an assessment and looks at past assessments; administrative rule maintenance is access controlled. There is no use case which automatically alters an advertising platform or advertising budget. User scope is set via authentication and administrator-only rule maintenance is done outside of normal recommendation generation thus allowing the administrator to change the reasoning policy without impacting a controlled run. The workflow view (Figure 4.10) is added: This illustrates the journey the decision case takes as it progresses through feature preparation, learned evidence, rule and intent branches, calibration, explanation and manager review.

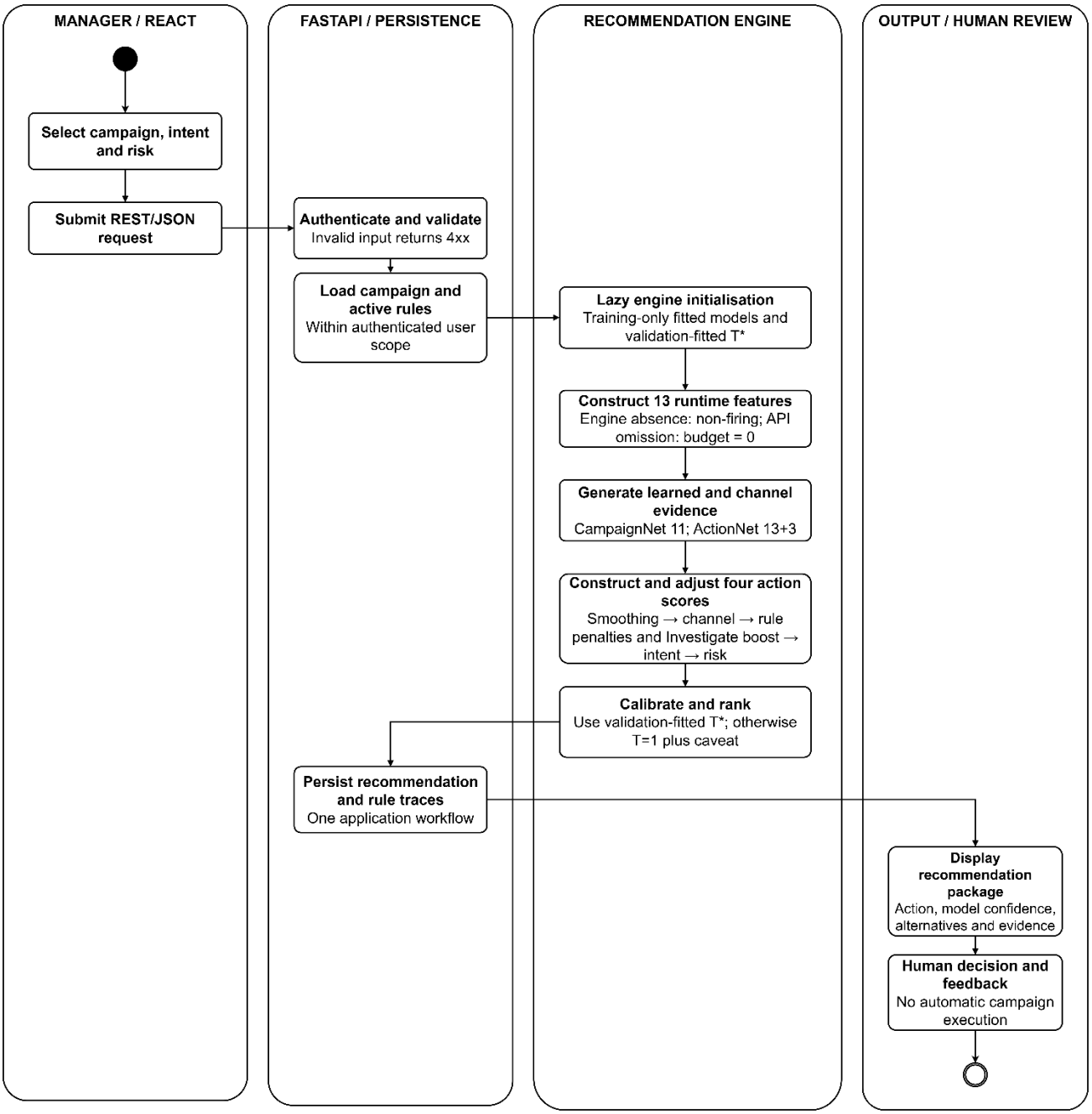


Figure 4.10 Recommendation Activity Workflow

The call-order view complementary to Figure 4.11 is shown. It then traces the request to the next service on the other side of the boundary (RecommenderEngine), components of it, and then on to persistence before returning to the response screen.

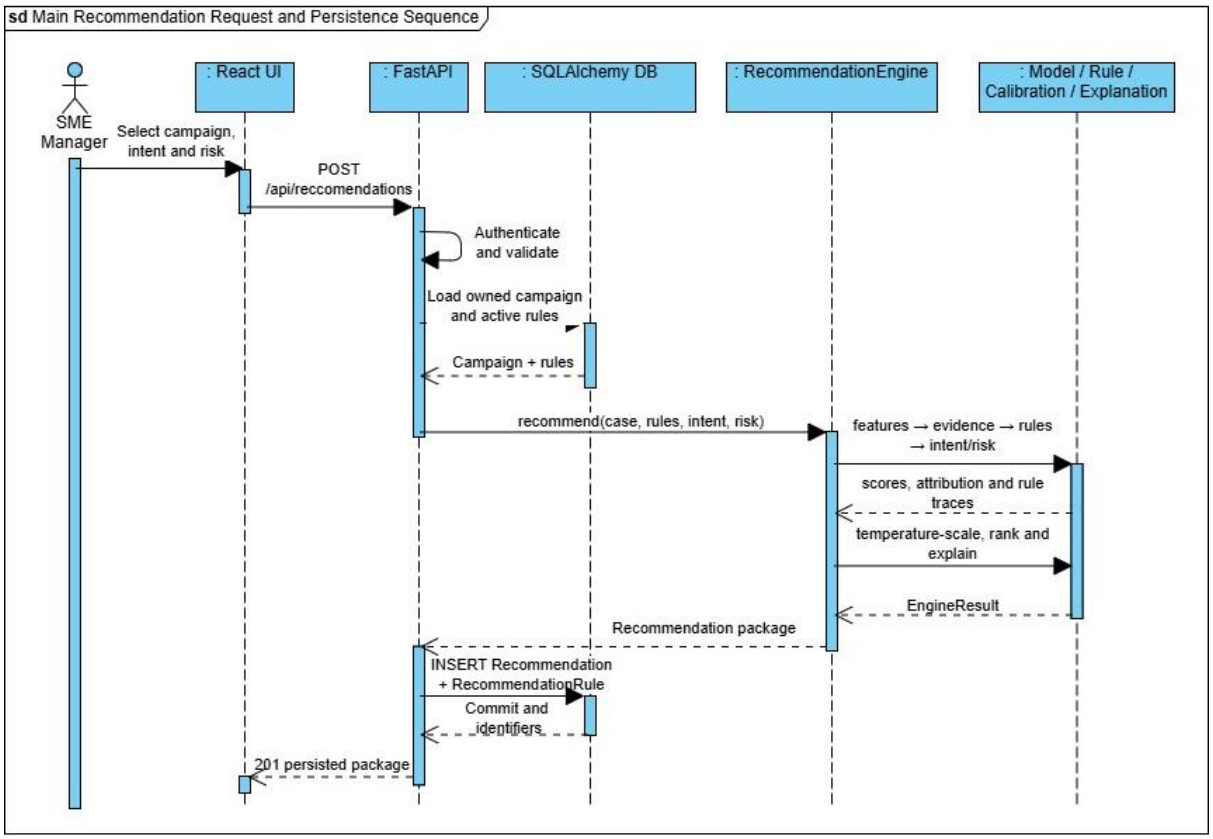


Figure 4.11 Main Recommendation Request and Persistence Sequence

The Activity view is the same as the Workflow view, indicating the workflow and alternate branches, and the Sequence view is the same as the Cross Boundary Call and Persistence order view. API validation stops if an invalid input is provided. Budget-feasibility rule is not applied if optional budget evidence is missing, and budget evidence of zero is considered to be a real value. Increases Investigate if conflicting evidence. You can tell if users have used ActionNet by looking at audited_actionnet or development_actionnet, and users that didn't have supervision are reported as campaignnet_only_fallback with a caveat. An unavailable audited calibration gate will show T = 1 with a caveat that it is not calibrated. The two figures are purposefully different from the whole system architecture in Figure 4.1 and the proposed-method system architecture in Figure 4.3, which is inside the system.

Design of interface and integration. The screens selected show the interface and integration design, and do not contain a user handbook. They display the evidence that is present prior to a request, the explicit decision controls, the returned decision package and administrator only rule configuration.

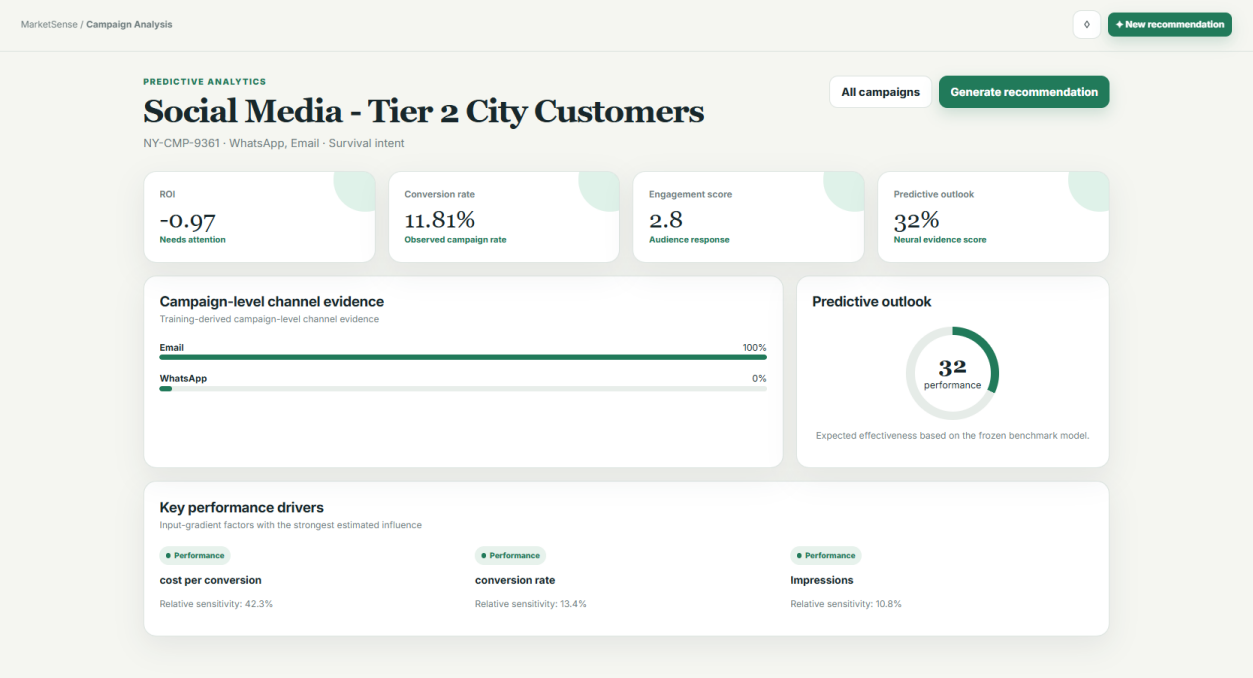


Figure 4.12 Campaign Analysis Interface

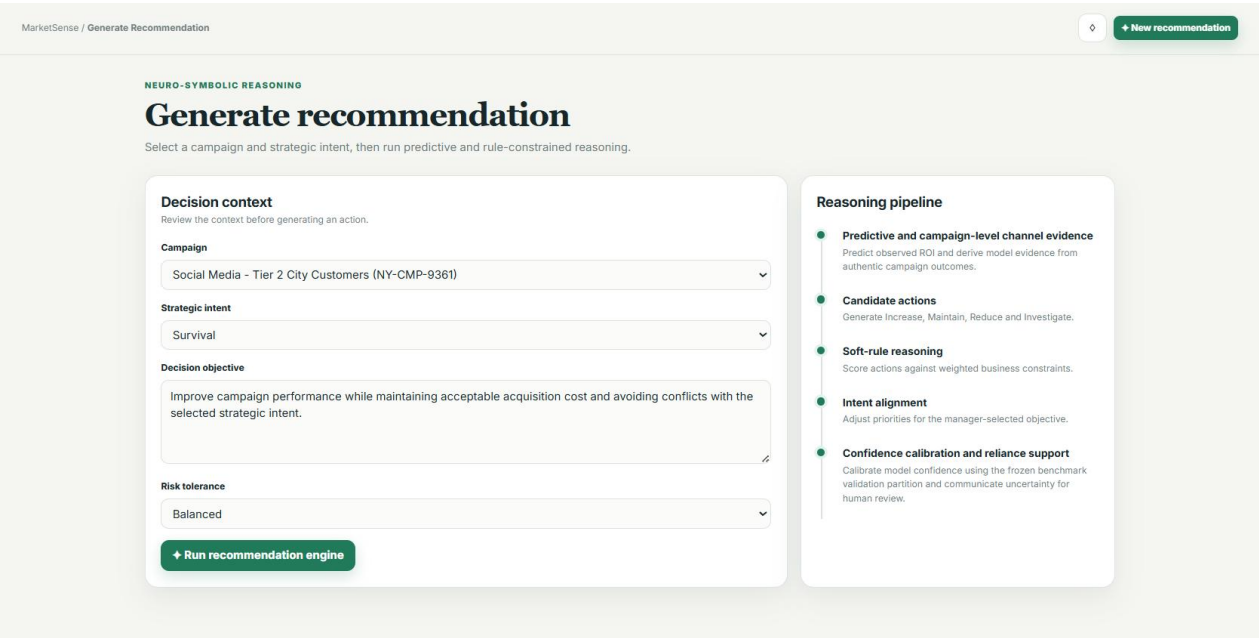


Figure 4.13 Recommendation Request Interface

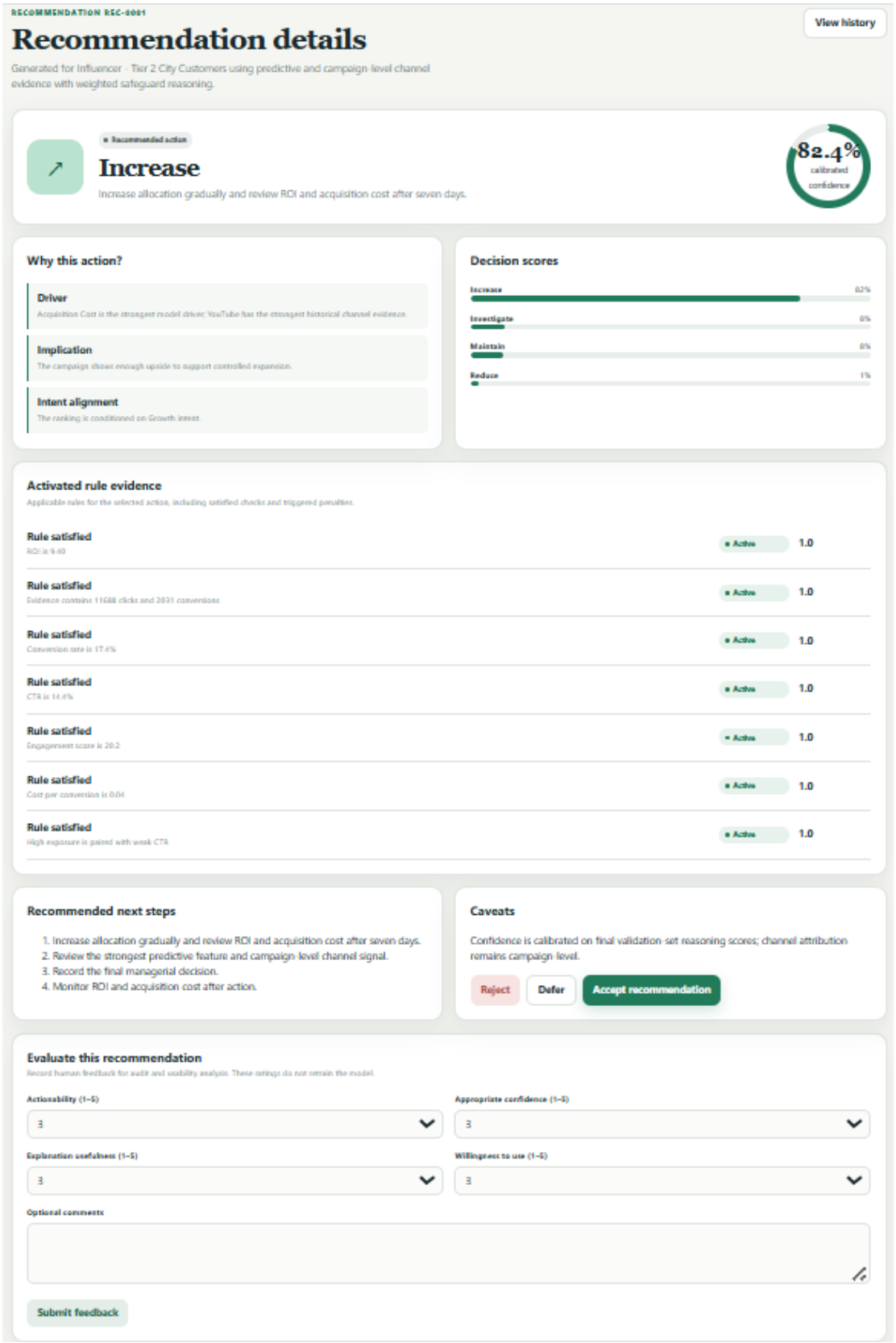


Figure 4.14 Recommendation Evidence Interface

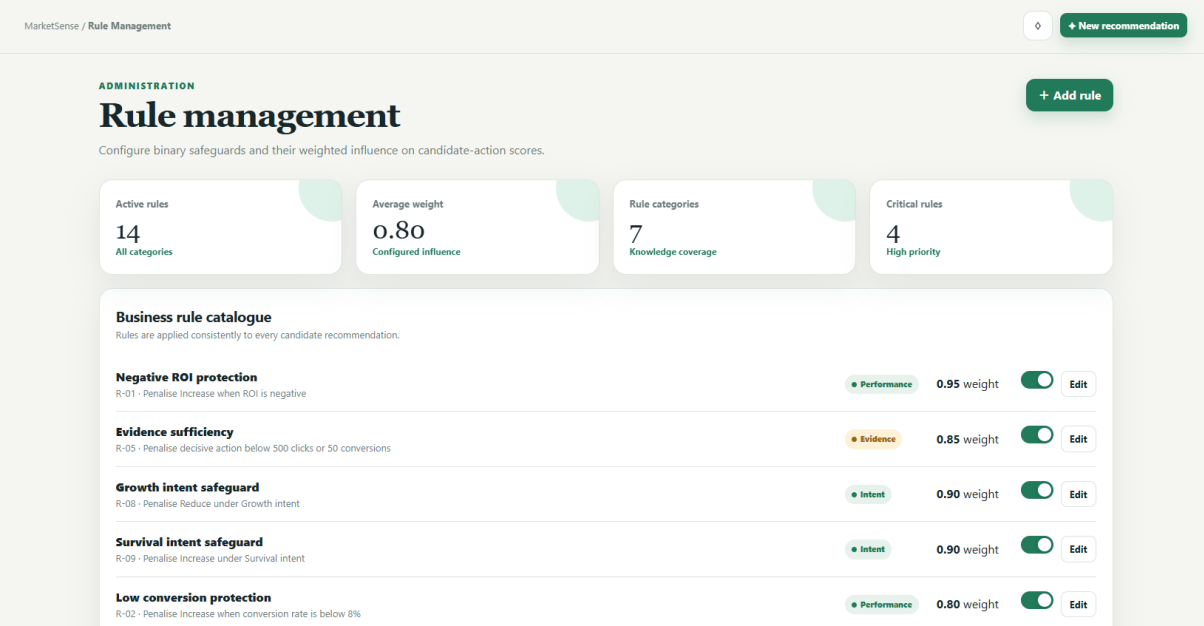


Figure 4.15 Rule Management Interface

Figures 4.12-4.15 show demonstration records of interface and integration design and are used to demonstrate specified interface and integration design, but not experimental results. They exhibit pre-request campaign evidence, explicit intent and risk inputs, calibrated model-confidence output, structured explanation and administrator only rule maintenance. The request pipeline, as well as the result view, refer to the temperature scaled output as confidence calibration, or calibrated model confidence, not human trust. Appropriate reliance, behavioural overreliance and low-confidence uncritical acceptance (as separate phenomena) are examined in Chapter 5.

4.3.2 Key Algorithms and Mathematical Formulation

This subsection formalises the consequential computations required by the proposed recommendation path. Standardisation, mean-squared error, cross-entropy, softmax and temperature scaling are established methods. The ROI-to-action transformation, evidence-fusion coefficients, channel adjustments, weighted safeguard penalties, Investigate boost, intent offsets and risk offsets are project-specific design choices. The equations define what shall be computed, whereas Algorithms 4.1 and 4.2 specify how the computations are organised during engine initialisation and recommendation inference. All equation objects in Equations (4.1)-(4.11b) shall be checked in the final PDF for visible operators, subscripts, action order and equation numbers; surrounding prose provides the machine-readable symbol definitions.

Let i index campaign cases, j index numerical features, a index actions in $A = \{\text{Increase}, \text{Maintain}, \text{Reduce}, \text{Investigate}\}$, r index active rules, g denote the selected business intent and

R denote the selected risk category. All action vectors in this subsection use the fixed order (Increase, Maintain, Reduce, Investigate). The public-source training partition, audited benchmark-training partition and audited benchmark-validation partition are distinguished explicitly throughout the equations and pseudocode.

Feature construction and training-only preprocessing

The implementation constructs four zero-safe ratio features. A zero denominator produces zero for click-through, conversion and lead-to-conversion rates. When the number of conversions is zero, cost per conversion retains the acquisition-cost value rather than producing an undefined value. These transformations are deterministic and are applied identically in training and runtime feature construction.

$$\begin{aligned}
\mathrm{CTR}_i &= \begin{cases} \dfrac{\mathrm{Clicks}_i}{\mathrm{Impressions}_i}, & \mathrm{Impressions}_i > 0 \\ 0, & \text{otherwise} \end{cases} \\
\mathrm{CVR}_i &= \begin{cases} \dfrac{\mathrm{Conversions}_i}{\mathrm{Clicks}_i}, & \mathrm{Clicks}_i > 0 \\ 0, & \text{otherwise} \end{cases} \\
\mathrm{LCR}_i &= \begin{cases} \dfrac{\mathrm{Conversions}_i}{\mathrm{Leads}_i}, & \mathrm{Leads}_i > 0 \\ 0, & \text{otherwise} \end{cases} \\
\mathrm{CPCV}_i &= \begin{cases} \dfrac{\mathrm{AcquisitionCost}_i}{\mathrm{Conversions}_i}, & \mathrm{Conversions}_i > 0 \\ \mathrm{AcquisitionCost}_i, & \text{otherwise} \end{cases}
\end{aligned} \tag{4.1}$$

For branch $b \in \{C, A\}$, the scaler parameters are fitted only from the training data used by that branch. CampaignNet uses $b = C$ and 11 predictors from the public-source training partition; ActionNet uses $b = A$ and all 13 campaign features from the audited benchmark-training partition. Validation and test records are transformed using the frozen training statistics.

$$z_{ij}^{(b)} = \frac{x_{ij} - \mu_{j,b}^{(\mathrm{tr})}}{\sigma_{j,b}^{(\mathrm{tr})}}, b \in \{C, A\} \tag{4.2a}$$

The CampaignNet ROI target is clipped at the first and ninety-ninth percentiles estimated from the public-source training partition and is then standardised using training-only target statistics.

$$\tilde{r}_i = min\left(max\left(r_i, Q_{0.01}^{(\mathrm{tr})}\right), Q_{0.99}^{(\mathrm{tr})}\right), y_i = \frac{\tilde{r}_i - \mu_y^{(\mathrm{tr})}}{\sigma_y^{(\mathrm{tr})}} \tag{4.2b}$$

In Equations (4.2a)-(4.2b), x_{ij} is the original value of feature j for case i; $z_{ij}^{(b)}$ is its standardised value for branch b; $\mu_{j,b}^{(\mathrm{tr})}$ and $\sigma_{j,b}^{(\mathrm{tr})}$ are training-only feature statistics; r_i is observed ROI; $Q_{0.01}^{(\mathrm{tr})}$ and $Q_{0.99}^{(\mathrm{tr})}$ are training-only clipping quantiles; and y_i is the clipped and standardised ROI target.

Learned evidence branches

CampaignNet is an 11-16-1 feed-forward network with one ReLU hidden layer. It is trained in full batches for 80 epochs using Adam, learning rate 0.015 and weight decay 10^{-4}. Its mean-squared-error objective is given in Equation (4.3a). ActionNet receives 13 standardised campaign features concatenated with a three-element one-hot intent vector, forming a 16-element input. It uses a 24-unit ReLU hidden layer and four output logits, and is trained for 220 epochs on the audited benchmark-training partition using Adam, learning rate 0.012, weight decay 10^{-4} and the cross-entropy objective in Equation (4.3b). No validation or test record updates either network.

$$\mathcal{L}_C(\theta_C) = \frac{1}{N_C}\sum_{i=1}^{N_C}\left(f_C\left(z_i^{(C)};\theta_C\right) - y_i\right)^2 \tag{4.3a}$$

$$\mathcal{L}_A(\theta_A) = -\frac{1}{N_A}\sum_{i=1}^{N_A}\sum_{a\in A} y_{ia}\,\log p_{ia}^{(A)} \tag{4.3b}$$

$$p_i^{(A)} = softmax\left(f_A\left([z_i^{(A)}; e(g_i)];\theta_A\right)\right) \tag{4.3c}$$

$$\hat{r}_i = \sigma_y^{(\mathrm{tr})} f_C\left(z_i^{(C)};\theta_C\right) + \mu_y^{(\mathrm{tr})} \tag{4.3d}$$

Here, N_C and N_A are the CampaignNet and ActionNet training-set sizes; y_i is the standardised clipped ROI target; y_{ia} is the one-hot audited action label; $e(g_i)$ is the one-hot intent vector; $p_i^{(A)}$ is the ActionNet action-probability vector; and θ_C and θ_A are the respective network parameters. Equation (4.3d) defines the denormalised ROI prediction $\hat{r}_i$ used by the ROI-evidence equations.

ROI evidence, branch fusion and predictive outlook

At inference, CampaignNet's output is returned to the original ROI scale. The implementation then maps the predicted ROI to four project-specific ROI logits in the fixed action order. The constants in Equations (4.4a)-(4.4d) are configured thresholds rather than learned parameters.

$$\ell_{i,\mathrm{Increase}}^{(\mathrm{ROI})} = \frac{\hat{r}_i - 1.25}{1.5} \tag{4.4a}$$

$$\ell_{i,\mathrm{Maintain}}^{(\mathrm{ROI})} = 0.8 - \frac{\left|\hat{r}_i - 1.0\right|}{1.5} \tag{4.4b}$$

$$\ell_{i,\mathrm{Reduce}}^{(\mathrm{ROI})} = \frac{0.5 - \hat{r}_i}{1.2} \tag{4.4c}$$

$$\ell_{i,\mathrm{Investigate}}^{(\mathrm{ROI})} = -0.1, p_i^{(\mathrm{ROI})} = softmax\left(\ell_i^{(\mathrm{ROI})}\right) \tag{4.4d}$$

When ActionNet is available, the two probability vectors are fused using fixed coefficients 0.25 and 0.75. If supervised ActionNet evidence is unavailable, the ROI distribution is retained and the ActionNet evidence-status field records a CampaignNet-only fallback. Uniform smoothing then reserves 0.10 probability mass across the four actions.

$$\tilde{p}_i = \begin{cases} 0.25p_i^{(\mathrm{ROI})} + 0.75p_i^{(A)}, & \text{ActionNet available} \\ p_i^{(\mathrm{ROI})}, & \text{otherwise} \end{cases} \tag{4.5a}$$

$$p_{ia} = 0.90\tilde{p}_{ia} + \frac{0.10}{4}, a \in A \tag{4.5b}$$

The campaign-analysis screen also reports a bounded predictive-outlook score. It is a project-specific weighted summary of the smoothed evidence vector, not a calibrated probability or realised business outcome.

$$O_i = 0.90p_{i,\mathrm{Increase}} + 0.55p_{i,\mathrm{Maintain}} + 0.25p_{i,\mathrm{Reduce}} + 0.45p_{i,\mathrm{Investigate}} \tag{4.5c}$$

Channel, rule, intent and risk score construction

For each channel, the implementation first computes a mean source-training performance value from observed ROI plus five times the conversion rate, then min-max normalises the channel means. The case-level channel signal is the mean over the campaign's channels; an unseen or absent channel receives the neutral value 0.5. The fixed channel adjustment affects Increase, Maintain and Reduce, while Investigate receives no channel adjustment.

$$\begin{aligned} m_h &= \frac{1}{|S_h|}\sum_{i\in S_h}(r_i + 5\mathrm{CVR}_i) \\ E_h &= \begin{cases} \dfrac{m_h - m_{min}}{m_{max} - m_{min}}, & m_{max} > m_{min} \\ 0, & m_{max} = m_{min} \end{cases} \\ c_i &= \begin{cases} \dfrac{1}{|H_i|}\sum_{h\in H_i} E_h, & |H_i| > 0 \\ 0.5, & |H_i| = 0 \end{cases} \end{aligned} \tag{4.6a}$$

$$\delta(c_i) = 0.10\begin{bmatrix} c_i - 0.5 \\ 0.25 - |c_i - 0.5| \\ 0.5 - c_i \\ 0 \end{bmatrix} \tag{4.6b}$$

In Equation (4.6a), S_h is the set of source-training cases using channel h; m_h is its mean performance value; m_{min} and m_{max} are the minimum and maximum channel means; E_h is normalised campaign-level channel evidence; H_i is the set of channels used by case i; and c_i is the resulting case-level channel signal.

Let A_{ir} equal one when rule r is active and applicable to case i, and zero otherwise. Let v_{ira} equal one when that applicable rule is violated for action a. Rule weight w_r is between 0.70 and 0.95, priority π_r is an integer from one to three, the rule-penalty coefficient is 0.20 and the

independent Investigate-boost coefficient is 0.10. A violation changes scores but never removes an action from consideration.

$$q_{ia} = p_{ia} + \delta_a(c_i) - 0.20 \sum_r A_{ir}\, v_{ira} w_r max(1, \pi_r) + 0.20 \iota_a(g_i) \quad (4.7a)$$

$$q_{i,\text{Investigate}} \leftarrow q_{i,\text{Investigate}} + 0.10 \sum_r \sum_{b \in \{\text{Increase,Maintain,Reduce}\}} A_{ir}\, v_{irb} w_r \quad (4.7b)$$

The fixed intent vectors are shown in Equation (4.7c). These downstream offsets form one intent pathway; the ActionNet one-hot input forms the other. A2: Full Method without Intent disables both.

$$\begin{gathered} \iota(\text{Growth}) = (1,\ 0.2,\ -1,\ -0.1) \\ \iota(\text{Retention}) = (0.2,\ 1,\ -0.5,\ 0.1) \\ \iota(\text{Survival}) = (-1,\ 0.4,\ 1,\ 0.3) \end{gathered} \quad (4.7c)$$

After channel, rule and intent processing, the selected risk category contributes a fixed action-score vector. Balanced contributes zero. Conservative reduces Increase and raises Investigate, whereas Aggressive raises Increase and reduces Investigate.

$$s_i = q_i + \rho(R), \rho(\text{Balanced}) = (0,0,0,0) \quad (4.8a)$$

$$\rho(\text{Conservative}) = (-0.18,0,0,0.12), \rho(\text{Aggressive}) = (0.14,0,0,-0.08) \quad (4.8b)$$

Validation-only confidence calibration and decision

Calibration is eligible only when the complete audited benchmark B satisfies the following benchmark-completeness gate:

$$|B| \geq 120, \qquad \forall i \in B: AuditStatus_i = Audited$$

The benchmark must also contain every required audit field. This gate concerns benchmark completeness and does not imply that at least 120 cases are available for temperature fitting. Let the calibration set contain the 19 locked Balanced-profile validation cases, and let each action score be the complete final score after evidence fusion, channel adjustment, weighted rules and intent adjustment. The formal temperature is fitted using limited-memory BFGS according to:

$$T_B^* = argmin_{T>0} L(T)$$

$$L(T) = -\left(\frac{1}{19}\right) \sum_{i \in V_{cal}} \ln \left(\frac{\exp\left(\frac{s_{i,y_i}^B}{T}\right)}{\sum_{a \in A} \exp\left(\frac{s_{i,a}^B}{T}\right)} \right)$$

Test labels remain untouched. The validation-support vector in the fixed action order (Increase, Maintain, Reduce, Investigate) is:

$$\left(n_{Increase}, n_{Maintain}, n_{Reduce}, n_{Investigate}\right) = (0,8,6,5)$$

Accordingly, the fitted Balanced-profile temperature is interpreted only as an aggregate adjustment for the observed validation distribution and does not establish calibration for Increase or general calibration across all four actions. Planned verification reports the eligible-case count, correct-case count, mean confidence, observed accuracy and confidence-accuracy gap for every pre-specified confidence bin, including bins with zero support.

$$\tau^* = \underset{\tau}{argmin}\left[-\frac{1}{N_V}\sum_{i=1}^{N_V} \log\left(\frac{\exp(v_{i,y_i}/\exp(\tau))}{\sum_{b\in A}\exp(v_{ib}/\exp(\tau))}\right)\right] \qquad (4.9)$$
$$T = min\left(5, max(0.5, \exp(\tau^*))\right)$$

In Equation (4.9), v_i is the final validation score vector generated without runtime risk offsets; y_i is the audited reference-action index; $N_V = 19$; τ is the optimised log-temperature parameter; and T is the exponentiated temperature clipped to [0.5, 5], exactly as specified. At runtime, T is applied after the selected risk offset. Because $T > 0$, temperature scaling changes probability sharpness but not action ranking. The selected action is the maximum calibrated probability and confidence is that maximum.

$$P_{ia}(T) = \frac{\exp(s_{ia}/T)}{\sum_{b\in A}\exp(s_{ib}/T)} \qquad (4.10a)$$

$$\hat{a}_i = \underset{a\in A}{argmax}\, P_{ia}(T), C_i = max_{a\in A} P_{ia}(T) \qquad (4.10b)$$

$$C_i^{(\text{raw})} = softmax(s_i)_{\hat{a}_i}, \text{Level}(C_i) = \begin{cases} \text{High}, & C_i \geq 0.75 \\ \text{Medium}, & 0.50 \leq C_i < 0.75 \\ \text{Low}, & C_i < 0.50 \end{cases} \qquad (4.10c)$$

Local predictive sensitivity

For explanation support, CampaignNet applies absolute gradient-times-input to the ROI-branch winning logit, not to the final rule-adjusted action. Let $a_i^{(\text{ROI})}$ be the action with the largest ROI-branch probability. In Equations (4.11a)-(4.11b), u_{ij} is the absolute gradient-times-input value for feature j and case i, while α_{ij} is its normalised local sensitivity. The sensitivities sum to one when the denominator is non-zero, and the five largest values are retained. They must not be interpreted as causal effects or customer-journey attribution.

$$a_i^{(\text{ROI})} = \underset{a\in A}{argmax}\, p_{ia}^{(\text{ROI})}, u_{ij} = \left| z_{ij}^{(C)} \frac{\partial \ell_{i,a_i^{(\text{ROI})}}^{(\text{ROI})}}{\partial z_{ij}^{(C)}} \right| \qquad (4.11a)$$

$$\alpha_{ij} = \begin{cases} \frac{u_{ij}}{\sum_k u_{ik}}, & \sum_k u_{ik} > 0 \\ 0, & \text{otherwise} \end{cases} \quad (4.11b)$$

Consequential fixed parameters

Table 4.5

Consequential Fixed Parameters and Freeze Status

Component	Specified value	Selection status	Planned verification
CampaignNet	11-16-1; Adam; learning rate 0.015; weight decay 10^{-4}; 80 epochs	Fitted on public-source training data only	Seed and ablation checks
ActionNet	16-24-4; Adam; learning rate 0.012; weight decay 10^{-4}; 220 epochs	Fitted on 87 audited training cases	Seed and A2 checks
Fusion	0.25 CampaignNet + 0.75 ActionNet; 0.90 evidence + 0.10 uniform mass	Fixed project-specific coefficients	Branch-ablation checks
Channel	Strength 0.10; unseen or absent channel signal = 0.50	Training-derived evidence and neutral fallback	Without-channel comparison
Rules	14 safeguards; weights 0.70-0.95; priorities 1-3; penalty 0.20	Predefined and frozen; not learned	Activation and A1 checks
Investigate boost	0.10 × violated-rule weight; no priority multiplier	Independent fixed fallback contribution	Trace audited and fallback roles
Intent	Strength 0.20 plus ActionNet one-hot intent input	Two specified pathways	Disable both in A2: Full Method without Intent
Risk	Conservative [-0.18, 0, 0, +0.12]; Aggressive [+0.14, 0, 0, -0.08]	Fixed project-specific offsets	Risk sensitivity check
Calibration	LBFGS learning rate 0.05; at most 60 iterations; T clipped to [0.5, 5]	Fitted on 19 audited validation cases after $\|B\| \geq 120$ gate	Uncalibrated comparison
Confidence bands	High ≥ 0.75; Medium 0.50-0.7499; Low < 0.50	Fixed display thresholds	Boundary tests

The fixed values in Table 4.5 are project-specific design controls rather than empirically optimised constants. The 0.25/0.75 CampaignNet-ActionNet fusion gives greater weight to ActionNet because it is trained directly on the audited four-action task while retaining ROI evidence from CampaignNet; the 0.90/0.10 smoothing prevents a single evidence branch from becoming completely deterministic. The campaign-level channel strength of 0.10 is deliberately smaller than the learned evidence contribution so that historical channel averages can adjust, but not dominate, the ranking. Safeguard weights and priorities encode documented relative importance, the 0.20 penalty permits a violated safeguard to influence ranking without hard rejection, and the smaller 0.10 Investigate boost makes review more likely without automatically forcing Investigate. The 0.20 intent strength is bounded below the main learned evidence contribution, while Conservative and Aggressive risk offsets are smaller contextual adjustments; Balanced is the zero-offset reference profile.

Sensitivity analysis shall test whether conclusions depend excessively on these fixed choices. The planned checks are: fusion weights 0.50/0.50 and 0.75/0.25 in addition to the specified 0.25/0.75; campaign-level channel strength 0.05 and 0.15 around 0.10; safeguard weights scaled by ±10% while preserving their ordering, rule-penalty coefficient 0.15 and 0.25 around 0.20, and Investigate boost 0.05 and 0.15 around 0.10; intent strength 0.10 and 0.30 around 0.20; and Conservative/Aggressive risk vectors scaled by 0.5× and 1.5×. These are robustness checks, not hyperparameter optimisation, and the held-out test labels shall not be used to select a preferred setting

Algorithm 4.1 Model Initialisation and Validation-Only Calibration

Require: versioned public campaign data D_{raw}; audited benchmark B; configured seed s; CampaignNet f_C; ActionNet f_A; fixed feature order, 14-rule catalogue and coefficients; Adam learning rates 0.015 and 0.012; weight decay 10^{-4}; 80 and 220 epochs; and LBFGS calibration with learning rate 0.05 and at most 60 iterations.

Ensure: fitted engine state containing frozen preprocessing statistics, trained models, campaign-level channel evidence, temperature and status fields; source and benchmark test partitions remain untouched.

1:	Set the NumPy and PyTorch random seeds to s	▷ Fix the initial state
2:	$D \leftarrow PrepareFrame(D_{\text{raw}}; 0.70)$	▷ Clean, order and impute
3:	Construct CTR, CVR, LCR, CPCV using Equation (4.1)	▷ Derive deterministic ratios
4:	$D_{\text{train}}: D_{\text{val}}: D_{\text{test}} = 70:15:15$	▷ Preserve temporal order
5:	Fit $\mu^{(\text{tr})}, \sigma^{(\text{tr})}, Q_{0.01}^{(\text{tr})}, Q_{0.99}^{(\text{tr})}$ on D_{train} [Eqs. (4.2a)-(4.2b)]	▷ Training-only fitting
6:	Initialise CampaignNet parameters θ_C	▷ 11-16-1 network

7:	**for** $e \in \{1, \ldots, 80\}$ **do**	▷ Full-batch optimisation
8:	$z_i^{(C)} \leftarrow Standardise_C(x_i)$	▷ Scale 11 inputs
9:	$\hat{y}_i \leftarrow f_C(z_i^{(C)}; \theta_C)$	▷ Predict normalised ROI
10:	Compute $\mathcal{L}_C$ using Equation (4.3a)	▷ Compute mean-squared error
11:	$\theta_C \leftarrow Adam(\theta_C, \nabla_{\theta_C}\mathcal{L}_C; \eta = 0.015, \lambda = 10^{-4})$	▷ Update CampaignNet
12:	**end for**	▷ Complete CampaignNet fitting
13:	$\{E_h\} \leftarrow ChannelEvidence(D_{\text{train}})$ [Eq. (4.6a)]	▷ Freeze channel statistics
14:	$B_{\text{train}} : B_{\text{val}} : B_{\text{test}} = 70 : 15 : 15$	▷ Keep benchmark test untouched
15:	Fit the ActionNet scaler on 13 features from B_{train} and one-hot encode intent	▷ Training-only action inputs
16:	Initialise ActionNet parameters θ_A	▷ 16-24-4 network
17:	**for** $e \in \{1, \ldots, 220\}$ **do**	▷ Full-batch optimisation
18:	$p_i^{(A)} \leftarrow softmax(f_A([z_i^{(A)}; e(g_i)]; \theta_A))$	▷ Predict four action probabilities
19:	Compute $\mathcal{L}_A$ using Equation (4.3b)	▷ Compute cross-entropy
20:	$\theta_A \leftarrow Adam(\theta_A, \nabla_{\theta_A}\mathcal{L}_A; \eta = 0.012, \lambda = 10^{-4})$	▷ Update ActionNet
21:	**end for**	▷ Complete ActionNet fitting
22:	Freeze the 14-rule catalogue, intent and risk vectors, and score coefficients	▷ Prevent post-hoc tuning
23:	**if** $\lvert B \rvert \geq 120$ **and all audit fields and statuses are valid then**	▷ Apply complete-benchmark gate
24:	$v_i \leftarrow Score(B_{\text{val},i}; \rho = 0)$ [Eqs. (4.4a)-(4.7c)]	▷ Exclude runtime risk offsets
25:	$T \leftarrow FitTemperature(\{v_i, y_i\}_{i=1}^{19})$ [Eq. (4.9)]	▷ Validation-only LBFGS fitting
26:	Record the calibration status as calibrated	▷ Expose calibration state
27:	**else**	▷ Handle unavailable calibration
28:	$T \leftarrow 1$; record the corresponding uncalibrated status	▷ Explicit fallback
29:	**end if**	▷ Complete calibration gate
30:	$\mathcal{E} \leftarrow \{\theta_C, \theta_A, \mu, \sigma, E, T, \text{statuses}, \text{hashes}\}$	▷ Save reproducible engine state

Algorithm 4.2 Intent-Aware Logic-Constrained Recommendation Inference

Require: validated campaign vector x_i; selected intent g_i; risk category R_i; active frozen rule catalogue; fitted engine state $\mathcal{E}$; and action set A.

Ensure: ranked recommendation package $\mathcal{P}_i$ containing the selected action, calibrated model confidence, alternatives, Driver-Implication-Action-Caveat explanation, Intent metadata, sensitivity evidence, rule trace and status fields; no automatic marketing execution.

1:	Validate x_i, g_i, R_i; construct the 13-feature vector using Equation (4.1)	▷ Maintain input parity

2:	$z_i^{(C)} \leftarrow Standardise_C(x_i)$	▷ Apply frozen 11-input scaler
3:	$\hat{r}_i \leftarrow \sigma_y^{(\mathrm{tr})} f_C(z_i^{(C)}; \theta_C) + \mu_y^{(\mathrm{tr})}$ [Eq. (4.3d)]	▷ Predict ROI on original scale
4:	$p_i^{(\mathrm{ROI})} \leftarrow softmax(\ell_i^{(\mathrm{ROI})}(\hat{r}_i))$ [Eqs. (4.4a)-(4.4d)]	▷ Construct ROI action evidence
5:	$\alpha_i \leftarrow TopFiveGradientInput(f_C, z_i^{(C)})$ [Eqs. (4.11a)-(4.11b)]	▷ Create local sensitivity trace
6:	**if audited or development ActionNet evidence is available then**	▷ Select supervised branch
7:	$z_i^{(A)} \leftarrow Standardise_A(x_i)$	▷ Apply frozen 13-input scaler
8:	$p_i^{(A)} \leftarrow softmax(f_A([z_i^{(A)}; e(g_i)]; \theta_A))$ [Eq. (4.3c)]	▷ Create intent-conditioned evidence
9:	$\tilde{p}_i \leftarrow 0.25 p_i^{(\mathrm{ROI})} + 0.75 p_i^{(A)}$ [Eq. (4.5a)]	▷ Fuse learned branches
10:	Record the ActionNet evidence status	▷ Preserve branch provenance
11:	**else**	▷ Handle unavailable supervision
12:	$\tilde{p}_i \leftarrow p_i^{(\mathrm{ROI})}$; record CampaignNet-only fallback and prepare its Caveat	▷ Make fallback explicit
13:	**end if**	▷ Complete evidence selection
14:	$p_{ia} \leftarrow 0.90\tilde{p}_{ia} + 0.10/4$ [Eq. (4.5b)]; compute O_i [Eq. (4.5c)]	▷ Smooth evidence and compute outlook
15:	$c_i, \delta(c_i) \leftarrow ChannelAdjustment(H_i, \{E_h\})$ [Eqs. (4.6a)-(4.6b)]	▷ Add training-derived evidence
16:	$q_{ia} \leftarrow p_{ia} + \delta_a(c_i)$	▷ Initialise channel-adjusted scores
17:	**for each active rule r and action a do**	▷ Evaluate frozen catalogue
18:	Evaluate applicability A_{ir} and violation v_{ira}; append the result to the rule trace	▷ Preserve auditability
19:	**if $A_{ir} v_{ira} = 1$ then**	▷ Apply soft constraint
20:	$q_{ia} \leftarrow q_{ia} - 0.20 w_r max(1, \pi_r)$ [Eq. (4.7a)]	▷ Penalise affected action
21:	**if $a \neq$ Investigate then**	▷ Check fallback contribution
22:	$q_{i,\mathrm{Investigate}} \leftarrow q_{i,\mathrm{Investigate}} + 0.10 w_r$ [Eq. (4.7b)]	▷ Boost Investigate independently
23:	**end if**	▷ Complete fallback update
24:	**end if**	▷ Complete rule update
25:	**end for**	▷ Complete rule reasoning
26:	$q_i \leftarrow q_i + 0.20 \iota(g_i)$ [Eqs. (4.7a) and (4.7c)]	▷ Apply downstream intent pathway
27:	$s_i \leftarrow q_i + \rho(R_i)$ [Eqs. (4.8a)-(4.8b)]	▷ Apply selected risk vector
28:	$P_i(T) \leftarrow softmax(s_i/T)$ [Eq. (4.10a)]	▷ Calibrate model confidence
29:	$\hat{a}_i \leftarrow argmax_{a \in A} P_{ia}(T), C_i \leftarrow max_{a \in A} P_{ia}(T)$ [Eqs. (4.10b)-(4.10c)]	▷ Select action and confidence
30:	Generate Driver, Implication, Action and Caveat; attach Intent as metadata	▷ Support human review
31:	$\mathcal{P}_i \leftarrow \{\hat{a}_i, C_i, \mathrm{alternatives}, \alpha_i, \mathrm{rules}, \mathrm{statuses}\}$	▷ Assemble recommendation package

32:	Persist $\mathcal{P}_i$ and manager-decision state; return without executing a campaign or budget change	▷ Retain traceability and human authority

4.4 Baseline Implementation Design

Comparable partitions, action labels, output fields and metric functions are used for B0, B1-B4, the Intent-Only and Weighted Deterministic baselines, the Unconstrained AI safety reference, the Post-hoc Hard-Rule Filter comparator, the three evidence-branch conditions and the proposed-method ablations. The action mapping is always 0 = Increase, 1 = Maintain, 2 = Reduce and 3 = Investigate. This mapping is preserved in every score vector, probability vector, centroid label list, top-two ranking, export and metric function. A pre-run assertion terminates the run when a component supplies a different action list. Exact score ties are resolved by the canonical order rather than alphabetically.

Table 4.6

Baseline Implementation and Comparison Parity

Condition	Implementation	Parity control
B1: Heuristic Scorecard	Fixed ROI, conversion, engagement and ambiguity thresholds produce action scores followed by softmax.	No fitted labels; identical held-out cases, canonical action order and output schema.
B2: Predictive Only	Training-only standardisation and nearest-class-centroid probabilities over the 13 prepared features.	Same benchmark training and test partitions; no campaign-level channel evidence, intent, rules or calibration.
B3: Predictive plus Channel Evidence	B2 logits adjusted by normalised training-derived campaign-level channel evidence.	Same campaign-level channel-evidence source and output schema; no proposed rule, intent or calibration mechanism.
B4: Integrated Learned Backbone	CampaignNet and ActionNet fixed fusion with the specified smoothing; downstream campaign-level channel adjustment, rules, intent offsets, risk offsets and calibration disabled.	Same fitted branches, frozen fusion coefficients, benchmark partitions, canonical action order and metric functions.
B0: Majority-Class Popularity	The most frequent action in the benchmark training partition is predicted for every test case; confidence equals its training prevalence.	Training labels only; deterministic; canonical tie order and identical test/metric code.

Intent-Only Deterministic Baseline	Maps Growth to Increase, Retention to Maintain and Survival to Reduce without learned campaign evidence.	Same supplied intent and action vocabulary; deterministic; no rules, channel evidence or calibration.
Weighted Deterministic Baseline	Combines frozen intent scores, campaign-signal increments, evidence/conflict adjustments and softmax without fitted model parameters.	Same prepared inputs and canonical order; coefficients frozen before test evaluation.
Unconstrained AI Baseline	Uses the learned evidence backbone while disabling safeguard reasoning, feasibility filtering and the reachable no-safe state.	Same fitted models, intent, cases and metrics as P; interpreted as a safety reference, not a deployable target.
Post-hoc Hard-Rule Filter	B2 probabilities are filtered after scoring when the shared conflict catalogue rejects an action.	Uses the same conflict definitions and canonical action order, but hard rejection rather than weighted integration.
Branch: CampaignNet only	CampaignNet four-action evidence with smoothing; ActionNet and fixed fusion disabled.	Same cases and four-action output; downstream channel, rule, intent, risk and calibration adjustments disabled.
Branch: ActionNet only	ActionNet four-class evidence conditioned on the audited or selected intent; CampaignNet and fusion disabled.	Same ActionNet scaler, training cases and action order; downstream modules disabled; evaluation fails explicitly if ActionNet is unavailable.
Branch: fixed fusion	0.25 CampaignNet evidence + 0.75 ActionNet evidence followed by the same 0.90 evidence + 0.10 uniform smoothing.	Same fitted branches and frozen coefficients; downstream modules disabled to isolate fusion.
Proposed-method configurations	A1: Full Method without Rules; A2: Full Method without Intent; A3: Full Method without Calibration; and P: Full Proposed Method use the specified engine switches.	All other settings, data, actions, traces and metrics remain fixed.

Table 4.6A lists the branch condition (A1-A3) and index-to-action mapping that is allowed for P: Full Proposed Method, B1-B4 and the Post-hoc Hard-Rule Filter comparator, CampaignNet-only, ActionNet-only and fixed-fusion branch conditions, the persistent score and probability vectors, the top-two outputs, and the JSON and CSV export and all metric functions. Before the evaluation the runner shall say that all components participating are displaying this same sequence. If there is a mismatch it should be a run terminator, NOT be silently remapped. In the event of a tie the team with the lowest Canonical Index will be declared the winner.

Table 4.6A

Canonical Index-to-Action Mapping

Canonical index	Action	Vector position	Tie precedence
0	Increase	First	1
1	Maintain	Second	2
2	Reduce	Third	3
3	Investigate	Fourth	4

Deterministic shall be run once per frozen partition since the output will be the same each time, but the neural conditions shall be run once with each seed (42, 52, 62) as there will be variation in the outputs between runs. The runner should record the latency and the above correctness, top-two, scenario-fit and logical-conflict and confidence inputs per case, for each method. The outputs of Conservative and Aggressive are exploratory and uncalibrated, unless they are separately calibrated. Outcomes completed should be put in Chapter 5.

B1: Heuristic Scorecard is a facsimile of 4 deterministic scores. With ROI = r, conversion rate = c and engagement = e: Reduce = 0.9 max(0, -r) + 1.2 max(0, 0.08 - c); Increase = 0.8 max(0, r - 1.8) + 1.1 max(0, c - 0.20) + 0.02e; Maintain = 1 - min(1, |r - 1| / 2) + max(0, 0.18 - |c - 0.15|); and Investigate = 0.25 only when 0.05 <= c <= 0.12 and 0.20 <= r <= 0.90, otherwise 0. The scores are converted to probabilities using Softmax. The four steps to exact ties is: Increase, Maintain, Reduce, Investigate.

B2: Only one 13-dimensional centroid/action observed should be used in benchmark-training cases and the means and population standard deviations should be reported. These are the values to use for a test case to be standardised. The class logit will be the negative euclidean distance to the respective centroid and softmax will output the probability as – Increase, Maintain, Reduce, Investigate. These are stored in centroid labels, score-vector indexes, predicted labels, exact-score ties, top two rankings, exports and metric functions.

B3: Predictive plus Channel Evidence is meant to build-on B2: Predictive Only. Channel performance will be 0.7 ROI + 8 conversion rate for each case of source-training with means calculated by each channel listed and min-max normalised. Average of all available campaign level channel signals or 0.50 if no signal is available shall be a case signal h. Log-probability adjustments shall be +1.3(h - 0.50) for Increase, +1.3(0.50 - h) for Reduce, +0.35(0.60 - |h -

If it was to be Maintain the score it would be 0.50|) and if it was to be Investigate it would be zero. The order of actions for softmax, exact ties, top two output and exports will be: Increase, Maintain, Reduce, Investigate.

B4: Integrated Learned Backbone shall use the fixed 0.25/0.75 fusion coefficients, and the above 0.90 evidence and 0.10 uniform smoothing. Adjust, protect reasoning, intent offsets and risk offsets and confidence calibration shall be off for channel level campaign. In B4, the backbone of the learned method, which is the same fitted branches, benchmark partitions, frozen coefficients and four action output schema and metric functions, will be separated from the whole proposed method.

The Post-hoc Hard-Rule Filter should begin with B2: Predictive Only and should filter out an action if either of the following is true: shared hard-conflict catalogue: Growth-Reduce, Survival-Increase or Retention-Reduce; ROI < 0, conversion rate < 0.08, CTR < 0.03 or engagement < 5; Reduce when at least two of ROI > 2, conversion rate > 0.20, CTR > 0.10 and engagement > 20 are true or clicks < 500 or conversions < 50. With the same canonical indexes, the other probabilities will then be re-normalised. Probability 1 Investigate is going to be returned, unless something is done. The order for the actions for the predicted labels, exact-score ties, top two, exports and metric functions shall be: Increase, Maintain, Reduce, Investigate by default.

The pre-run assertion will verify that all the baseline and evaluation components are using the canonical mapping (0 = Increase, 1 = Maintain, 2 = Reduce and 3 = Investigate) before evaluation. Not having action list or reordering action list or incompatible action list will not remap the list and will terminate the run.

Algorithm 4.3 Reproducible Baseline and Comparator Inference with Parity Controls

Require: frozen benchmark-training and test partitions; 13-feature order; canonical four-action vocabulary; source-training campaign records; fitted CampaignNet and ActionNet states; fixed fusion and smoothing coefficients; exact B0 and B1-B3 specifications; deterministic intent and weighted-score definitions; unconstrained and post-hoc comparator switches; and the shared conflict catalogue.

Ensure: one prediction, confidence, top-two ranking, explanation field and per-case latency for B0, B1-B4, the Intent-Only Deterministic Baseline, Weighted Deterministic Baseline, Unconstrained AI Baseline and Post-hoc Hard-Rule Filter under identical test cases, the canonical action mapping and the same metric functions.

1:	Fit feature means, standard deviations and class centroids on benchmark training only.	▷ Freeze shared B2/B3/post-hoc predictive state
2:	Fit source-training campaign-level channel means and min-max bounds.	▷ Freeze B3 channel-evidence state
3:	Load the frozen CampaignNet and ActionNet states required by B4.	▷ Fail explicitly if either learned branch is unavailable
4:	for each held-out test case x do	▷ Use identical cases
5:	Compute B1's four fixed scores from r, c and e.	▷ Apply the exact heuristic equations
6:	Arrange B1 scores in canonical action order and apply softmax.	▷ Increase, Maintain, Reduce, Investigate
7:	Standardise x using the frozen benchmark-training statistics.	▷ No test fitting
8:	Compute B2 logits as negative distances to the action centroids.	▷ Predictive-only evidence
9:	Apply softmax to B2 logits in canonical action order.	▷ Deterministic canonical tie rule
10:	Copy B2 log-probabilities for B3.	▷ Preserve predictive parity
11:	Set h to the mean available campaign-level channel value, otherwise 0.50.	▷ Neutral unseen-channel fallback
12:	Apply the three specified B3 channel adjustments; leave Investigate unchanged.	▷ Training-derived campaign-level channel evidence
13:	Apply softmax to B3 logits in canonical action order.	▷ Deterministic canonical tie rule
14:	Compute CampaignNet and ActionNet evidence and apply the fixed fusion for B4.	▷ Integrated learned backbone
15:	Apply the specified B4 smoothing and disable downstream channel, rule, intent, risk and calibration adjustments.	▷ Isolate the learned backbone
16:	Copy B2 probabilities for the Post-hoc Hard-Rule Filter.	▷ Preserve predictive parity
17:	Reject every action with one or more shared hard conflicts.	▷ Post-hoc replacement logic
18:	if at least one action remains then	▷ Check survivor set
19:	Renormalise surviving probabilities and select the lowest canonical index under an exact tie.	▷ Deterministic canonical tie rule
20:	else return Investigate with confidence 1.	▷ All-rejected fallback
21:	Record prediction, confidence, top two, explanation and latency for B1-B4 and the post-hoc comparator.	▷ Common output contract
22:	end for	▷ Complete held-out inference
23:	Evaluate every configuration with the same class, calibration, conflict and latency functions.	▷ Preserve comparison parity

4.5 Development and Evaluation Environment

The selected development and evaluation environment, model identifiers, class order, seeds and principal parameters and persistence boundary are given in Table 4.7. The actual environment

that has been executed and any material deviation from this specification shall be reported in Chapter 5.

Table 4.7

Development and Evaluation Environment

Area	Selected configuration
Host	Windows 11 64-bit local development workstation; CPU-only execution target; at least 16 GB RAM. Exact host capture is reported in Chapter 5.
Backend	Python 3.11.9 in the active project environment; FastAPI 0.139.0; Uvicorn 0.51.0; SQLAlchemy 2.0.46; PyMySQL 1.1.2.
Model and data	PyTorch 2.7.1+cu118; pandas 2.1.4; scikit-learn 1.7.2; NumPy 1.26.4 in the executed project environment.
Frontend	React/React DOM 19.2.7; TypeScript 7.0.2; Vite 8.1.4; Node.js 24.18.0; pnpm 11.16.0; committed pnpm lockfile.
Persistence	MySQL image mysql:8.4 configured in docker-compose.yml; Docker CLI was unavailable during the final environment capture. Temporary SQLite is executed for isolated API tests.
Execution controls	Chronological 70/15/15 split; seeds 42, 52 and 62; five ECE bins; 2,000 bootstrap samples in the evaluation runner.
External services	None required at inference time; no cloud AI, advertising-platform, CRM or ERP connection.

4.6 Implementation Constraints

The principal implementation constraint is the imbalance between the 55,555-record public campaign source and the 125-case audited decision benchmark. ActionNet receives only 87 training cases, and the held-out action supports are 3 Increase, 7 Maintain, 5 Reduce and 4 Investigate. The validation subset contains no Increase case. Per-class results, confusion matrices and uncertainty intervals are therefore mandatory, and calibration and four-class generalisation claims must remain limited. The source also does not establish multi-SME provenance, while benchmark labels are audited operational judgements under project-defined scenarios rather than universal ground truth.

The development target is a CPU workstation with no dedicated production infrastructure. Compact hidden layers, bounded actions and fixed epoch budgets are selected to keep repeated experiments feasible. MySQL 8.4 is the operational database target, while temporary SQLite may

be used for isolated ORM and API tests. This portability choice does not cover production-scale concurrency, failover or database operations. No live campaign execution or data synchronisation is specified; managers apply reviewed recommendations outside the system.

A second constraint concerns channel granularity. Although records list up to three channels, outcome fields are campaign-level rather than channel-specific. The channel module therefore averages training-derived channel values for the channels named in a case. It cannot estimate channel interaction, sequence, incremental effect or causal contribution, and all surrounding text and Chapter 5 conclusions must preserve that boundary.

The security specification supports bounded local research use rather than an Internet-facing production service. Passwords use salted scrypt hashes, session tokens are HMAC-signed with an eight-hour expiry, administrative rule operations require an administrator role, campaign payloads are validated and browser origins are restricted to the configured frontend. A non-development installation must use a long random token secret. TLS termination, secret rotation, rate limiting, account recovery, central audit monitoring, managed backup and penetration testing remain outside the specified prototype boundary.

Privacy and human-control requirements shape both storage and interface behaviour. The method does not require customer names, social-media credentials or behavioural tracking identifiers. Human-study records use pseudonymous participant codes, store consent status and omit free-text comments from analysis export. The manager explicitly selects Survival, Retention or Growth; intent is not inferred. The interface displays calibrated model confidence rather than human trust, while trust alignment and overreliance remain participant-evaluation constructs.

Fallback and boundary behaviour are specified explicitly. Invalid principal inputs stop at Pydantic validation; unseen channels receive neutral evidence; unavailable calibration uses $T = 1$ with uncalibrated status; absent budget remains null and skips the budget rule; and no marketing action is executed automatically. ActionNet provenance is returned and persisted through action_evidence_status, with a structured caveat for CampaignNet-only fallback. Human-study routes remain disabled by default and may be enabled only when the official TAR UMT approval or exemption reference is supplied through ETHICS_APPROVAL_REFERENCE and the benchmark-eligibility gate is satisfied.

Evaluation traceability is part of the development specification. R001-R006 identify the source audit, technical comparison, calibration and reliance study, logic experiment, intent experiment and explanation experiment. Every run shall store timestamp, source identity, dataset and rule-

catalogue hashes, split, seeds, software environment, calibration status and ActionNet evidence status in run_metadata.json. predictions.csv shall store the run ID, source identity, Campaign_ID, date, intent, reference action, system, seed, prediction, confidence, top two, conflict reasons, action-evidence status and latency. The application database separately retains interactive recommendation evidence, rule traces and managerial decisions. Chapter 5 reports the executed run identities and results.

4.7 Chapter Summary and Development Handoff

This chapter has taken the methodology of Chapter 3 and made it into a full development blueprint. The whole-system architecture, storage boundaries and human-review control is defined in Figure 4.1, the data-processing and evidence pipeline (training-only fitting, validation-only calibration and untouched test boundary) is defined in Figure 4.2, and the internal proposed method and its intent-aware, logic-constrained score construction is defined in Figure 4.3. Supporting figures give details of the minimum database, class, interaction and interface required for coding.

The development handoff will include the architecture, precise feature and split contracts, chosen model structures, equations, algorithms, 14 predefined safeguards, fallback semantics, baseline parity rules, environment settings and trace schema and constraints described in this chapter. These contracts shall be adhered to when coding. The system developed, actual run configuration and evidence of verification and evaluation carried out will be recorded in chapter 5. All formal tests rely on the assumption that the model, split, target, preprocessing, leakage control, decision logic, parameter policy, fallback and output semantics or privacy boundary are the same as specified in the corresponding Chapter 3 or Chapter 4 content, but any material deviation impacting the model, split, target, preprocessing, leakage control, decision logic, parameter policy, fallback, output semantics or privacy boundary must be corrected transparently in the content before the corresponding formal test is interpreted.

Chapter 5

Experiments, Results and Discussion

5 Experiments, Results and Discussion

5.1 Introduction

This chapter reports the completed technical and formative human evaluation of the intent-aware, logic-constrained decision-support prototype. It separates implementation evidence from measured outcomes and uses the same frozen action vocabulary, partitions, comparator definitions and metrics specified in Chapters 3 and 4. The chapter first reconciles the planned and executed setup, then reports the main comparison, component evidence, robustness, calibration, selective decision behaviour, human-review findings and failure analysis. Results are interpreted within the 125-case audited benchmark, 19-case held-out test partition, three predefined random seeds and 18-participant formative study; they are not presented as population-level SME or causal marketing evidence.

The primary technical question is not whether the system maximises reference-action agreement at any cost, but whether it provides a defensible balance among agreement, scenario fit, logical consistency, calibrated uncertainty, traceability and human review. Accordingly, the unconstrained learned condition is retained as a safety reference even when it achieves higher raw predictive scores. Calibration and human-centred findings are reported independently so that a positive explanation result cannot be used to conceal a negative calibration result.

5.2 Experimental Setup Recap

Table 5.1 reconciles the pre-specified design with the executed evaluation. The public campaign source and 125-case audited benchmark remained distinct. The benchmark was split chronologically into 87 training, 19 validation and 19 test cases. All neural configurations used seeds 42, 52 and 62, whereas deterministic comparators were executed once because their outputs do not vary by seed. Temperature scaling used only the 19 validation cases; the test labels were not used for fitting. The held-out test support was Increase = 3, Maintain = 7, Reduce = 5 and Investigate = 4. The validation partition contained no Increase case, which limits class-specific calibration claims.

Table 5.1

Reconciliation of Planned and Executed Evaluation

Evaluation element	Executed configuration	Interpretation boundary
Audited benchmark	125 cases; Train = 87, Validation = 19, Test = 19; no ID overlap or detected near-duplicate pair	The independent technical unit is the campaign case; the held-out sample is small.
Model repetition	Neural configurations: seeds 42, 52 and 62; deterministic comparators: one run	Seed-case outputs are repeated predictions of the same 19 test cases, not 57 independent cases.
Calibration	Temperature fitted on validation only; five fixed confidence bins; zero Increase validation cases	Aggregate, profile-specific evidence only; no Increase-class or population calibration claim.

Evaluation element	Executed configuration	Interpretation boundary
Safeguards and selective policy	Six hard safeguards, eight soft safeguards; RECOMMEND, CHECK ADVISED, HUMAN REVIEW REQUIRED and NO SAFE RECOMMENDATION	No marketing action is executed automatically.
Human study	18 complete participants, nine trials each, 162 responses; eight marketing practitioners and ten non-practitioners	Descriptive formative evidence; participant-clustered intervals; no practitioner-population claim.
Verification	57 automated Python tests and the TypeScript/Vite production build passed in the human-study evidence freeze	Evidence applies to the frozen local prototype rather than production deployment.

The actual evaluation extends the original B1-B4 progression with four supplementary reference comparators. B0: Majority-Class Popularity tests a frequency-only prediction; the Intent-Only Deterministic Baseline tests the supplied intent in isolation; the Weighted Deterministic Baseline tests fixed scoring without learned parameters; and the Unconstrained AI Baseline retains learned evidence while removing safeguard and feasibility controls. These additions do not redefine B1-B4. They provide lower-bound, deterministic and safety-reference views needed to interpret the proposed method fairly.

5.3 Results

5.3.1 Main Comparison

Table 5.2 reports the principal held-out results. Neural point estimates pool the three predefined runs over the same 19 test cases, while deterministic comparators use one prediction per case. The unit of generalisation remains the campaign case. Accuracy differences are therefore described in percentage points and accompanied, for the main learned comparisons, by case-clustered bootstrap intervals that resample the 19 campaign identifiers and retain all seed-specific predictions for a sampled case.

Table 5.2

Held-Out Recommendation Results for Baselines, Comparators and the Proposed Method

System	Accuracy	Macro F1	Top-2	Scenario fit	Conflict	ECE
B0 Majority-Class	36.84%	0.1346	63.16%	100.00%	10.53%	0.0109
Intent-Only Deterministic	68.42%	0.5938	78.95%	100.00%	10.53%	0.3158
Weighted Deterministic	57.89%	0.5071	84.21%	100.00%	10.53%	0.0620
B1 Heuristic Scorecard	52.63%	0.4722	78.95%	100.00%	10.53%	0.2874
B2 Predictive Only	63.16%	0.6003	94.74%	89.47%	21.05%	0.0553
B3 Predictive + Channel	63.16%	0.6003	100.00%	89.47%	21.05%	0.0684
B4 Integrated Backbone	78.95%	0.7831	94.74%	89.47%	21.05%	0.3984
Post-hoc Hard-Rule	63.16%	0.6444	84.21%	100.00%	0.00%	0.0742
Unconstrained AI	92.98%	0.9308	100.00%	100.00%	10.53%	0.6482
P Full Proposed Method	82.46%	0.8279	89.47%	100.00%	0.00%	0.5187

Relative to B4, the full method improved pooled recommendation accuracy from 78.95% to 82.46%, an increase of 3.51 percentage points, and Macro F1 from 0.7831 to 0.8279, an increase of 0.0448. Scenario fit increased by 10.53 percentage points and logical conflict decreased by 21.05 percentage points to zero. Relative to B1, accuracy increased by 29.82 percentage points and Macro F1 by 0.3557. The case-clustered 95% bootstrap intervals were [64.91%, 98.25%] for full-method accuracy and [61.40%, 94.74%] for B4 accuracy; the overlap reflects the small 19-case test set and prevents a broad superiority claim.

The unconstrained learned baseline achieved the highest raw accuracy (92.98%) and Macro F1 (0.9308), exceeding the full method by 10.53 percentage points in accuracy and 0.1029 in Macro F1. It nevertheless produced a 10.53% logical-conflict rate, whereas the proposed method produced none. Its case-clustered conflict interval extended to 26.32%, while the proposed-method interval remained [0, 0]. This is the central trade-off: safeguards sacrificed some exact primary-label agreement to avoid actions classified as unsafe or infeasible. The result supports a constrained decision-support interpretation, not a claim that the full method is the strongest unconstrained classifier.

B0's very small ECE must not be mistaken for high-quality calibration. It predicts a nearly constant majority-class probability, so low calibration error can coexist with only 36.84% accuracy and a Macro F1 of 0.1346. Conversely, B2 and B3 had stronger prediction performance but a 21.05% conflict rate. The post-hoc filter removed measured conflicts but remained at 63.16% accuracy, showing that late rejection is not equivalent to integrated score construction and selective review.

5.3.2 Quantitative Analysis

The branch comparison identifies where predictive performance originates. CampaignNet alone achieved 43.86% accuracy and a Macro F1 of 0.3697. ActionNet alone achieved 91.23% accuracy and 0.9087 Macro F1, while fixed CampaignNet-ActionNet fusion achieved 92.98% accuracy and 0.9308 Macro F1. The action-supervised branch therefore supplied most reference-action discrimination. CampaignNet still contributes ROI-oriented evidence and sensitivity information, but the results do not support portraying it as the dominant classifier.

The selective policy was evaluated on the 19 held-out cases using a minimum automatic confidence of 0.35 and a minimum top-two margin of 0.02. Nine cases were automatically classified as RECOMMEND, one as CHECK ADVISED, eight as HUMAN REVIEW REQUIRED and one as NO SAFE RECOMMENDATION. Automatic coverage was therefore 47.37% and review demand was 52.63%. The nine automatically accepted recommendations had 100% reference-action agreement and a 0% conflict rate. One case was flagged as outside benchmark coverage, and the no-safe rate was 5.26%. These results show that abstention and review states are reachable; they do not demonstrate production safety beyond the benchmark.

Table 5.3

Calibration and Selective-Decision Evidence

Evidence	Observed result	Conclusion
Validation ECE, raw to calibrated	0.2181 to 0.3087, averaged across seeds	Calibration did not improve reliability.
Validation NLL, raw to calibrated	2.2923 to 2.4006	The negative calibration result is corroborated by NLL.
Validation Brier score, raw to calibrated	0.6027 to 0.6770	The negative calibration result is corroborated by Brier score.

Evidence	Observed result	Conclusion
Automatic recommendation coverage	9/19 = 47.37%	Only cases passing confidence, margin, feasibility and coverage gates were automatic.
Accepted recommendation accuracy	9/9 = 100.00%	Descriptive held-out evidence; the support is nine cases.
Automatic conflict rate	0/9 = 0.00%	No accepted automatic recommendation violated the conflict definition.
No-safe outcome	1/19 = 5.26%	The reachable fallback was exercised rather than forcing an unsafe action.

Temperature scaling was fitted only on validation data, but it worsened every recorded validation seed: ECE changed from 0.2025 to 0.3083 for seed 42, from 0.2602 to 0.3619 for seed 52 and from 0.1917 to 0.2559 for seed 62. The mean change was therefore an increase of 0.0906 rather than an improvement. Validation support was Maintain = 8, Reduce = 6, Investigate = 5 and Increase = 0. The fitted temperature is consequently retained as an evaluated implementation component, but the thesis does not claim successful four-class calibration or calibration of the Increase action.

Per-intent reliability analysis was likewise descriptive. Each intent subset contained only four to nine validation cases and no Increase reference action. Some seed-intent cells changed in different directions, but aggregate ECE, NLL and Brier score did not justify a calibration-benefit claim. This finding directly limits RQ2 and RO3 rather than being treated as an implementation error or omitted from the analysis.

5.4 Component or Ablation Analysis

Table 5.4 separates learned evidence, intent, safeguards, calibration, explanation and review-gate effects. Removing rules produced the same predictive scores as the unconstrained AI condition: accuracy rose to 92.98%, but logical conflict rose from 0% to 10.53%. Removing intent reduced accuracy to 73.68%, Macro F1 to 0.7514 and scenario fit to 94.74%. Removing calibration did not change action predictions or conflict rate, but reduced test ECE from 0.5187 to 0.2942; this reinforces the negative calibration finding. Removing the structured explanation or review gate did not alter action predictions because these components operate after score construction.

Table 5.4

Component and Ablation Results

Configuration	Accuracy	Macro F1	Scenario fit	Conflict	Main interpretation
CampaignNet only	43.86%	0.3697	94.74%	12.28%	ROI evidence alone is insufficient for four-action prediction.
ActionNet only	91.23%	0.9087	100.00%	10.53%	Action supervision supplies most predictive discrimination.
Fixed fusion	92.98%	0.9308	100.00%	10.53%	Fusion improves raw prediction but does not enforce safeguards.
A1 without rules	92.98%	0.9308	100.00%	10.53%	Rules trade some primary-label agreement for zero conflict.
A2 without intent	73.68%	0.7514	94.74%	5.26%	Intent contributes to contextual fit and action agreement.
A3 without calibration	82.46%	0.8279	100.00%	0.00%	Predictions are unchanged; ECE improves to 0.2942.

Configuration	Accuracy	Macro F1	Scenario fit	Conflict	Main interpretation
A4 without explanation	78.95%*	-	-	0.00%*	Explanation presence changes, not score construction.
A5 without review gate	78.95%*	-	-	0.00%*	Automatic coverage rises from 47.37% to 52.63%; accepted accuracy falls to 90%.
P full method	82.46%	0.8279	100.00%	0.00%	Best constrained balance among the specified B1-B4 comparison.

The asterisked A4 and A5 values come from the seed-42 selective-policy audit and are not directly pooled with the three-seed values in the other rows. Under this audit, P and A4 both obtained 78.95% accuracy and 47.37% automatic coverage, confirming that explanation generation does not alter the action label. A5 increased automatic coverage to 52.63% but reduced accepted accuracy from 100% to 90%. The review gate therefore has a distinct safety-support role even though it does not change the underlying recommendation scores.

The safeguard intervention audit examined all 125 benchmark cases as a diagnostic rather than an unbiased test. Thirteen cases changed from an unacceptable unconstrained action to a reviewed safe-set outcome, including reachable Investigate and no-safe states. All 13 satisfied the independently reviewed acceptable-action-set criterion, while none matched the single primary reference label. This apparent discrepancy is expected: the primary label measures exact action agreement, whereas the intervention audit asks whether the constrained outcome is acceptable when the unconstrained action is unsafe. Both measures are retained and clearly separated.

Table 5.4A

Safeguard Intervention and Consistency Closure

Measure	Result	Rate	Interpretation
Benchmark-wide constrained interventions	13/125	10.40%	Diagnostic intervention rate; not an unbiased held-out estimate.
Independently reviewed acceptable-set outcomes	13/13	100.00%	Every changed constrained outcome belonged to the reviewed acceptable action set.
Exact primary-label matches among interventions	0/13	0.00%	Exact-label equality and set-based safety answer different evaluation questions.
Held-out unconstrained logical conflicts	2/19	10.53%	Conflict observed in the safety-reference condition.
Held-out proposed-method logical conflicts	0/19	0.00%	No conflict was detected under the frozen 14-rule catalogue.

Table 5.4A closes Block 3 by reporting both the benchmark-wide intervention rate and held-out conflict evidence. The 10.40% intervention rate is diagnostic because it uses all 125 audited cases. The held-out comparison remains the unbiased performance evidence: conflict decreased from 10.53% for the unconstrained condition to 0% for the proposed method. The 13/13 acceptable-set result supports safeguard consistency, whereas the 0/13 exact-label result confirms that acceptability and primary-label agreement must remain separate.

Table 5.4B

Per-Intent Effect of Removing Intent Conditioning

Intent (cases)	A2 accuracy	P accuracy	Difference	Scenario fit A2 to P	Conflict A2 to P
Growth (5)	93.33%	93.33%	0.00 pp	100.00% to 100.00%	0.00% to 0.00%
Retention (9)	70.37%	88.89%	+18.52 pp	88.89% to 100.00%	11.11% to 0.00%
Survival (5)	60.00%	60.00%	0.00 pp	100.00% to 100.00%	0.00% to 0.00%

Table 5.4B closes Block 4 using averages across seeds 42, 52 and 62. Intent conditioning produced its clearest contribution for Retention: accuracy increased by 18.52 percentage points,

Macro F1 increased from 0.4969 to 0.8889, scenario fit increased by 11.11 percentage points and conflict decreased by 11.11 percentage points. Growth and Survival outcomes were unchanged. The evidence therefore supports a localised Retention benefit within the 19-case test partition, not a uniform improvement across all three intents. Because A2 was not presented as a separate human-study condition, participant actionability or willingness effects cannot be attributed specifically to intent conditioning.

5.5 Robustness or Sensitivity Analysis

Robustness was assessed through three fixed seeds, case-clustered bootstrap intervals, branch removal, component ablation, per-intent calibration tables, partition-leakage checks and selective-policy thresholds. The 125-case integrity audit found 125 unique campaign identifiers, zero exact feature-duplicate pairs, zero near-duplicate pairs under the predefined 0.05 normalised-distance rule and no identifier overlap between the 87/19/19 partitions. Every record used a valid action from Increase, Maintain, Reduce and Investigate.

The main learned results were directionally stable enough to show the trade-off but statistically imprecise. The full-method case-clustered 95% intervals were [64.91%, 98.25%] for accuracy and [0.5786, 0.9715] for Macro F1. The B4 intervals were [61.40%, 94.74%] and [0.5580, 0.9308], respectively. The unconstrained intervals were [80.70%, 100.00%] and [0.7080, 1.0000]. These overlaps and the small class supports mean that differences should be interpreted as benchmark-specific descriptive evidence. The zero-conflict result is stronger within the observed cases but remains bounded by the completeness of the predefined rule catalogue.

Local latency remained well below the two-second interactive target. The proposed method had a mean latency of 6.30 ms, p50 of 6.04 ms and p95 of 8.56 ms across the recorded technical runs. These timings support RQ4 for the local CPU prototype, but they exclude network delay, production database contention, multi-user load and external-service latency.

5.6 Error and Failure Analysis

The held-out failure review identified four primary-reference disagreements for the unconstrained learned condition: two Reduce recommendations also triggered evidence-sufficiency conflicts, one disagreement affected Investigate and one affected Maintain. For the full selective system, three held-out cases disagreed with the primary reference and seven otherwise correct cases were escalated because confidence, margin, feasibility or coverage requirements were not satisfied. Escalation is therefore not identical to predictive failure; it records insufficient evidence for automatic use.

Table 5.5

Failure Modes, Controls and Residual Risks

Observed issue	Evidence and implemented response	Residual interpretation
Unconstrained safeguard conflict	10.53% conflict rate; rules and feasibility filtering reduce it to 0% for P	The result depends on the documented 14-rule catalogue; unknown constraints remain possible.
Insufficient evidence	Two unconstrained Reduce cases violated the evidence-sufficiency rule; selective review prevents automatic use	Additional representative data may reduce unnecessary abstention.
No safe candidate	One held-out case produced NO SAFE RECOMMENDATION	The system correctly abstains, but a manager must investigate or revise the case.
Outside benchmark coverage	One held-out case was marked outside coverage	This is a distance-based warning, not a complete real-world out-of-distribution detector.
Calibration degradation	ECE, NLL and Brier score all worsened after temperature scaling	Confidence must not be described as successfully calibrated; larger class-balanced validation data are required.
Primary label versus acceptable set	13 diagnostic interventions were set-valid but 0/13 matched the single primary label	Exact agreement and safety acceptability answer different questions and must not be merged.

The structured explanations passed all four automated faithfulness checks for all 19 held-out cases: source-fact support, triggered-rule consistency, material-perturbation response and unused-metadata invariance. This provides traceability evidence for the implemented templates and rule traces. It does not prove that every natural-language explanation is complete, persuasive or cognitively optimal, which is why the human evaluation remains separate.

5.7 Discussion

The technical results support the project's main design rationale with an important qualification. Integrating intent and predefined safeguards improved the constrained balance relative to B1-B4: the full method achieved 82.46% accuracy, 0.8279 Macro F1, 100% scenario fit and zero measured conflict. However, the unconstrained learned condition achieved higher raw accuracy. The contribution is therefore not a universally superior predictor. It is an auditable decision-support construction that makes the safety-versus-agreement trade-off explicit and permits abstention when evidence or feasibility is inadequate.

The human study produced 162 complete trial responses from 18 participants, including eight marketing practitioners and ten non-practitioners. Each participant reviewed nine cases, and every condition was seen by all 18 participants, although trial counts varied from 25 to 29. Table 5.6 reports participant-weighted means with participant-clustered 95% bootstrap intervals. H6: Complete Calibrated DIAC obtained the highest actionability, explanation usefulness and trust-alignment values, but these results must be interpreted alongside calibration failure and data-quality limitations.

Table 5.6

Formative Human-Study Results by Presentation Condition

Condition	Trials	Actionability	Explanation usefulness	Trust alignment	Willingness to review
H1 Raw-Uncalibrated	25	2.97 [2.78, 3.17]	3.14 [2.83, 3.47]	0.535 [0.472, 0.597]	3.39 [3.19, 3.61]
H2 Calibrated-Confidence Only	29	1.31 [1.00, 1.78]	1.28 [1.00, 1.67]	0.236 [0.111, 0.375]	4.83 [4.56, 5.00]

Condition	Trials	Actionability	Explanation usefulness	Trust alignment	Willingness to review
H3 Structured-Uncalibrated	27	4.19 [4.06, 4.39]	3.86 [3.39, 4.28]	0.667 [0.542, 0.778]	2.31 [1.97, 2.72]
H4 Terse-Calibrated	29	2.17 [2.00, 2.44]	2.08 [1.92, 2.33]	0.326 [0.250, 0.424]	4.89 [4.67, 5.00]
H5 Structured-Calibrated No Caveat	25	3.42 [3.14, 3.67]	3.31 [2.94, 3.61]	0.590 [0.514, 0.667]	2.89 [2.50, 3.31]
H6 Complete Calibrated DIAC	27	4.81 [4.53, 5.00]	4.67 [4.11, 5.00]	0.924 [0.840, 1.000]	1.39 [1.11, 1.67]

The pre-specified H6-H3 contrast helps distinguish the complete presentation from structured text without calibration. H6 increased participant-weighted actionability by 0.611 [0.306, 0.861], explanation usefulness by 0.806 [0.389, 1.278] and trust alignment by 0.257 [0.153, 0.375]. However, willingness to review decreased by 0.917 [-1.222, -0.583]. H2-H1 also showed that confidence-only presentation reduced trust alignment by 0.299 and actionability by 1.667 while increasing willingness to review by 1.444. Thus, confidence information alone made participants more cautious but less positive, whereas the complete DIAC package was rated more useful but also reduced review willingness. This mixed pattern does not justify an unrestricted appropriate-reliance claim.

Eighteen trials contained an incorrect recommendation; three were accepted without checking, giving an incorrect uncritical-acceptance rate of 16.67%. Across 132 trials classified as incorrect or low confidence, 28 were accepted without checking, giving an at-risk uncritical-acceptance rate of 21.21%. In addition, counterbalancing was uneven, CB-3 had no participant, two participant blocks pre-dated the main October collection period, 13 participants recorded less than 120 seconds across nine trials, the largest repeated response vector occurred 24 times and no free-text comment was submitted. These quality signals require the human findings to remain formative and descriptive.

5.8 Answers to Research Questions and Chapter Summary

Table 5.7 answers each research question using the evidence reported in this chapter. The verdicts deliberately distinguish implementation success from empirical support and retain the project's scope boundaries.

Table 5.7

Research-Question Verdict Matrix

Research question	Principal evidence	Verdict	Boundary
RQ1. Is there more alignment with audited reference actions, actionability and rule consistency when both predictive and campaign-level channel evidence are combined with binary safeguard evaluation and weighted soft influence on candidate-action scores, versus defined partial baselines?	P exceeded B1-B4 in the declared constrained comparison; versus B4, accuracy increased 3.51 percentage points, Macro F1 increased 0.0448, scenario fit increased 10.53 percentage points and conflict decreased from 21.05% to 0%. Unconstrained AI retained higher raw accuracy but 10.53% conflict.	Supported within the audited benchmark	Small 19-case test; wide overlapping intervals; safeguard completeness and exact-label/set-valid distinctions limit generalisation.
RQ2. What is the impact of calibrated model confidence and structured Driver–Implication–Action–Caveat explanations on confidence reliability, explanation usefulness and appropriate reliance during human review?	Calibration worsened validation ECE, NLL and Brier score. DIAC improved actionability and explanation usefulness, but willingness to review decreased and at-risk uncritical acceptance was 21.21%.	Partially supported	Explanation benefit is formative; confidence calibration and unrestricted appropriate reliance are not supported.
RQ3. Are there project-defined Survival, Retention and Growth conditions that enhance fit and usefulness of scenarios as judged by the project?	Averaged across seeds 42, 52 and 62, intent conditioning improved Retention accuracy from 70.37% to 88.89%, scenario fit from 88.89% to 100%, and logical conflict from 11.11% to 0%; Growth and Survival outcomes were unchanged. The human design did not independently randomise the presence of intent.	Partially supported	Technical fit is supported; an independent human usefulness effect of intent was not isolated.

Research question	Principal evidence	Verdict	Boundary
RQ4. Is the deployed prototype able to give traceable recommendations in reproducible configurations with an acceptable local response latency?	All 19 explanations passed automated faithfulness checks; tests/build passed; mean/p50/p95 local latency was 6.30/6.04/8.56 ms; audit and selective statuses were persisted.	Supported for the local prototype	Not a production deployment, concurrency, network or live-campaign result.

In summary, the full framework met the declared constrained comparison against B1-B4 and produced traceable, low-latency recommendations with zero observed conflicts. The unconstrained model achieved higher raw predictive scores, demonstrating the cost of enforcing safeguards. Intent contributed to scenario fit, while the selective policy achieved 47.37% automatic coverage with 100% accepted accuracy on nine cases. The structured DIAC presentation received strong formative ratings, but temperature scaling failed to improve reliability and the human evidence revealed reduced willingness to review. Chapter 6 therefore concludes that the artefact is technically complete for its prototype scope while calibration and broad human-effect claims remain incomplete.

Chapter 6

Conclusion and Recommendations

6 Conclusion and Recommendations

6.1 Summary of Findings

This study addressed the gap between campaign analytics and bounded managerial action by developing an intent-aware, logic-constrained neuro-symbolic decision-support prototype. The artefact combines a predictive ROI branch, an action-supervised branch, training-derived campaign-level channel evidence, a four-action candidate space, explicit Survival, Retention or Growth intent, six hard safeguards, eight weighted soft safeguards, selective review states and Driver–Implication–Action–Caveat explanations. It supports human review only and does not execute marketing actions.

The 125-case audited benchmark was frozen into 87 training, 19 validation and 19 test cases with no detected identifier leakage or near-duplicate pair under the specified rule. Against the formal B1-B4 sequence, the complete method achieved the best constrained balance: 82.46% pooled accuracy, 0.8279 Macro F1, 100% scenario fit and zero measured logical conflict. The unconstrained learned condition achieved higher raw accuracy of 92.98% but produced a 10.53% conflict rate. This establishes the principal finding as a safety-agreement trade-off rather than universal predictive superiority.

Validation-only temperature scaling did not improve confidence reliability. Mean validation ECE increased from 0.2181 to 0.3087, and NLL and Brier score also worsened. The calibration objective was therefore only partially achieved. The selective policy nevertheless produced reachable review and no-safe outcomes: nine of 19 test cases were automatically recommended with 100% agreement and zero measured conflict, while the remaining ten required checking, human review or no-safe handling.

The formative human study included 18 complete participants and 162 trials. H6: Complete Calibrated DIAC received the strongest participant-weighted actionability, explanation-usefulness and trust-alignment ratings, but it also reduced willingness to review relative to H3. Incorrect or low-confidence recommendations were accepted without checking in 21.21% of at-risk trials. Together with counterbalancing and response-quality limitations, this finding supports the usefulness of structured explanation but not a broad claim of successful trust or reliance calibration.

6.2 Achievement of Research Objectives

Table 6.1 evaluates the four objectives using the exact wording stated in Chapter 1. An objective is marked achieved only when its implementation and required evidence are complete; a negative empirical result produces partial achievement rather than being converted into success.

Table 6.1

Achievement of Research Objectives

Research objective	Evidence	Status	Reason
RO1. To develop and freeze an audited decision benchmark based on public campaign records, project defined intent scenarios and independently reviewed reference actions.	125 valid cases; 87/19/19 split; action-set and safeguard review; provenance, hashes, duplicate and leakage checks.	Achieved	The benchmark and its audit evidence were frozen before the reported evaluation.
RO2. Implementing a predictive and channel-evidence backbone, a four-action candidate space, predefined weighted safeguards and explicit intent conditioning.	CampaignNet, ActionNet, campaign-level evidence, four actions, hard/soft safeguards, no-safe state and intent pathways were implemented and ablated.	Achieved	Every named component is traceable to code, diagrams, algorithms and evaluation evidence.
RO3. To adopt validation only model confidence calibration and Driver–Implication–Action–Caveat explanations for human review.	Validation-only scaling and DIAC were implemented; explanations passed 19/19 faithfulness audits and received strong formative ratings; calibration worsened ECE, NLL and Brier score.	Partially achieved	Adoption and explanation support were achieved, but the intended calibration benefit was not.
RO4. To assess reference-action agreement, actionability, scenario fit, rule consistency, calibration, reliance, auditability and latency with respect to defined baselines and ablations.	B0/B1-B4, deterministic and safety references, branch/component ablations, selective audit, human study, faithfulness checks and latency were reported.	Achieved within prototype/formative scope	The planned dimensions were evaluated, but small-sample and local-execution boundaries remain.

Overall, three objectives were achieved within the declared prototype scope and RO3 was partially achieved. This conclusion preserves the distinction between building a calibration mechanism and demonstrating that it improves calibration. It also avoids treating favourable interface ratings as proof of production effectiveness.

6.3 Research Contributions

The first contribution is the artefact itself: a reproducible action-level decision-support prototype that transforms campaign evidence and one manager-selected primary intent into Increase, Maintain, Reduce, Investigate or an explicit no-safe/review outcome. The artefact is not a channel-selection engine, an autonomous campaign optimiser or a complete marketing-strategy generator. Its contribution lies in bounded and reviewable recommendation construction.

The second contribution is methodological. The project separates learned evidence from explicit business safeguards, distinguishes hard feasibility rejection from weighted soft influence, uses a canonical action order across every comparator, and evaluates exact primary-label agreement separately from acceptable-action-set safety. The addition of B0, deterministic references, B1-B4, an unconstrained safety reference, branch ablations and post-hoc filtering makes the evidence chain more interpretable than a single full-versus-baseline comparison.

The third contribution is an auditable evaluation package. It includes benchmark hashes and leakage checks, validation-only calibration records, five-bin reliability evidence, case-level predictions, selective-policy states, rule traces, explanation-faithfulness tests, failure cases, case-clustered technical uncertainty and participant-clustered human-study intervals. This package enables later researchers to reproduce both positive and negative findings.

The fourth contribution is empirical but deliberately limited. Within the audited benchmark, integrated safeguards removed measured conflict while preserving stronger accuracy and Macro F1 than B1-B4. Within the formative study, structured DIAC explanations were rated as more actionable and useful, yet calibration and appropriate-reliance outcomes were mixed. Reporting these limits is part of the contribution because it identifies where additional evidence is required.

6.4 Limitations

The benchmark contains only 125 project-constructed decision cases derived from one public campaign source, with 19 cases in each validation and test partition. The test support per action is small, and the validation set contains no Increase case. Consequently, class-specific reliability, statistical precision and external generalisation are limited. The source records campaign-level outcomes rather than channel-specific incremental effects, so campaign-level channel evidence must not be interpreted as causal attribution or channel synergy.

The rule catalogue is predefined and reflects documented project assumptions. Zero observed conflict means no conflict was detected under these 14 rules; it does not prove safety under every business, legal or operational constraint. The all-case intervention audit provides diagnostic set-based evidence and is not an unbiased held-out performance estimate. Likewise, the benchmark-coverage flag is a local distance-based detector, not a comprehensive production out-of-distribution system.

The confidence-calibration mechanism failed to improve ECE, NLL or Brier score on the observed validation distribution. Because the calibration set is small and lacks Increase examples, the calibrated probabilities should not be presented as reliable population probabilities. Selective review reduces automatic coverage to 47.37%, which is appropriate for this prototype but may be operationally costly in practice.

The human study is formative. Although it contains eight marketing practitioners and ten non-practitioners, the subgroups are too small for practitioner-specific inference. Counterbalancing was uneven, CB-3 was absent, two blocks pre-dated the main collection period, response times were short for many participants, repeated response patterns occurred and no free-text comments were obtained. The study measures reactions to supplied recommendations rather than real business outcomes, long-term behaviour or unaided-versus-assisted performance.

Finally, the prototype was evaluated locally. Latency does not include network, concurrent-user, production MySQL, failover or external-service effects. The design accepts one explicit primary intent and produces an action-level recommendation; mixed intents, channel-first recommendations, budget allocation, content design and campaign execution are outside the completed scope.

6.5 Recommendations and Future Work

The first priority is to build a larger, independently sourced and class-balanced audited benchmark. A larger validation set must include adequate Increase support before refitting or comparing temperature scaling, vector scaling, isotonic calibration or class-conditional methods. Any replacement should be selected on validation data only and evaluated with ECE, NLL, Brier score and reliability support, not merely retained because it produces visually smoother confidence values.

The second priority is external validation. Future work should recruit a larger practitioner sample, enforce complete counterbalancing, record recruitment and session-quality evidence, include an unaided decision condition and analyse incorrect-recommendation behaviour as a primary safety endpoint. Longer field studies should measure decision quality, workload, adoption and downstream campaign outcomes without allowing the system to execute actions autonomously.

The third priority is to expand business coverage without changing the current claims retrospectively. A separate channel-first research version could recommend a channel and then an action-level configuration, but it would require new channel-specific labels, baselines, diagrams and evaluation. Mixed or weighted intents could be studied only after defining conflict resolution and ground-truth procedures. Firm size, maturity, prohibited actions and operational capacity should be added only from genuine sources rather than inferred or invented.

The fourth priority is production engineering: authenticated deployment, role and permission review, database migration testing, concurrency and load testing, monitoring, rollback, audit retention and privacy-impact review. Coverage detection should be extended beyond a single distance rule, and safeguard maintenance should include versioned approval, expiry and post-deployment incident review.

6.6 Final Conclusion

The project demonstrates that predictive evidence, explicit business intent and predefined safeguards can be organised into a traceable action-level decision-support workflow for SME-oriented campaign cases. Within the audited benchmark, the complete method outperformed the formal B1-B4 progression on the declared constrained comparison, achieved zero measured conflict and supported a reachable no-safe outcome. The prototype also generated faithful structured explanations at interactive local latency.

The study does not establish that the proposed framework is the most accurate unconstrained classifier, that temperature scaling successfully calibrated confidence, that users always relied appropriately, or that the system improves real SME marketing performance. Instead, its defensible result is narrower: the framework exposes and manages a measurable trade-off between action agreement and safeguard consistency, while retaining human review and complete evidence traces. With larger external benchmarks, stronger calibration data and a more rigorous practitioner study, this foundation can support a future publication-ready evaluation.

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Appendices

TARUMT
TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

ETHICS COMMITTEE (Checklist)

The nature of this project is most appropriately described as research involving: (more than one may apply):

	Y	N
– behavioural observation	/	
– self-report questionnaire(s)		/
– interview(s)		/
– qualitative methodologies (e.g. focus groups)		/
– psychological experiments		/
– epidemiological studies		/
– psychiatric or clinical psychology studies		/
– human physiological investigation(s)		/
– animal model(s)		/
– human tissue		/
- waiver of consent for the use of personal information in medical research, or personal health information		/
– human genetic analysis		/

/

– Other (please specify in the box below)

This study investigates human interaction with a decision-support system through structured behavioural task completion. Participants were informed of the purpose of the study, their rights, and confidentiality arrangements before participation. Written informed consent was obtained, and all collected data were anonymised and stored securely. The study involved minimal risk and was conducted in accordance with the approved ethics protocol.

Submitted by:
Name : Ng Jing Yong
Date : 10/11/2026

In order to enhance better understanding of the project, students should as far as possible include all directly relevant materials, figures or diagrams in the **main body** rather than in the Appendix. The appendix is reserved only for items which may not directly be relevant or essential to enhance a reader's understanding of the project, or which may interrupt the smooth reading of the project document (for example being too voluminous).

Appendices should only include supportive materials **directly referred** to in the writing and should be kept to a **minimum**, e.g. selected pages of an annual report, not the entire document. Examples of items included in Appendices are:

- Company's report and documentation, such as sample invoice, purchase order form, etc.
- Project meeting documentation e.g. minutes of meetings, tracking documents, memos etc.
- Questionnaires and results, interview questions and results, pilot test and results, observation sheet and results, experiment test plan and results, etc.
- Analysis/design diagrams (only those not incorporated in the main body of the report).

If there is more than one appendix, they should be identified as A, B, etc (e.g. Appendix A). Formulae and equations in appendices should be given separate numbering: Eq. (A.1), Eq. (A.2), etc.; in a subsequent appendix, Eq. (B.1) and so on. Similarly for tables and figures: Table A.1; Fig. A.1, etc.

IMPORTANT NOTE TO STUDENTS

APPENDIX *n* User Guide

- List the username and password of multiple roles (if applicable)
- Provide clear screen shots of each page, explain the functions of each button

As a rough guide, the user guide should include the following sections:

System Document

In this section, students should provide the following pieces of information:

- **System (hardware and software requirements)**. Students should describe the minimum hardware and software requirements to install the software application which has been developed by the students, for example, DBMS, OS, program development tools etc.
- **Installation**. Under this section, students should create an 'installation' CD and provide a brief step-by-step guide on how a new user can install the software on a computer system. Students should indicate any special setup information, such as the specific location of placing the database files, the SQL statements to add tables, etc. Software source code should also be included in this CD. Students are not required to print out the software code.

Operation Document

Under this section, students are required to provide a brief step-by-step guide on how to use the installed software. The guides should teach the user how to run the software and use its major functions and features. For example, steps show guide the users on how to run the system, e.g. to use an executable file or to use the IDE. The login information such as username and password for each of the different users (or roles) must be provided. Some screen interfaces would be useful.

APPENDIX *n+1* Developer Guide*

*To be included for ***Smart Campus Projects*** and ***Real-Life Projects***. This section should include the following:

- List the necessary software, installer, API, library that must be installed
- List the authentication details, such as username and password for all security

Other Appendices:

- Supporting documents
- Non-disclosure agreement (if applicable)
- Other detailed documentation, such as important references, interview results, survey results, etc.

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